Chapter 5

Composites

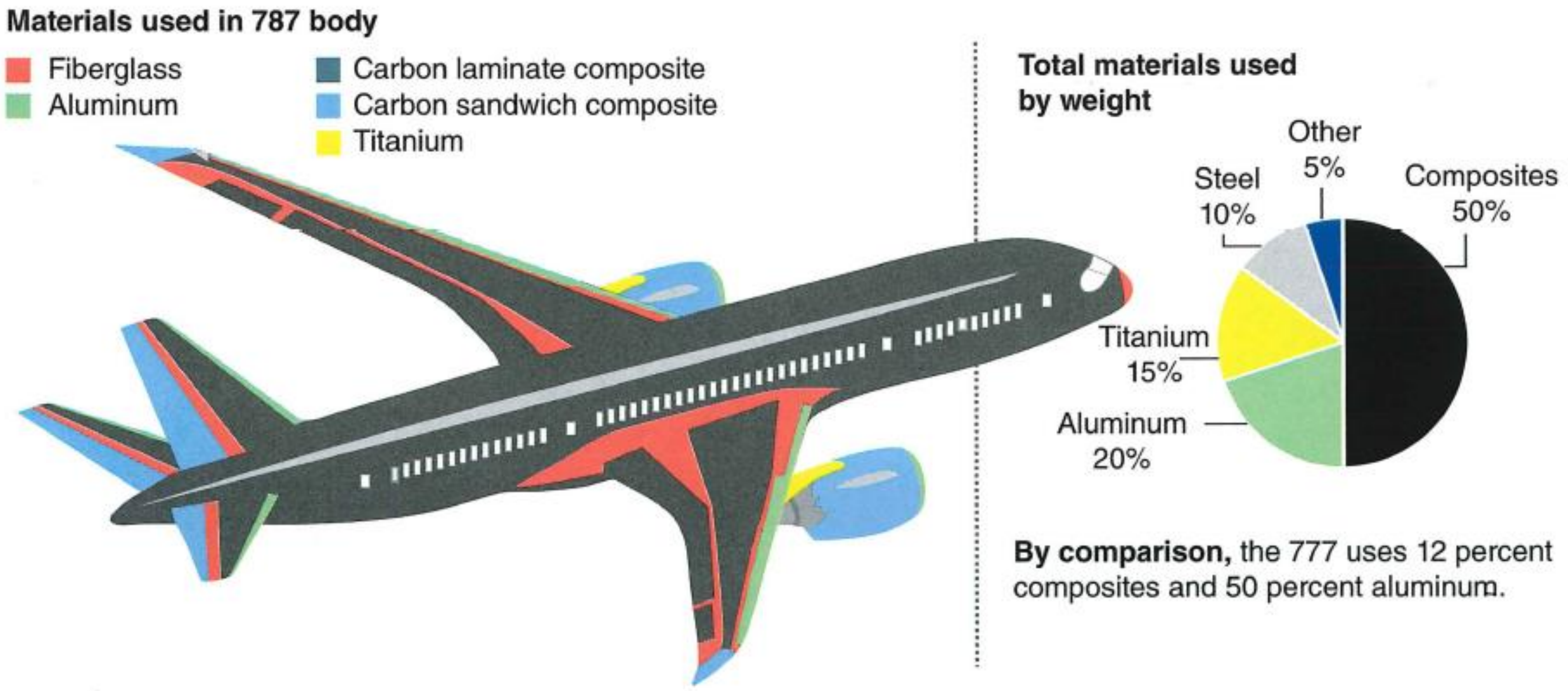


FIGURE 11-1 Material composition of modern transport aircraft.

Objectives

Students will be able to

1. Describe and Identify the construction of composite (laminated structure).
2. List and Describe the properties of laminated structure.
3. Describe and List the typical properties and characteristic of the materials (Fibers and Matrix) used for aircraft composite.
4. Describe and List the uses of Adhesives used for composite manufacturing.
5. Describe and identify the construction of sandwich structure.
6. List the properties of sandwich structure.

Objectives

Students will be able to

7. Describe and List the properties and characteristic of the materials used for sandwich structure.
8. Describe and List the Manufacturing and In-Service Damages of composites.
9. Describe and List the Nondestructive Inspections for defects in a composite.
10. Describe and identify the different damage categories
11. List the vacuum bagging components and Describe how a typical vacuum bagging is done.

Objectives

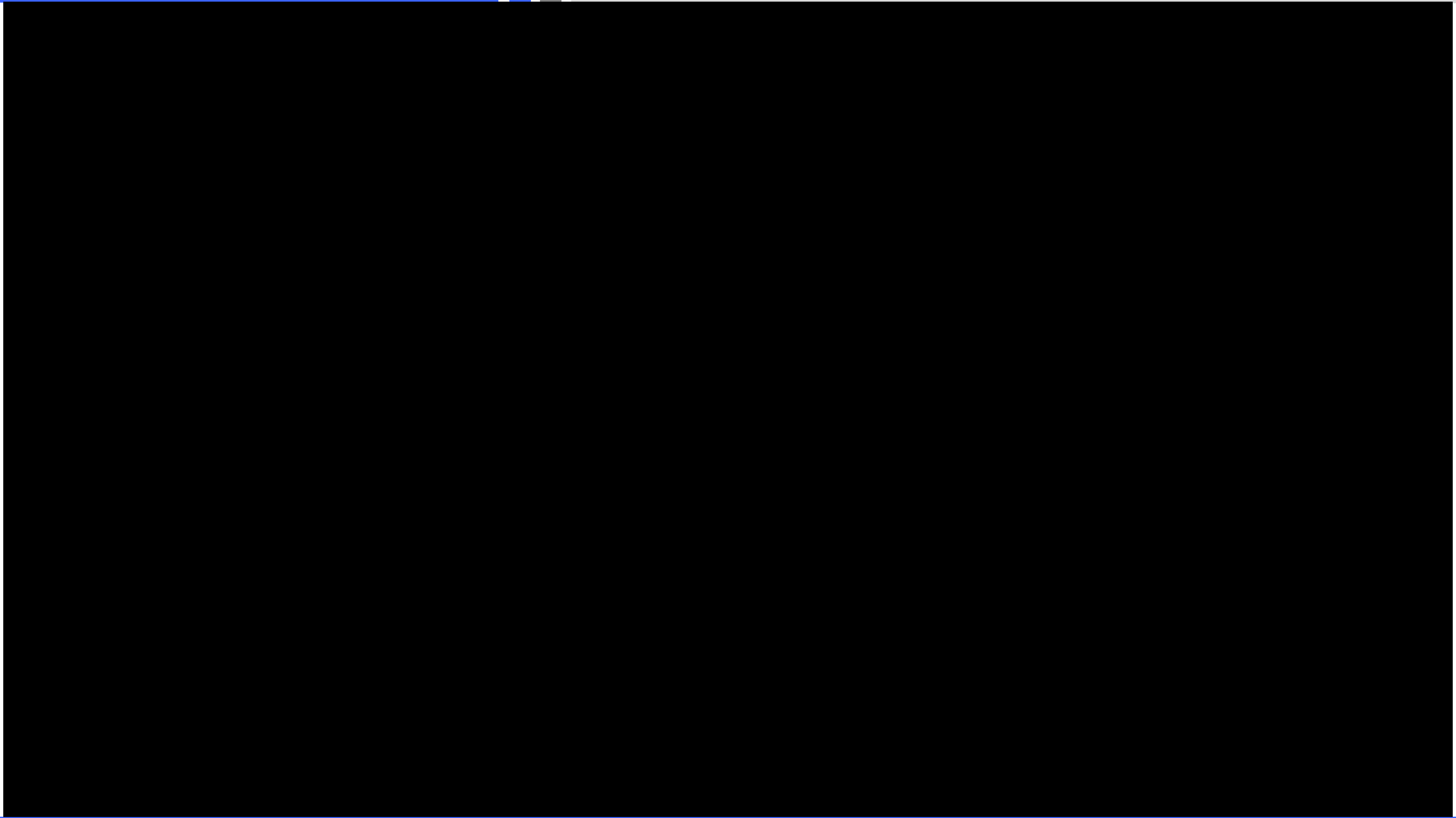
Students will be able to

12. Describe how the bonded repair is to be performed typically.
13. Describe and identify the type of composite repair: Bonded VS Bolted.
14. Describe the requirement of fasteners, repar doubler and tools for a bolted repair.

Content

I. Introduction

Why Composites???*



Why Composites???*

Composites	Metals
Lightweight compared to metals for the same strength to weight ratio.	Weight saving is obtained by machining like lightening holes, skin thickness variation etc.
Typically have a lower thermal expansion coefficient than metals.	Typically have a higher thermal expansion coefficient than composites.
Composite is able to have a complex shape as one piece, this reduces bolted joint assemblies.	Metallic bolted joints are generally more efficient than composite bolted joints
Corrosion resistant except for carbon plies with aluminium.	Susceptible to corrosion.
More fatigue resistant than metal.	Metals are susceptible to fatigue.

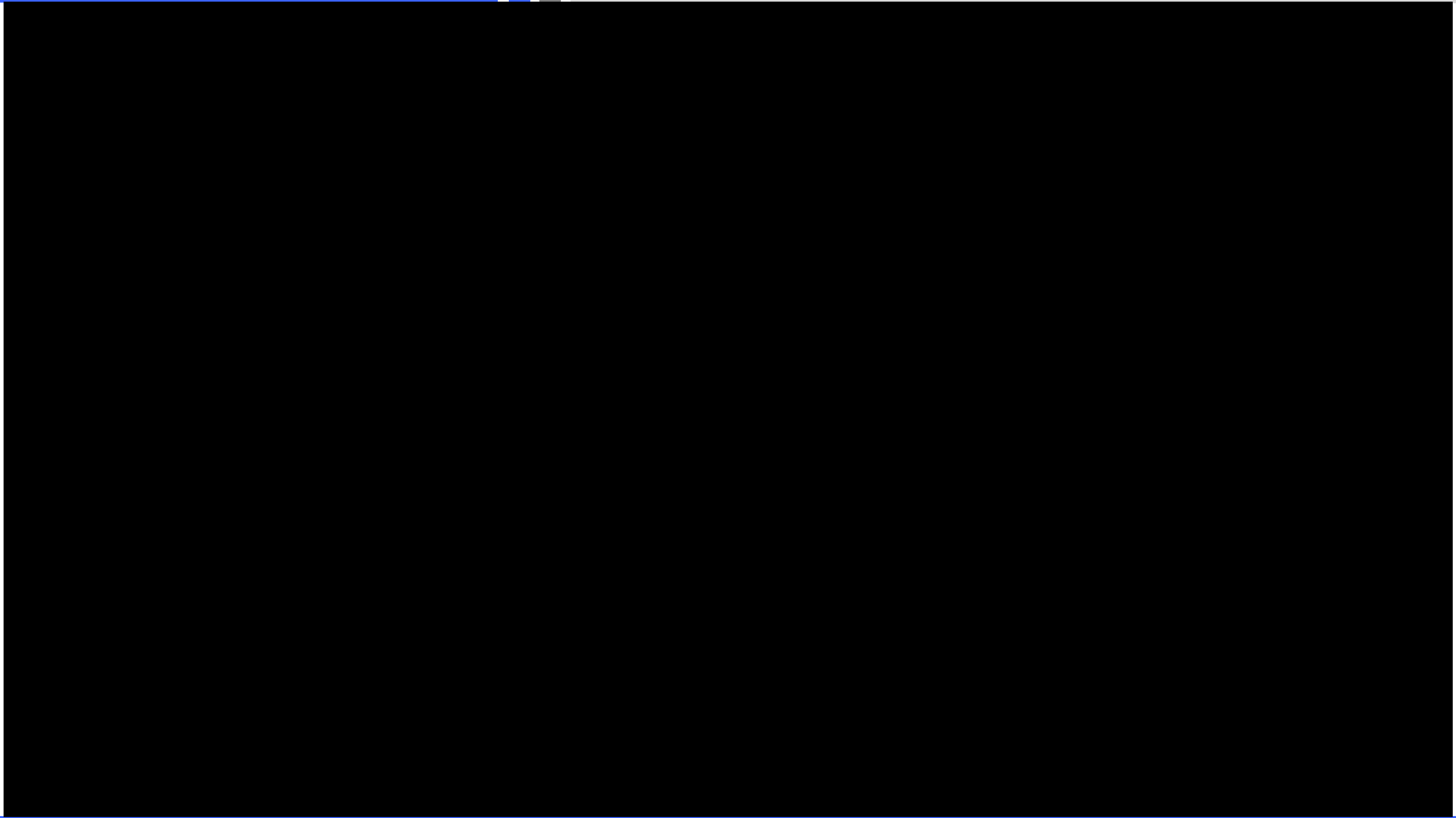
Table 1: Comparison of properties of composite & metal

I. Introduction

- Laymen definition of composite:
  - Mix two or more different materials together, the individual materials' properties and identities remained and the mixture has superior properties.
- Examples of composites around us:
  - Plants and Trees - Fibres & Saps
  - Concrete Wall - Cements & Steel Reinforcements
- Increased use of composite material on aircraft, example B787 and A350, due to rising oil price.
- Plastic is the industry terminology for composite material on the aircraft.

II. Laminated Structures

Introduction



Quiz

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Welcome to the quiz (Why Composite???)

Click the "Start Quiz" button to proceed

Start Quiz

II. Laminated Structures

II. Laminated Structures

Definition

- Composite materials consists of a combination of materials mixed together to achieve specific structural properties.
- Individual materials, such as resin and fibers do not dissolve or merge completely.
- The properties of the composite material are typically superior to that of the individual materials.
- Advanced composites are used in aerospace contains strong, stiff, engineered fibers embedded in a high performance matrix.

Major Components of a Laminate

- Individual materials often comes in the form of resin and fibers.
- Fibres are laid in layers using a resin system (matrix) and, therefore, composites manufactured this way are known as laminates.

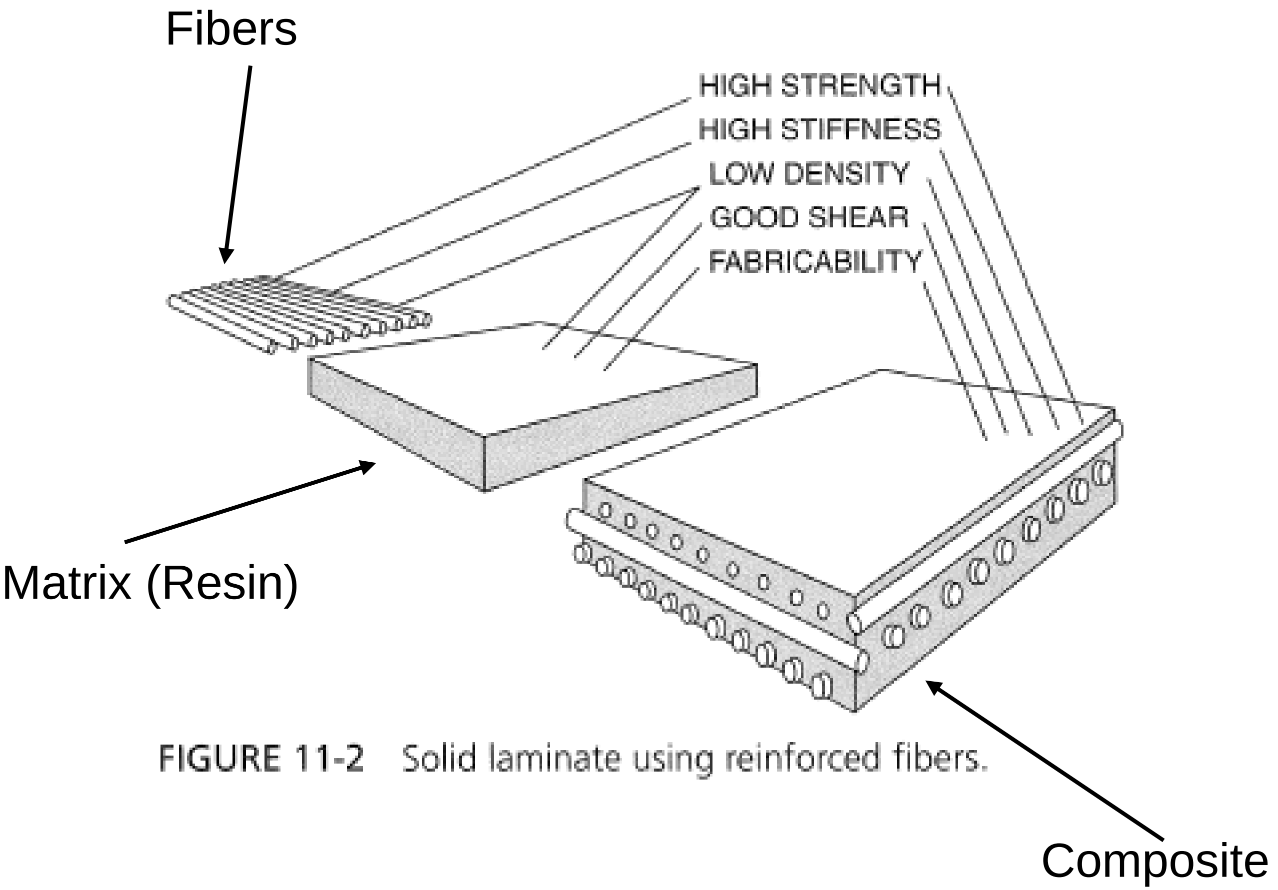
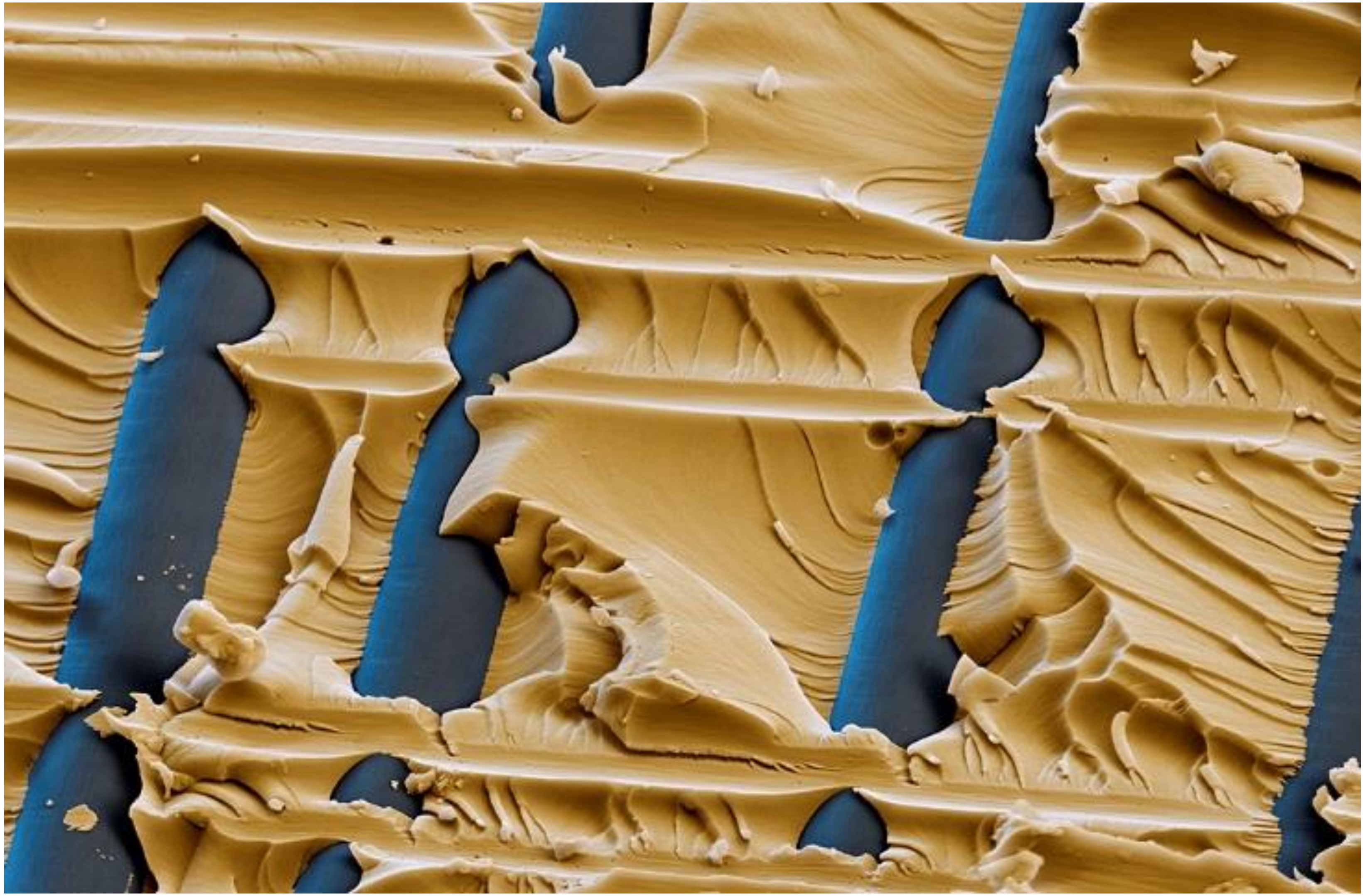


FIGURE 11-2 Solid laminate using reinforced fibers.

Major Components of a Laminate



Microscopic view of Fiber & Matrix.
Source: https://www.flickr.com/photos/basf/6765751461/in/photostream/

Major Components of a Laminate

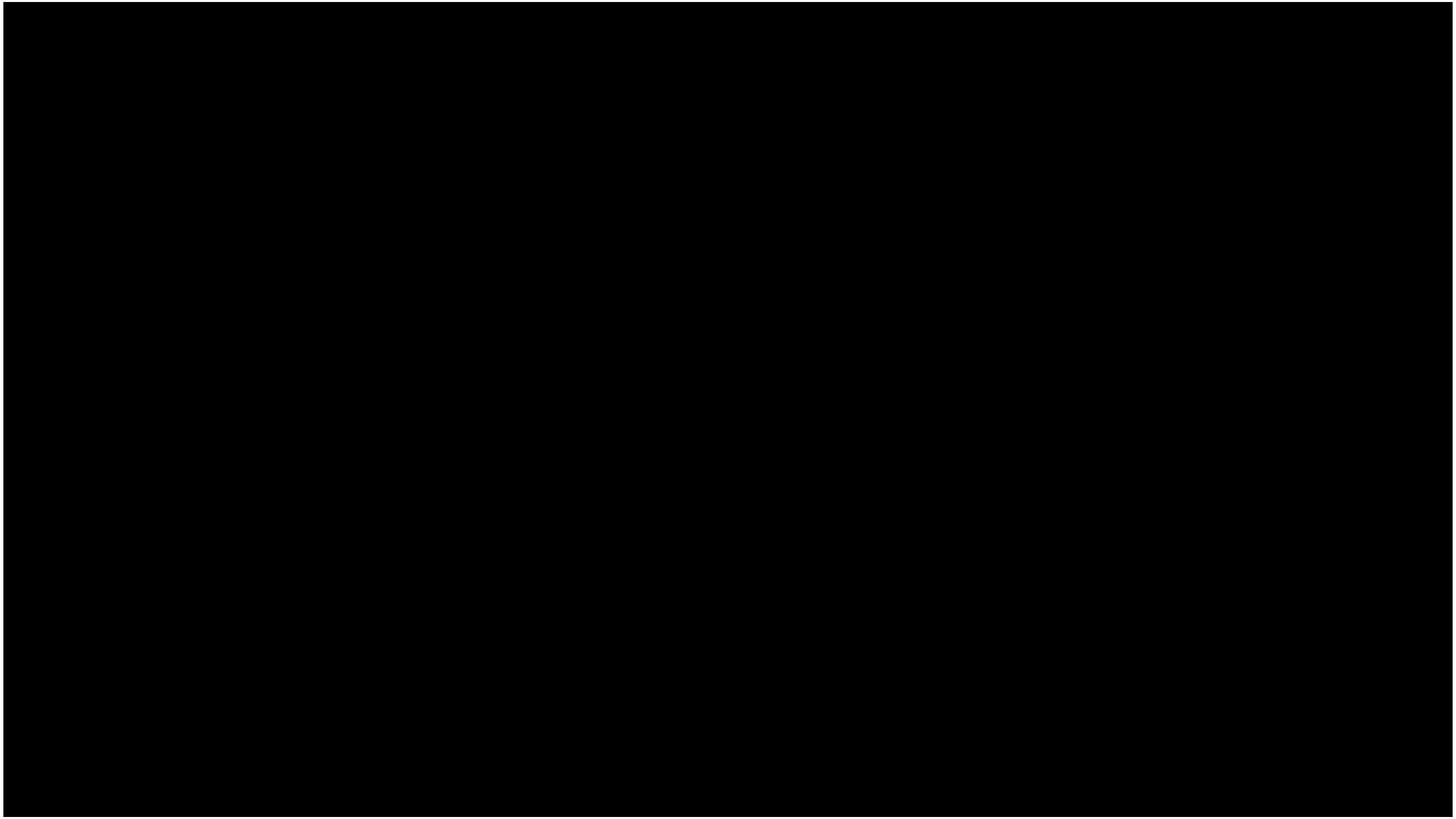
- **Fiber**:
  - The fibers in a composite are the primary load-carrying element of the composite material.
  - The composite material will be strong and stiff in the direction of the fibers.
  - Unidirectional composites have predominant mechanical properties in one direction and are said to be anisotropic. Components made from fiber-reinforced composites can be designed so that the fiber orientation produces optimum mechanical properties, but they can only approach the true isotropic nature of metals.

Major Components of a Laminate

- **Matrix**:
  - The matrix supports the fibers and bonds them together in the composite material.
  - The matrix transfers any applied loads to the fibers, keeping the fibers in their position and chosen orientation.
  - The matrix also gives the composite environmental resistance and determines the maximum service temperature of a composite structure.

II. Laminated Structures

Strength Characteristic: Fibre Orientation

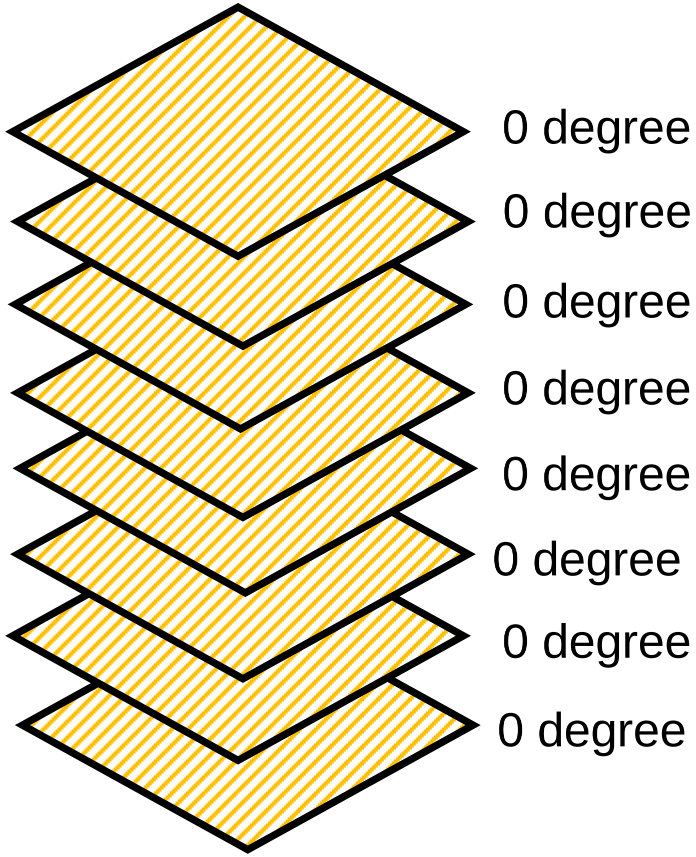


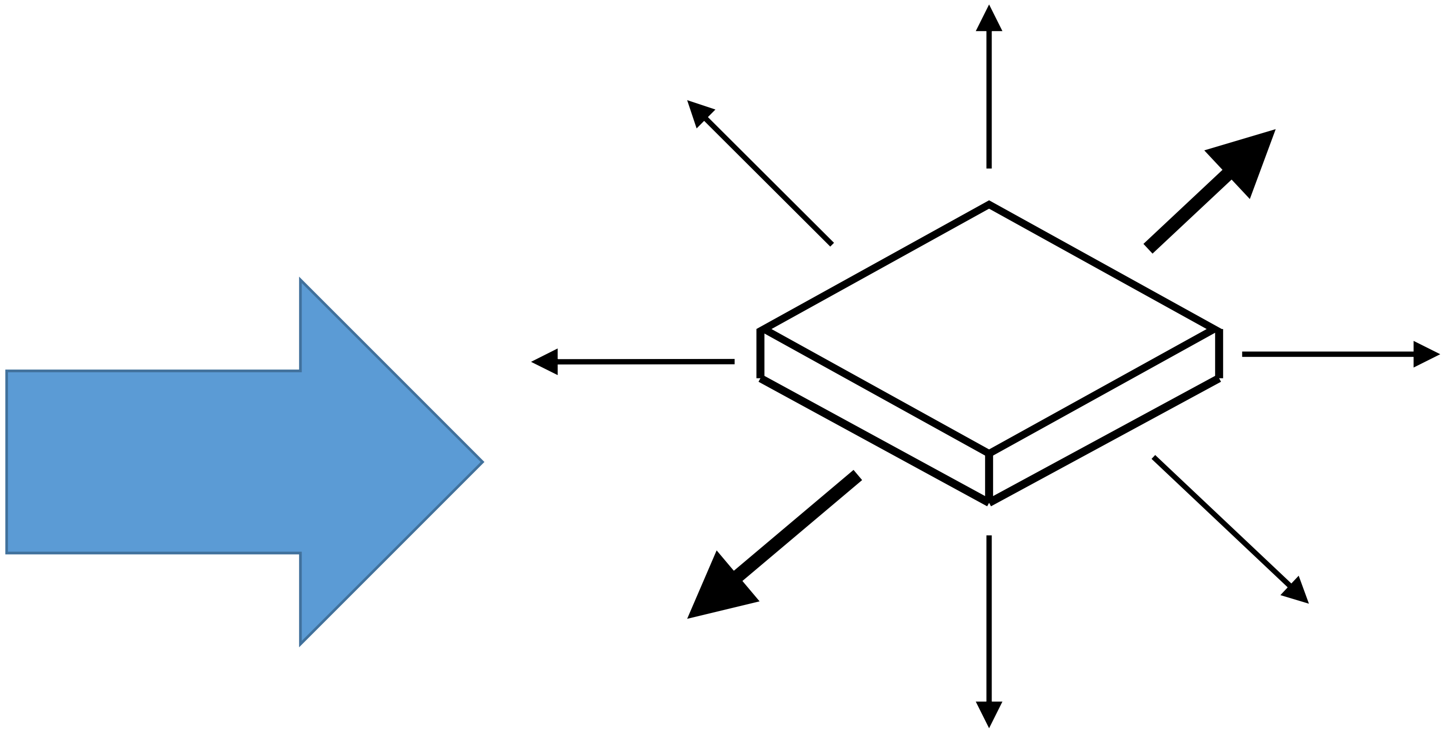
Strength Characteristic: Fibre Orientation

- Strength characteristics of composites differ from metals in that:
  - Dependent to a large extent by the fiber orientation.
  - Engineer, therefore, is able to design a material with different strength characteristic relative to some natural axes inherent to the material.
- Metals are isotropic materials.
  - Same strength characteristic in all directions.
- Many composites e.g. laminates are designed to be quasi-isotropic.
  - Such material has approximately the same properties as an isotropic material.
  - For example, Laminates with fibres oriented in 0°, 90° ,+45° and -45° simulate isotropic properties.

Strength Characteristic: Fibre Orientation

- **Unidirectional**: The fibers run in one direction and the strength and stiffness are in the fiber direction only. Prepreg tape is an example of a unidirectional ply orientation.

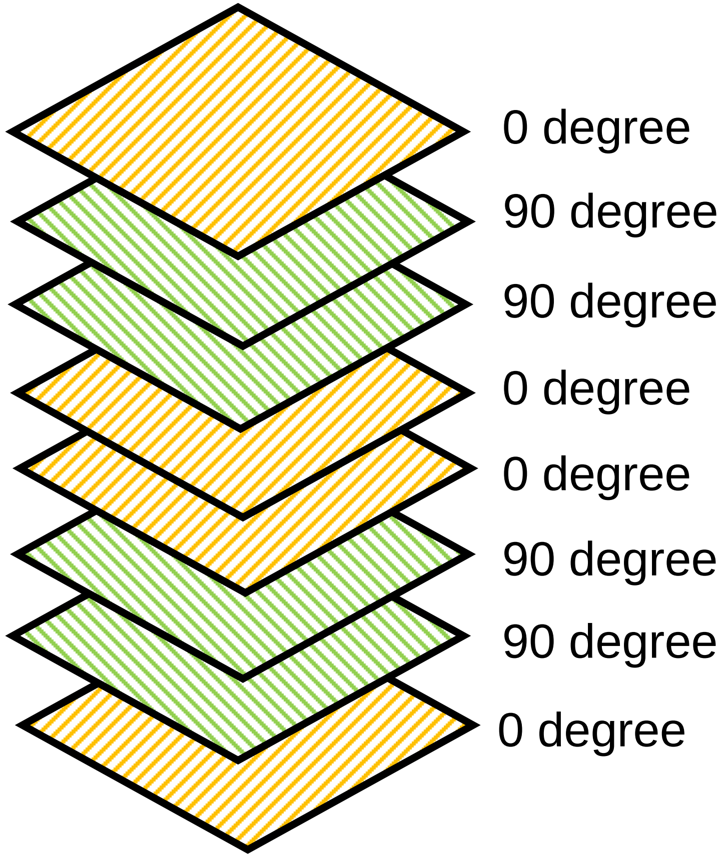


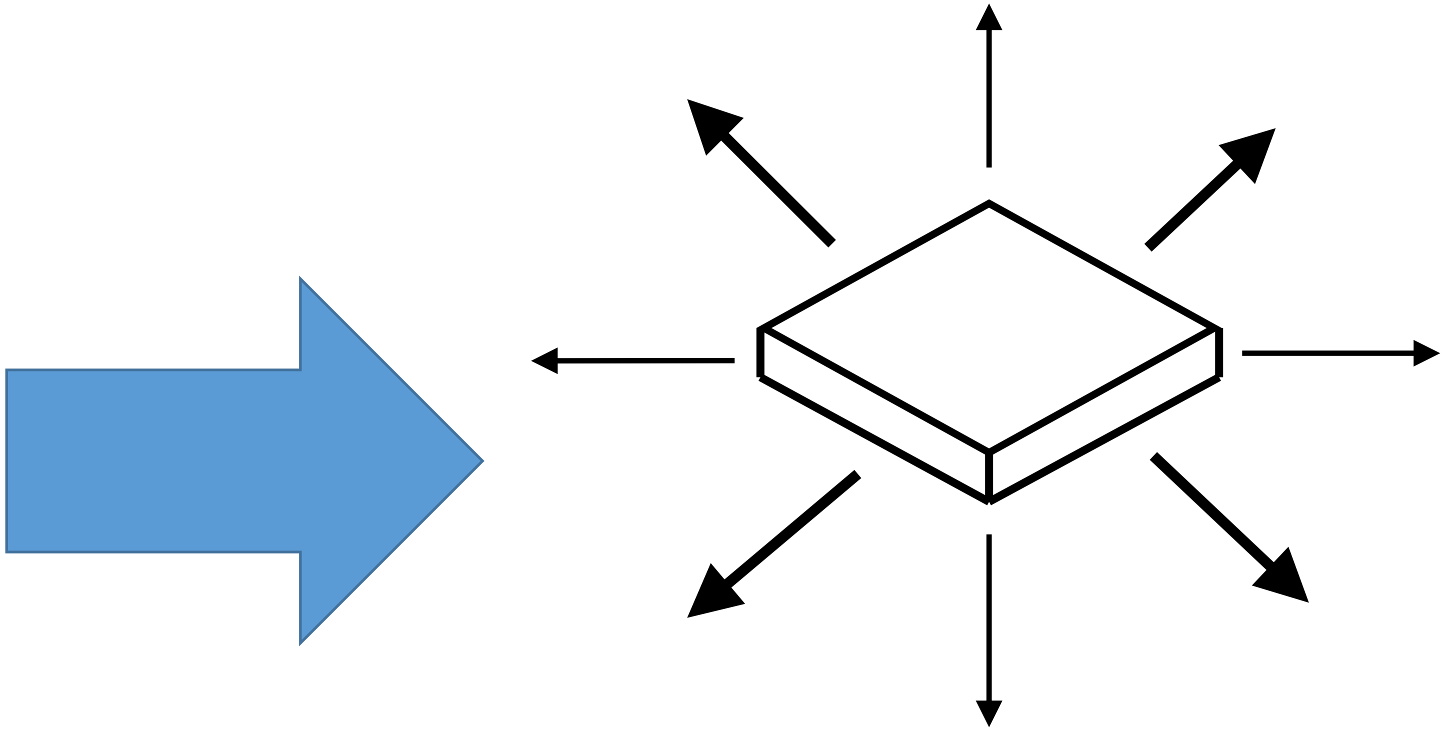


Strength and stiffness only in 0 degree direction

Strength Characteristic: Fibre Orientation

- **Bidirectional**: The fibers run in two directions, typically 90° apart. These ply orientations have strength in both directions but not necessarily the same strength. A plain weave fabric is an example of a bidirectional ply orientation.

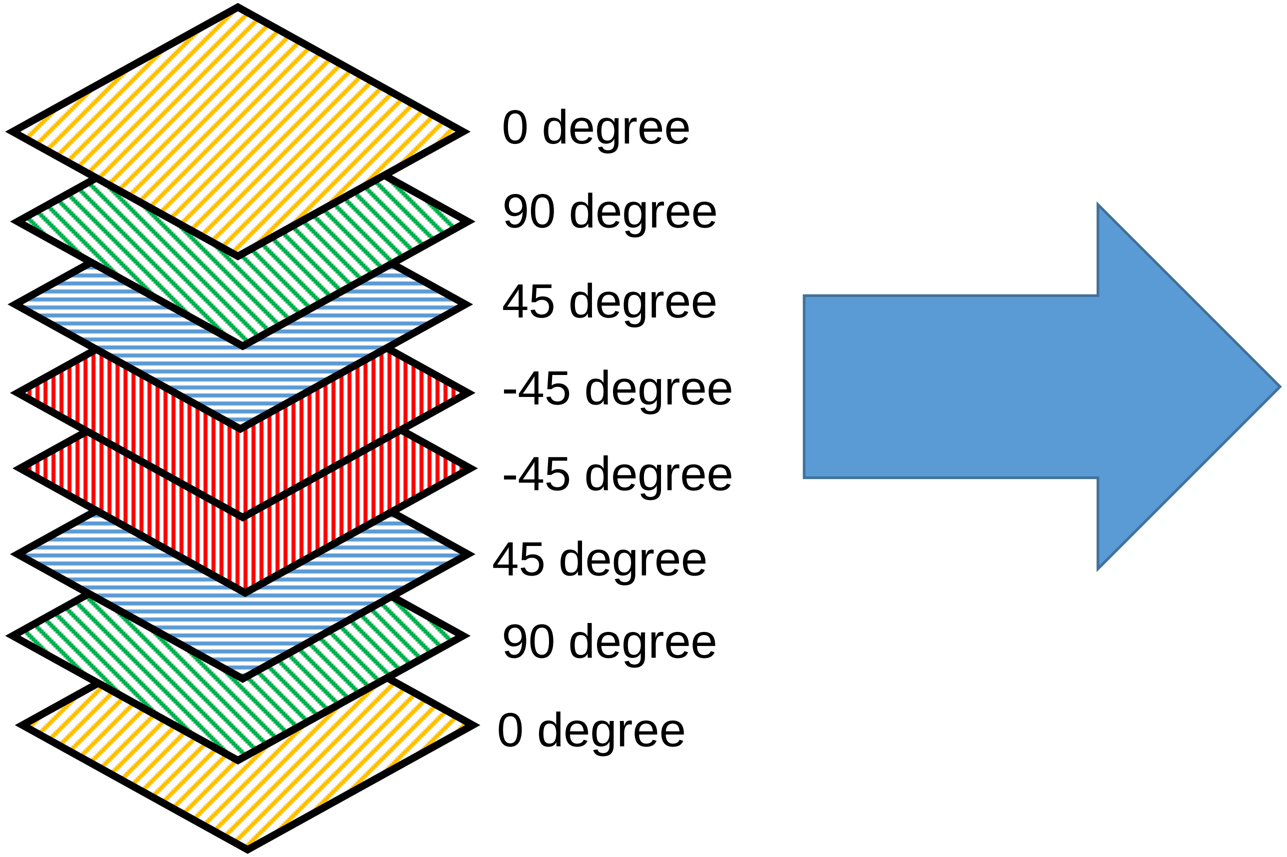


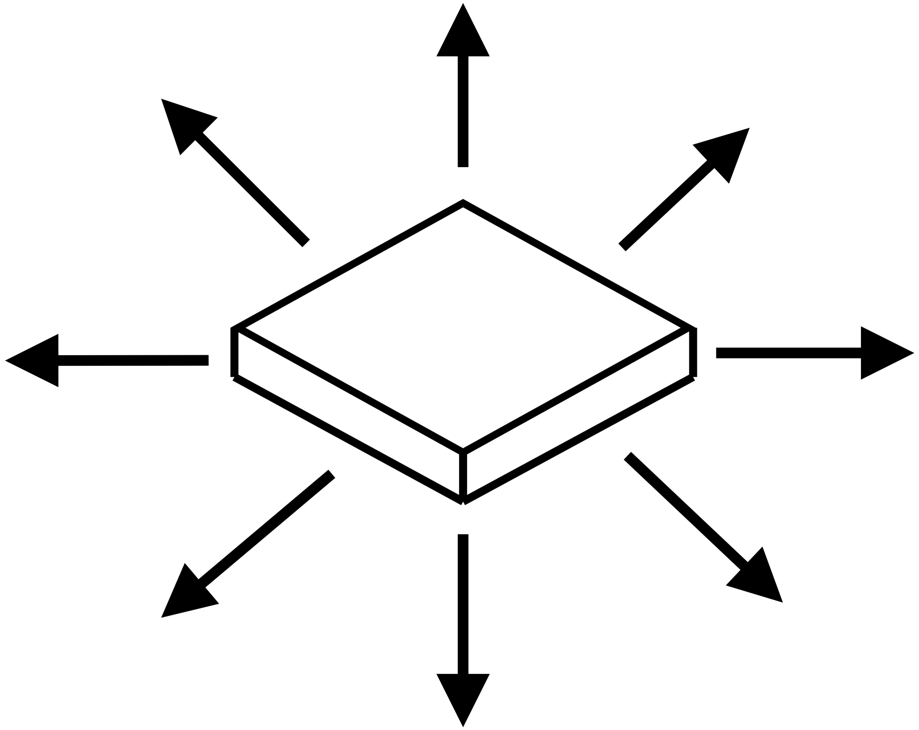


Strength and stiffness only in 0 and 90 degree directions

Strength Characteristic: Fibre Orientation

- **Quasi-isotropic**: The plies of a quasi-isotropic lay-up are stacked in a 0°, -45°, 45°, and 90° sequence or in a 0°, -60°, and +60° sequence. These types of ply orientations simulate the properties of an isotropic material such as aluminium or titanium.





Approximately the same strength in all directions

Strength Characteristic: Warp Clock

- Positive angles can be measured in either clockwise or counterclockwise direction.
- The ASTM and MIL-HDBK-17 standard is to measure positive angles in the clockwise direction when looking at the layup surface.
- Warp is the longitudinal fibers of a fabric. The warp is the high-strength direction, due to the straightness of the fibers.
- 90° to zero will be across the width of the fabric. 90° to zero is also called the fill direction.

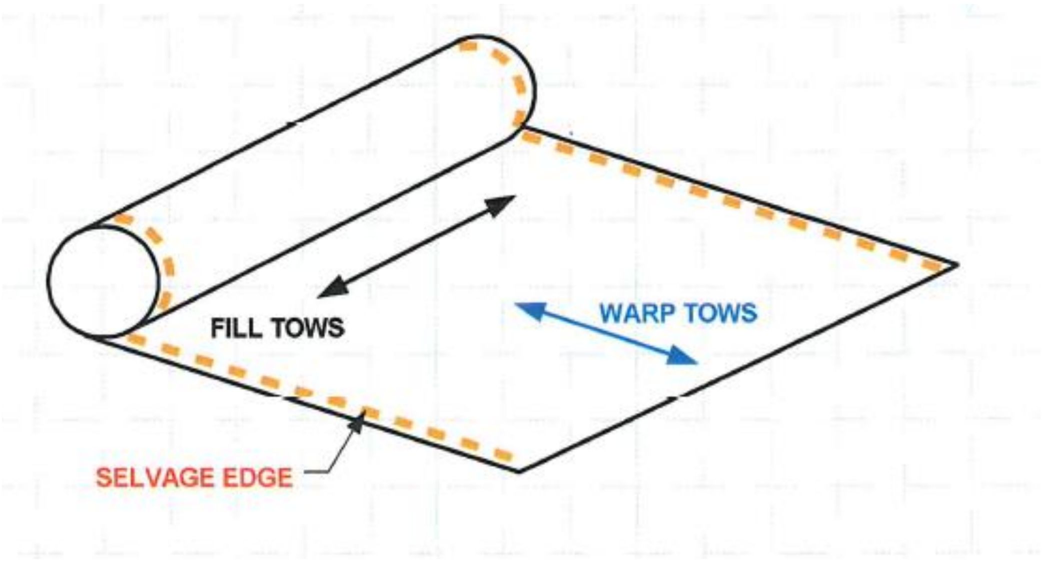


Fig 9-1: Warp direction (0 degree) & Fill direction (90 degree)

Strength Characteristic: Warp Clock

- A warp clock is used to describe direction of fibers on a diagram, spec sheet, or manufacturer's sheets.
- If the warp clock is not available on the fabric, the orientation will be defaulted to zero as the fabric comes off the roll.

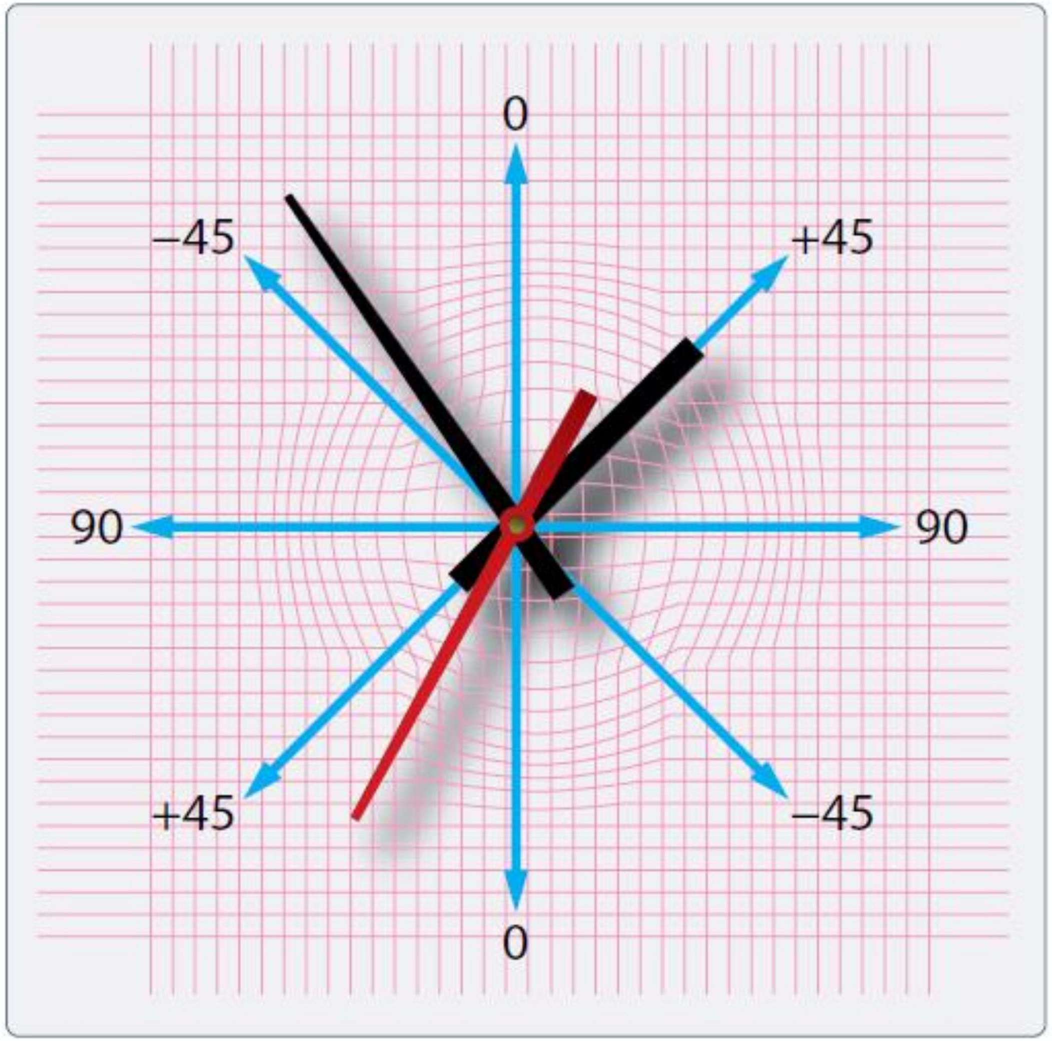


Figure 7-3. *A warp clock.*

Strength Characteristic: Warp Clock

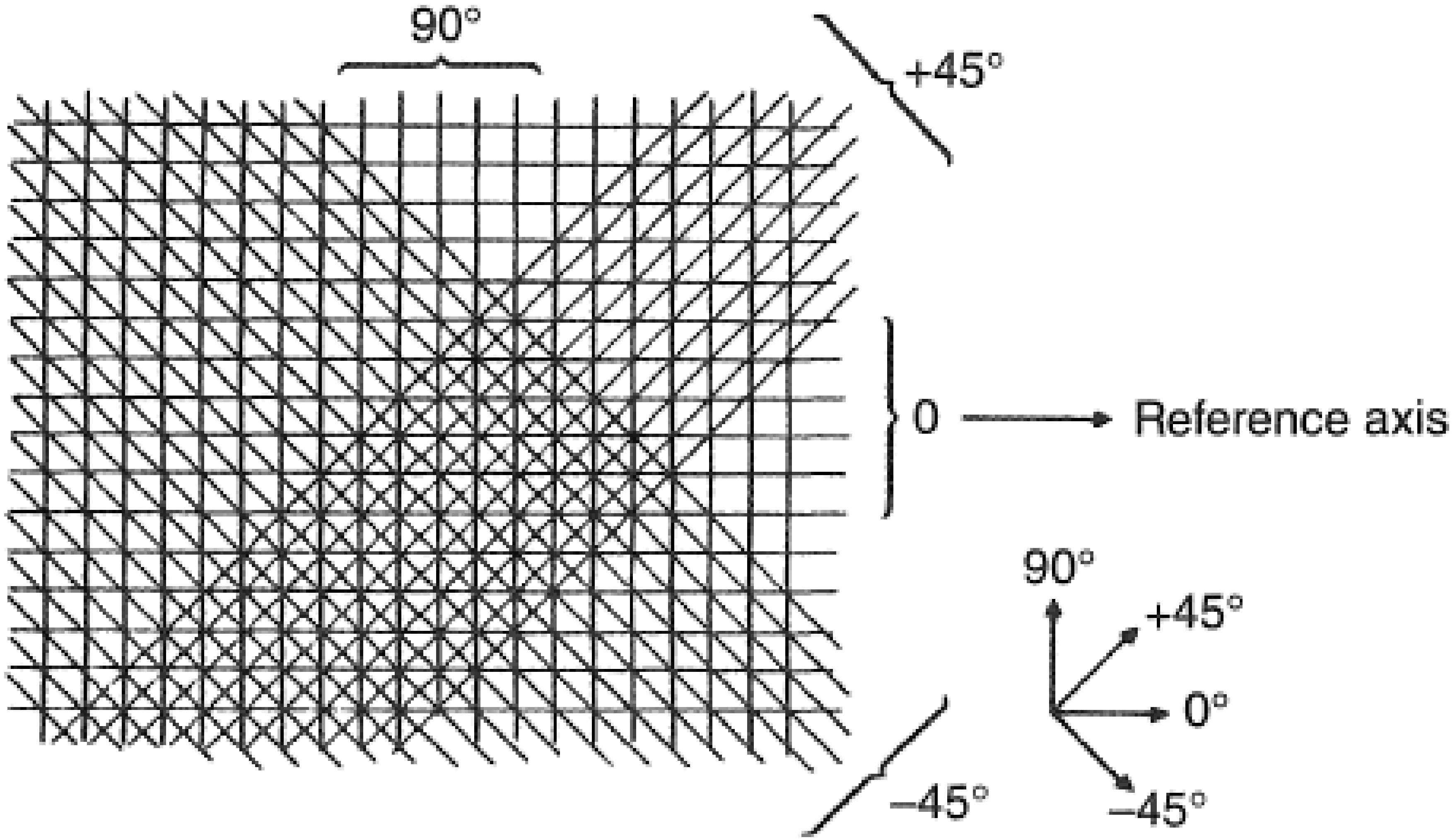


FIGURE 11-4 Warp clock to indicate ply orientation.

Strength Characteristic: Symmetry and Balance

- Most composite laminates have a symmetrical and balanced lay-up to reduce internal stress and avoid distortion during the curing process.
- Symmetry of lay-up means that for each layer above the midplane of the laminate there must exist an identical layer (same thickness, material properties, and angular orientation) below the midplane.
- To achieve balance, for every layer centered at some positive angle there must exist an identical layer oriented at a negative angle with the same thickness and material properties.

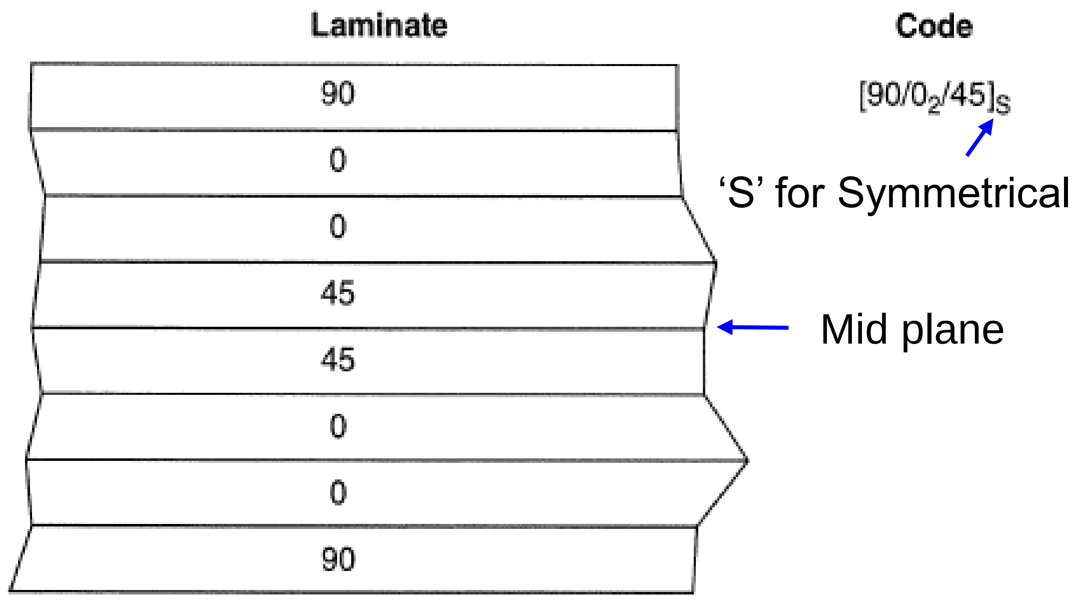


FIGURE 11-5 Symmetrical lay-up.

Quiz

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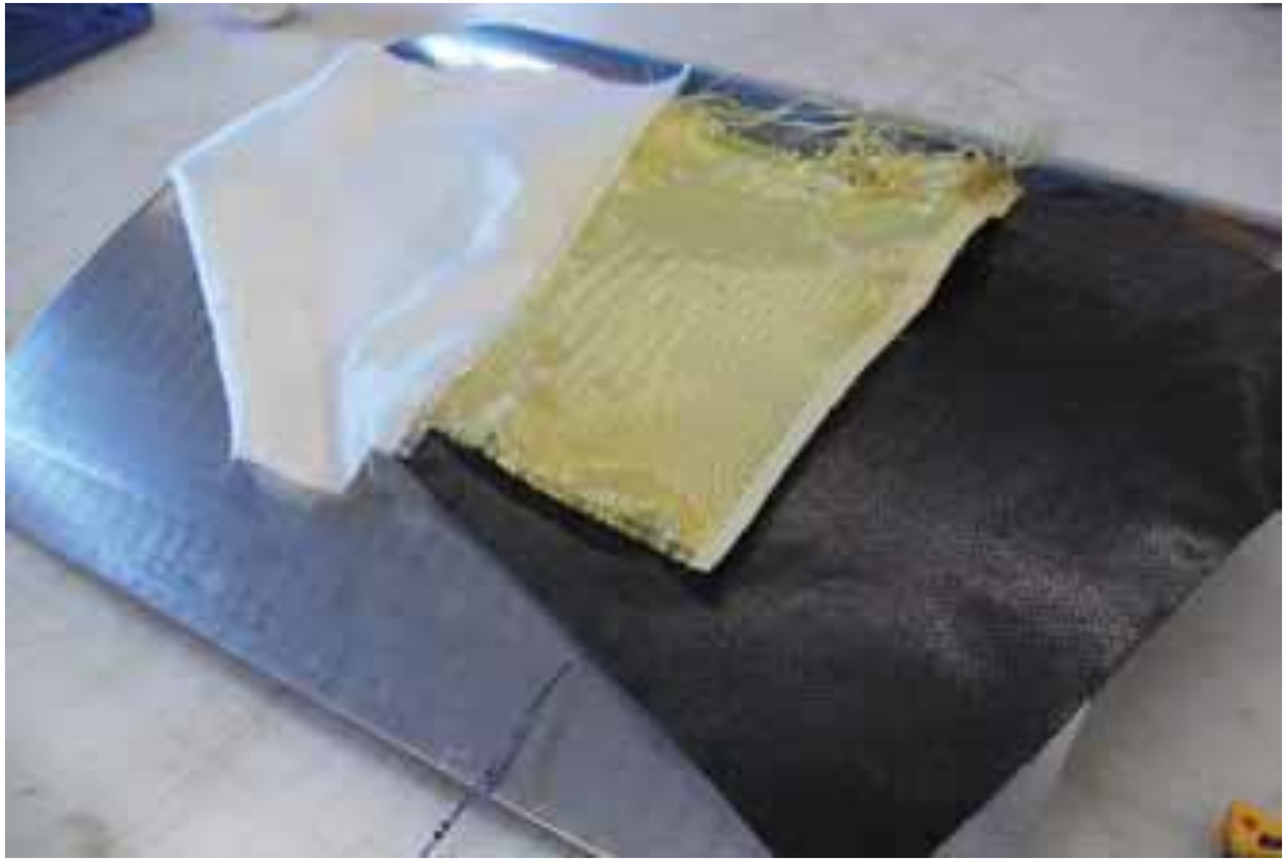
Welcome to the quiz (Symmetry VS Balanced)

Click the "Start Quiz" button to proceed

Start Quiz

Types of Fiber: Fiberglass

- The most commonly used fiber in the aerospace industries are :
  - Fiberglass
  - Aramid (Kevlar)
  - Carbon fibre



Fiberglass (left), Kevlar® (middle), and carbon fiber material (right).

(2016). *Faa.gov*. Retrieved 15 September 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/media/FAA-H-8083-32-AMT-Powerplant-Vol-1.pdf

Types of Fiber: Fiberglass

- **Fiberglass**:-
  - Typically used for secondary structures on aircraft.
  - Properties of fiberglass are less than carbon but lower cost than other carbon fiber.
  - Good chemical and galvanic corrosion resistance.
  - E Glass - Electrical glass (electrical waves could pass through it), Inert Electrical properties.
  - S and S2 Glass - Stronger than E glass but more brittle.
  - Distinguished by its white colour but also available in different colours

Types of Fiber: Kevlar*

- **Kevlar (Trademark name for Aramid)**:-
  - An organic fiber that is 43% lighter and twice in tensile strength as fiberglass.
  - Generally weak in compression and difficult to cut due to fuzzing.
  - High strength and good impact resistance.→ Rolled around fan case in the past to protect blade-out.*
  - Hygroscopic and can absorb up to 8% of their weight in water.
  - Sensitive to UV and will degrade under exposure.*
  - Combustible.*
  - Recognised by its distinct yellow.

Fig 9-2: Kevlar wrapped around the engine to contain blade-out
Source: http://science.howstuffworks.com/transport/flight/modern/turbine.htm

FIGURE 11-8 Aramid (Kevlar) fabric material.

Types of Fiber: Carbon/Graphite*

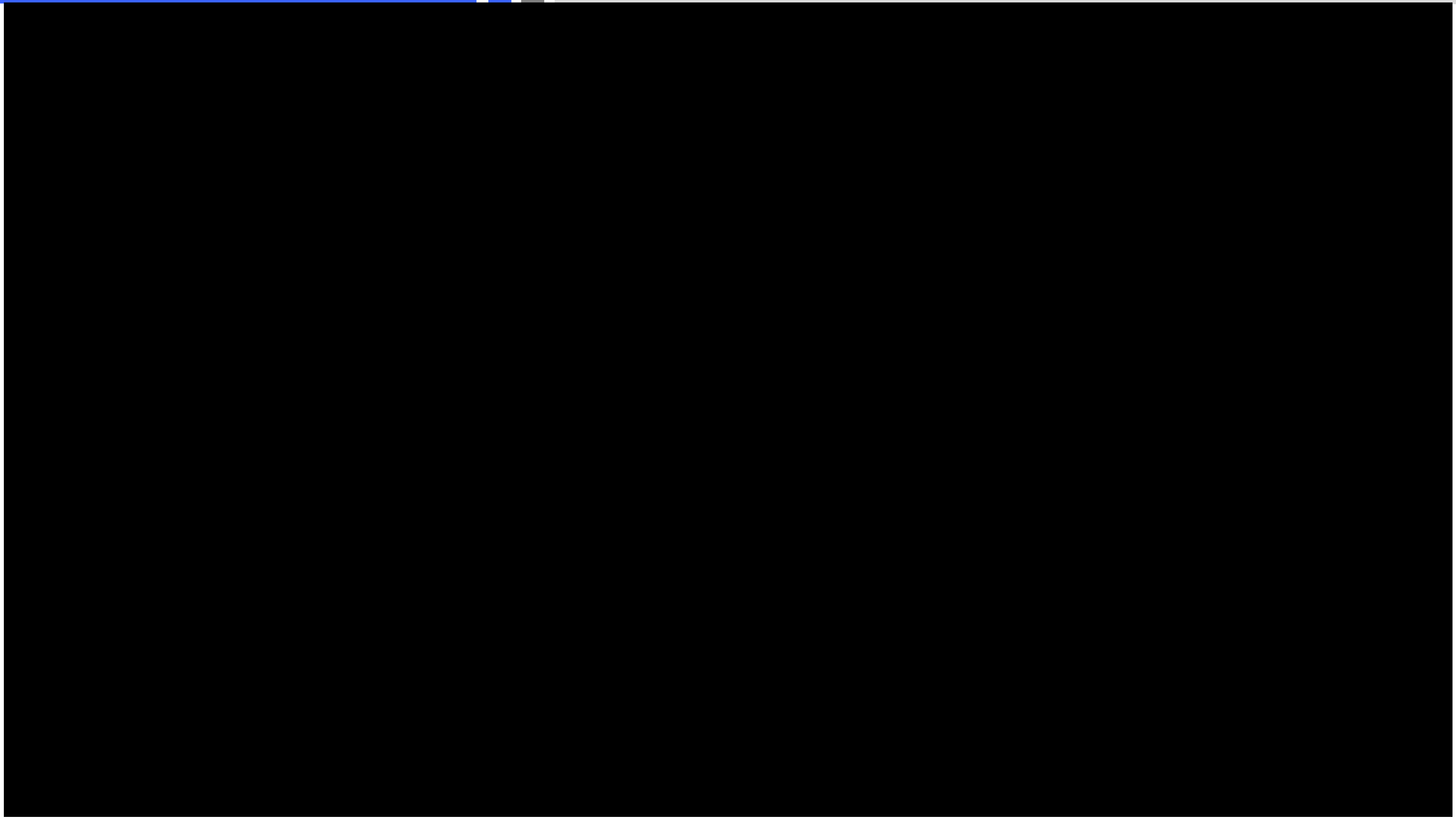
- **Carbon / Graphite**:-
  - High strength compared to other metallic or composite materials.
  - Superior fatigue property.
  - High resistance to corrosion and creep.*
  - Galvanic corrosion with metallic fasteners and structures.
  - Lower electric conductivity compared to aluminium (1000 times less).

Types of Fiber: Carbon/Graphite*

Let’s see how composites fare under impact load when compared to other materials in the video.

Remember that composite is only strong in the fiber direction.

Types of Fiber: Carbon/Graphite



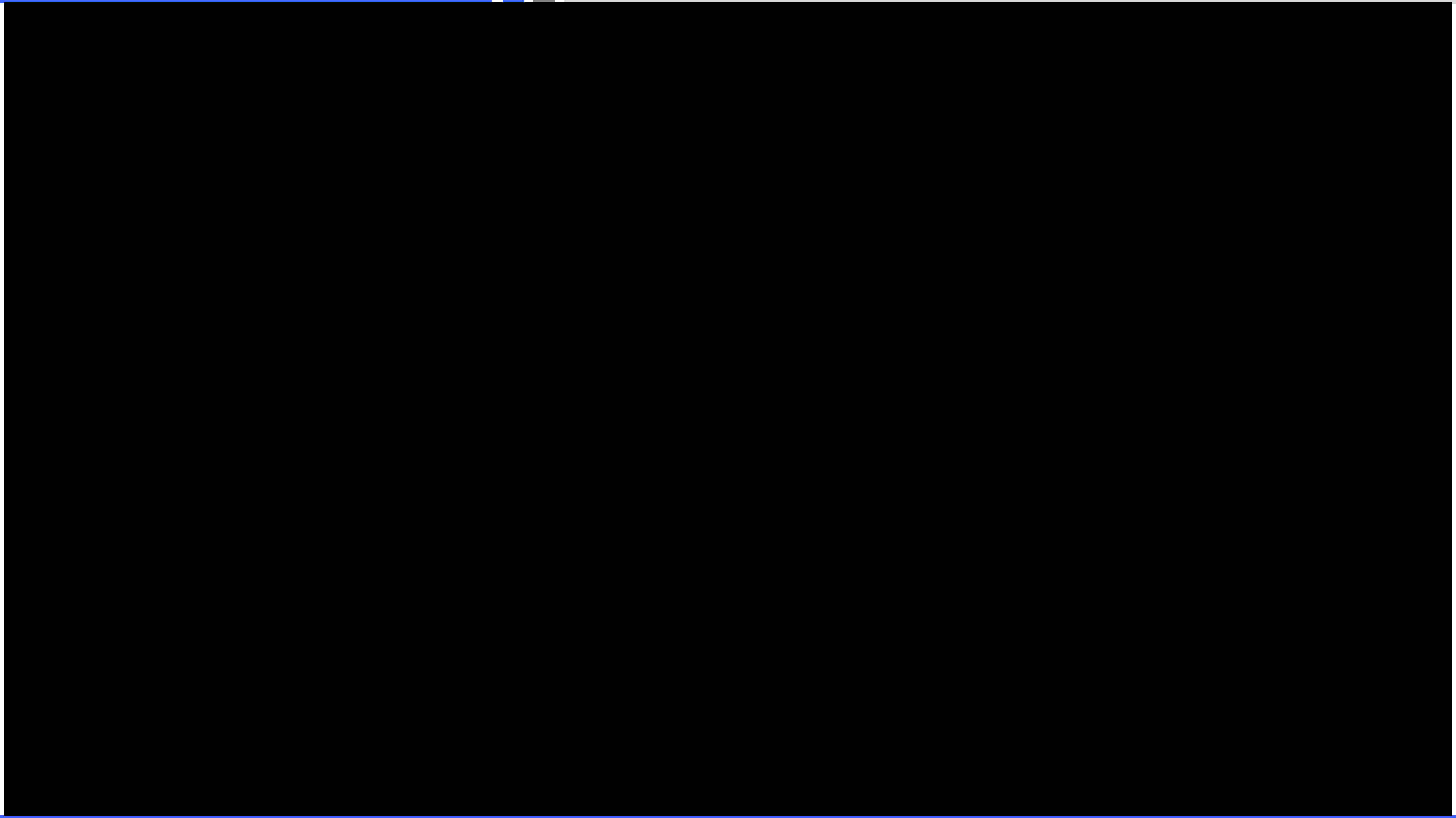
Drop Testing: Carbon Fiber, Steel, Aluminum Comparison

Drop Testing: Carbon Fiber, Steel, Aluminum Comparison. (2016). *YouTube*. Retrieved 15 September 2016, from https://www.youtube.com/watch?v=khJQgRLKMU0

Types of Fiber: Boron and Ceramic

- **Boron**:-
  - Very stiff and high tensile and compressive strength.
  - Do not flex well and difficult to use on contoured shape.
  - Expensive and hazardous.
- **Ceramic**:-
  - High-temperature application up to 2200 degree F.

Types of Fiber: Lightning Protection Fiber



Types of Fiber: Lightning Protection Fiber

- Not really a fiber, but a metallic mesh or foil.
- Bronze and Copper used instead of Aluminium for carbon composite.
- Typically on top of the composite laminate to increase electrical conductivity.

Figure 7: LSP-001 immediately after lightning strike test

Figure 9: LSP-026 close-up view, identical to LSP-001 except no protection

Fig 9-3: Composite with lightning protection fiber (left) composite without protection (right)

Source: http://www.dexmet.com/1_pdf/LSP%20for%20Carbon%20fibre%20Aircraft.pdf

Fiber Forms

- All product forms begin with spooled unidirectional raw fibers packaged as a continuous strand.
- An individual fiber is called a filament.
- Strand is also used for an individual glass fiber.
- Bundles of filaments are known as tows, yarns or rovings.
- A roving or tow is a single grouping of filament or fiber ends, such as 20-end or 60-end glass rovings. All filaments are in the same direction and they are not twisted.
- Carbon tows are usually identified as 3K, 6K, or 12K tows. K means 1000 filaments.
- When individual fiberglass filaments are twisted they are called yarns.
- Carbon fiber and Kevlar fibers are not twisted.

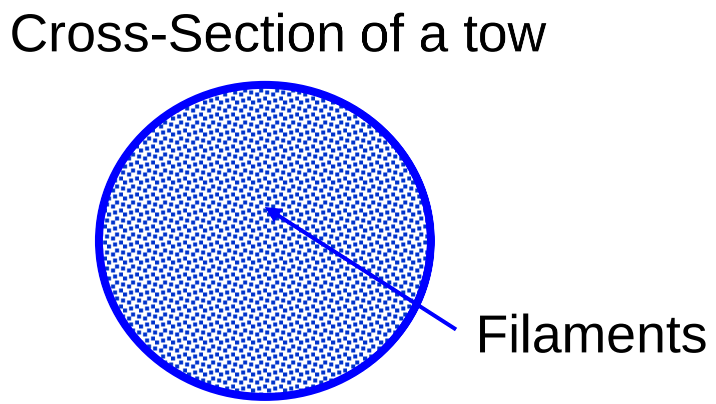


Fig 9-4: Cross section of a tow

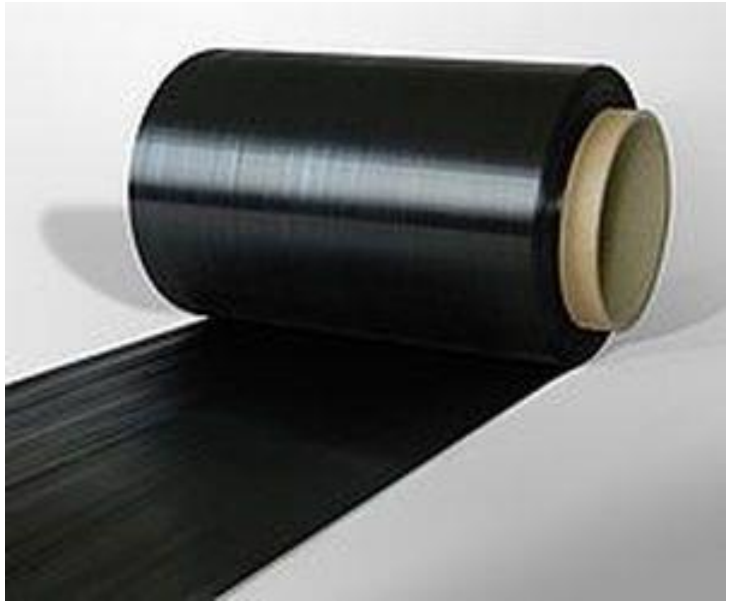


Fig 9-5: Example of a Tape

Source: http://www.directindustry.com/prod/toho-tenax-europe/prepreg-carbon-fibre-high-performance-37818-1422771.html

Fiber Forms: Unidirectional (Tape) & Bidirectional (Fabric)

- **Unidirectional (Tape):**
  - High-strength structural applications are often manufactured from unidirectional material.
  - All the fibers are running in the same direction and the material has only strength properties in the fiber direction.
  - Unidirectional materials are typically available as pre-preg materials.
- **Bidirectional (Fabric):**
  - Bidirectional materials, often called fabric, are woven materials that use tows of composite material.
  - A weaving process similar to what is used to manufacture textile products is used to make the fabric material.

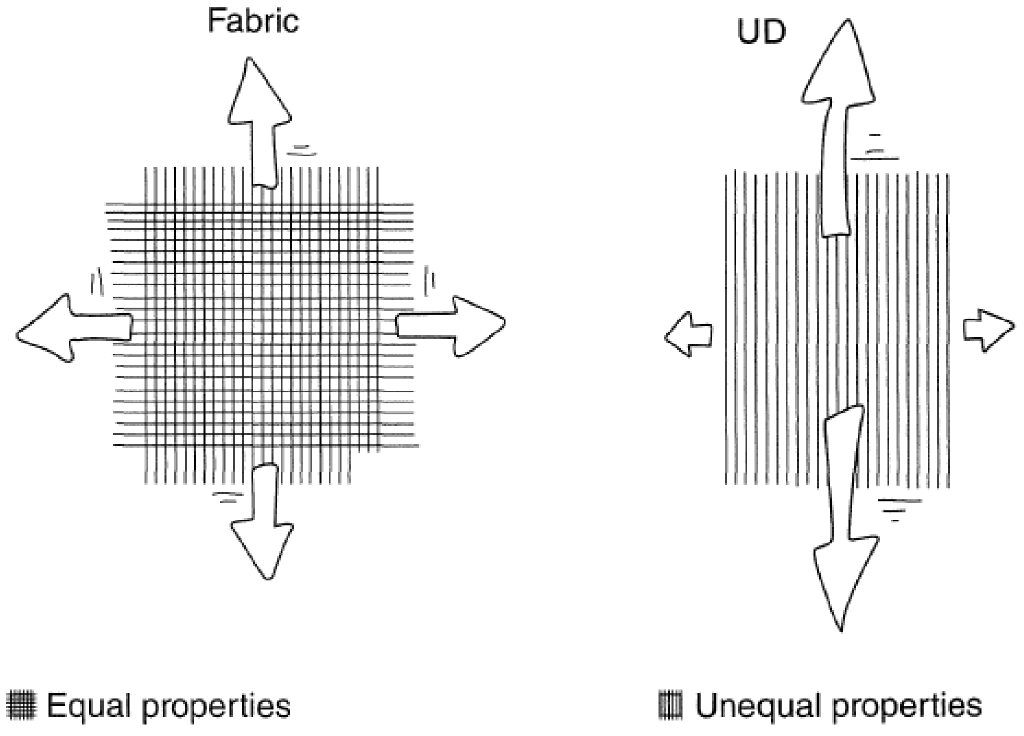
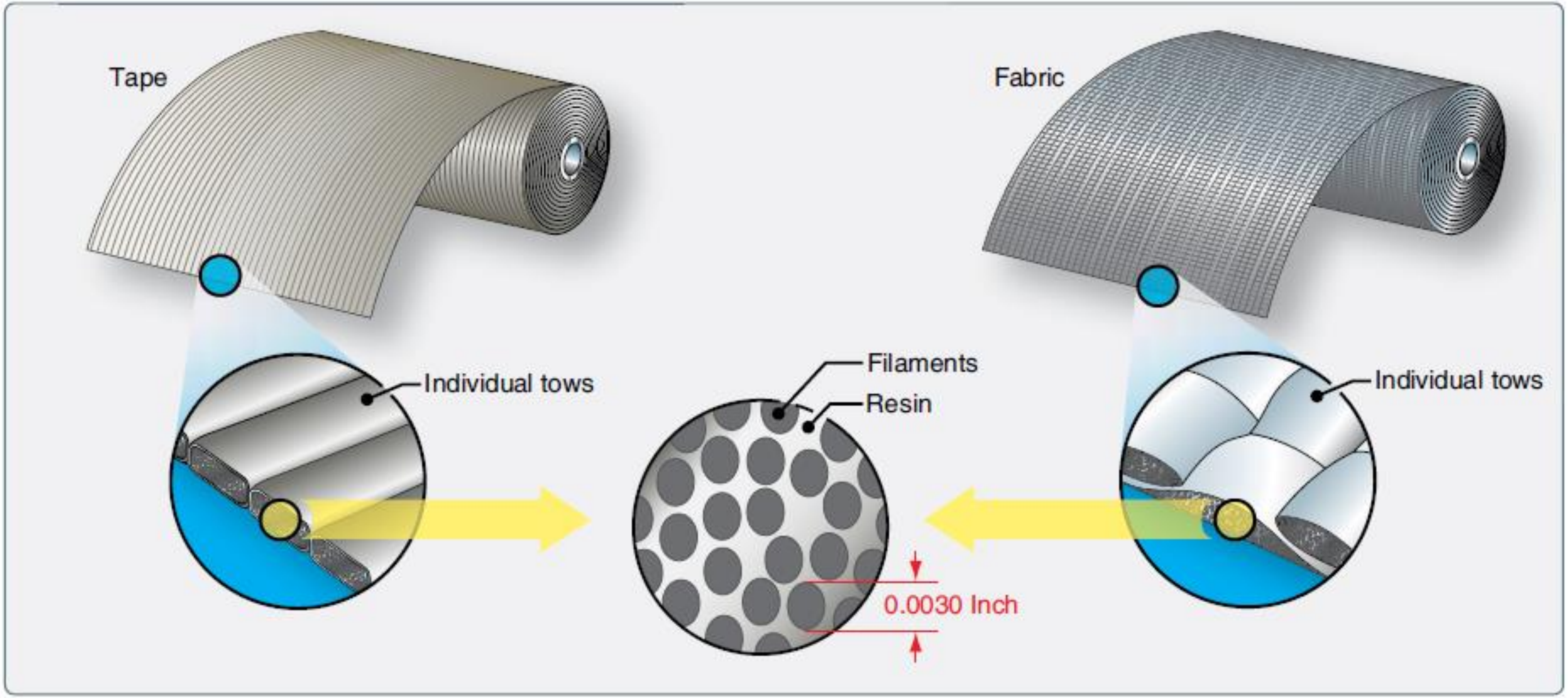


FIGURE 11-13 Bidirectional (fabric) and unidirectional (tape) properties.

Fiber Forms



Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Fiber Forms: Bidirectional (Fabric)

- For aerospace structures, tightly woven fabrics are usually the choice to save weight, minimizing resin void size, and maintaining fiber orientation during the fabrication process:
  - Plain weave is most highly interlaced and tightest of the basic fabric designs and most resistant to in-plane shear movement.
  - Basket weave, a variation of plain weave, has warp and fill yams that are paired: two up and two down.
  - Satin weaves have the fiber bundles traverse both in warp and fill directions changing over/under position less frequently. Satin weaves have less crimp and are easier to distort and are used for more complex shapes.
- Fabrics are widely available as dry fiber or pre-preg materials.

Fiber Forms: Bidirectional (Fabric)

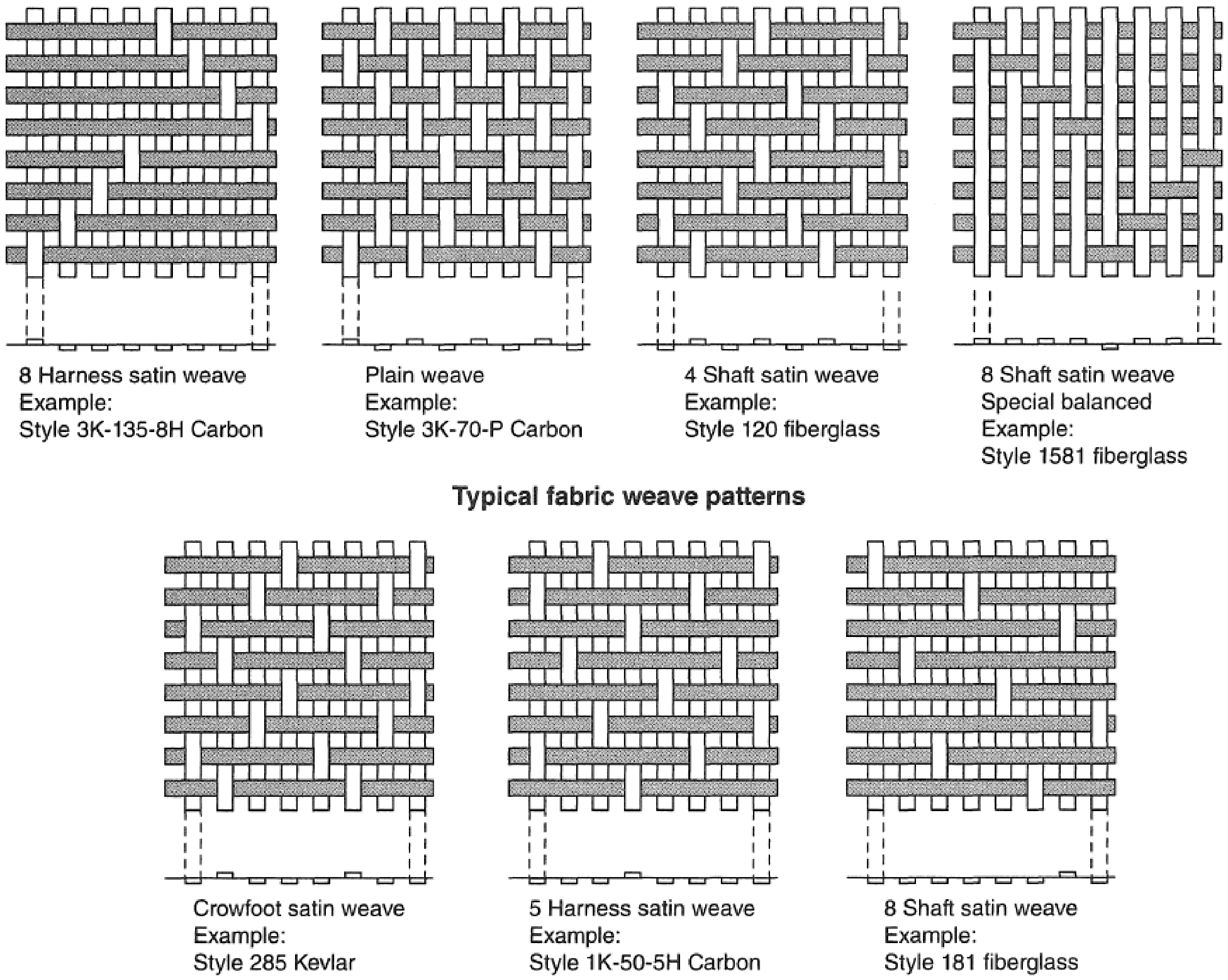


FIGURE 11-14 Basic weave styles for composite bidirectional products.

Fibre Forms: Nonwoven (Stitched) and 3D Knitted*

- Multi-axial fabrics consist of one or more layers of continuous fibres held in place by a secondary, nonstructural stitching thread.
- A composite component/structure can be formed by knitting the fibers together three dimensionally (3D knitting/weaving)*.

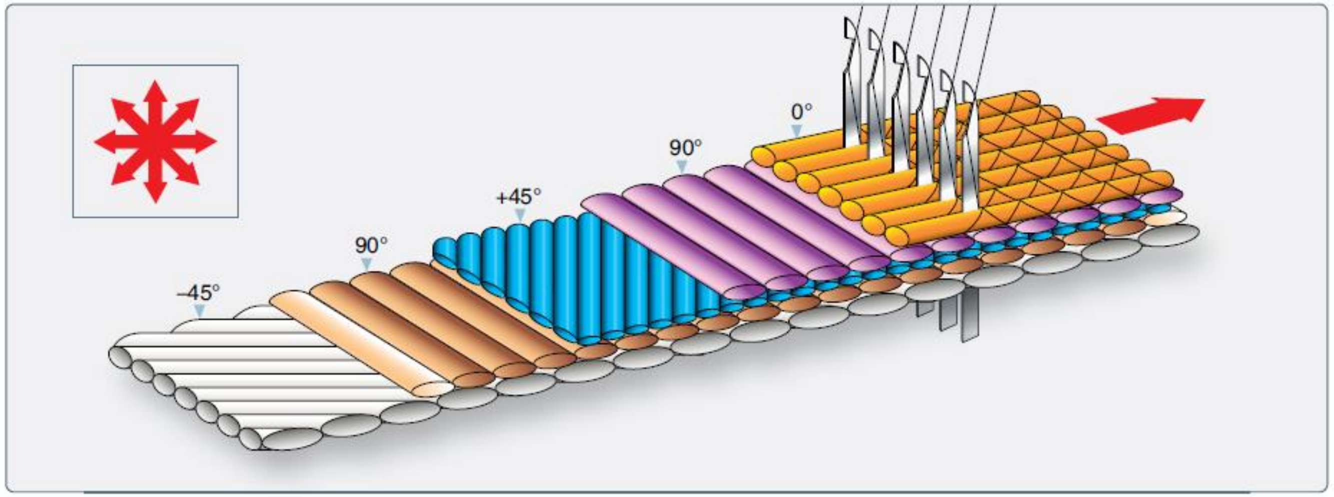


Figure 7-6. *Nonwoven material (stitched).*

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

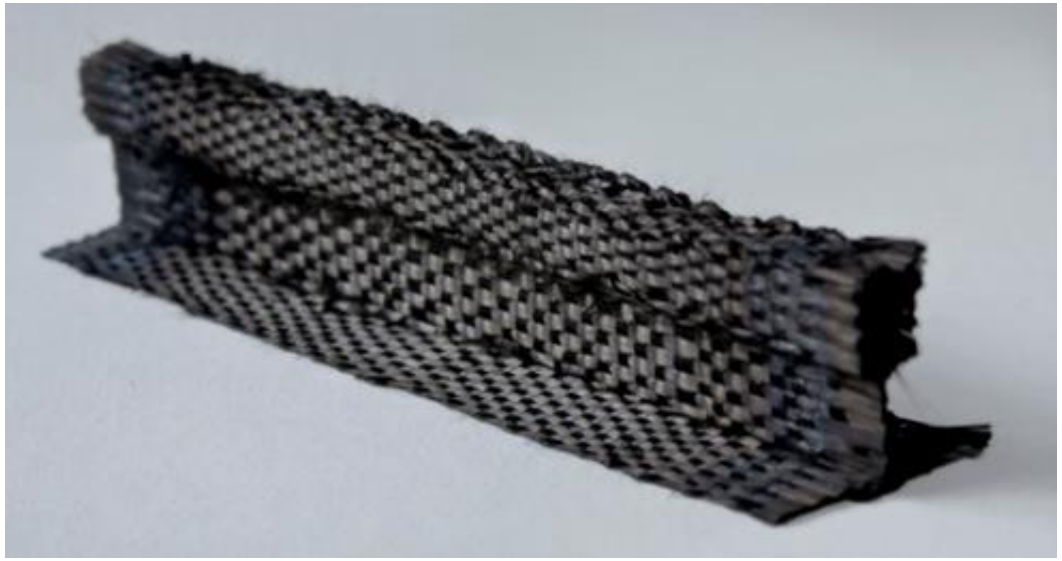


Fig 9-6: 3D Knitted Component
Source: http://www.jeccomposites.com/news/features/rtm-infusion/t-bar-and-cross-member-jointing-system

Matrix Materials

- Matrix is to support the fibers and bond them together in the composite material.
- It transfers any applied loads to the fibers and keeps the fibers in their position and chosen orientation.
- The matrix also gives the composite environmental resistance and determines the maximum service temperature of the finished component.
- The two main classes of resin system used are: thermoset and thermoplastic.
- Curing is a term which refers to the toughening or hardening of a polymer material by cross-linking of polymer chains, brought about by electron beams, heat or chemical additives.

Matrix Materials: Thermosetting Resins

- Thermoset materials take a permanent set or shape when cured.
- Cure temperatures range from room temperature to 350°F for polyesters, vinyl esters, epoxies, and cyanate esters, around 450°F for bismaleimide (BMI) materials, and close to 700°F for polyimides.
- Cure pressures are usually less than 100 pounds per square inch (psi).

Two part wet layup epoxy resin system with pump dispenser

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Matrix Materials: Thermosetting Resins

- **Epoxy** resins:
  - Extended service temperature range
  - Excellent mechanical properties: High strength and modulus
  - Low levels of volatiles
  - Excellent adhesion
  - Low shrinkage (Little change in volume after cured)
  - Good chemical resistance
  - Ease of processing
  - Brittleness
  - Reduction of properties in the presence of moisture.

Matrix Materials: Thermosetting Resins

- **Bismaleimide (BMI)** resins (A type of polyimide):
  - Are used when a higher service temperature is required, for example aero engines and high-temperature components.
  - Require higher cure temperatures, typically 375 to 450°F.
  - Epoxy-like processing characteristics yet have a higher temperature use limit.

- **Phenolic (Phenolformaldehyde)** resins:
  - Less expensive alternative for lower temperature component.
  - Used for interior components because of their low smoke and flammability characteristics (Good Fire Smoke Toxicity (FST) properties).

Matrix Materials: Thermosetting Resins

- **Polyimide** resins:
  - Excel in high thermal resistance, where their oxidative stability, low coefficient of thermal expansion and solvent resistance benefit the design.
  - Primary uses are circuit boards and hot engine and aerospace structures.
  - May be either a thermoset resin or a thermoplastic.
  - Require high cure temperatures, usually in excess of 550°F [~ 290°C].

Matrix Materials: Thermosetting Resins

Thermoset resins have three types of curing stages, A, B and C stage:

- **A Stage**:
  - The components of the resin (base material and hardener) have been mixed but the chemical reaction has not started.
  - The resin is in the A stage during a wet lay-up procedure.
- **B Stage**:
  - The components of the resin have been mixed and the chemical reaction has started.
  - Material has thickened and is tacky.
  - Pre-preg materials are in the B stage.
  - To prevent further curing, the resin is frozen. Any increase in temperature will start the cure cycle.

Matrix Materials: Thermosetting Resins

Thermoset resins have three types of curing stages, A, B and C stage:

- **C Stage**:
  - The resin is fully cured.
  - Some resin systems cure at room temperature and others need an elevated temperature cure cycle to fully cure.

Matrix Materials: Thermoplastic Resins

- Thermoplastic materials are non-curing systems that can be reshaped or reformed after the part has been processed or consolidated.
- Reshaping is accomplished by reheating the part until the material softens and applying pressure to produce a newly molded shape.
- Thermoplastics are consolidated at temperatures up to 800°F and pressures up to 200 psi.
- **Polyetheretherketone (PEEK)**:
  - An often used resin system for the aerospace industry to make structural parts.
  - Inherent flame resistance
  - Superior toughness
  - Good mechanical properties at elevated temperatures and after impact
  - Low moisture absorption.
  - Used in secondary and primary aircraft structures.

Matrix Materials: Thermoplastic Resins

- Amorphous thermoplastics are available in several physical forms, including films, filaments, and powders.
  - The specific advantages of amorphous thermoplastics depend upon the polymer.
  - Typically, the resins are noted for their processing ease and speed, high temperature capability, good mechanical properties, excellent toughness and impact strength, and chemical stability.
  - The stability results in unlimited shelf life, eliminating the cold storage requirements of thermoset pre-pregs.
  - Generally, are more expensive than thermoset resins.

Amorphous means non-crystalline solid, which is a solid that lacks the long-range order characteristic of a crystal.*

- **Pre-pregs**:
  - Consists of a combination of a matrix and fiber reinforcement.
  - Resin is then no longer in a low-viscosity stage, but has been advanced to a B-stage level of cure for better handling characteristics.
  - Different pre-preg form: unidirectional tapes (one direction of reinforcement), woven fabrics (several directions of reinforcement), continuous strand rovings, and chopped mat.
  - Must be stored in a freezer at a temperature below 0°F to retard the curing process (Stage B).
  - Typically cured with an elevated temperature. For example pre-preg materials are impregnated with epoxy resin, are cured at either 250 or 350°F.

Matrix Materials: Preimpregnated Products (Pre-pregs)

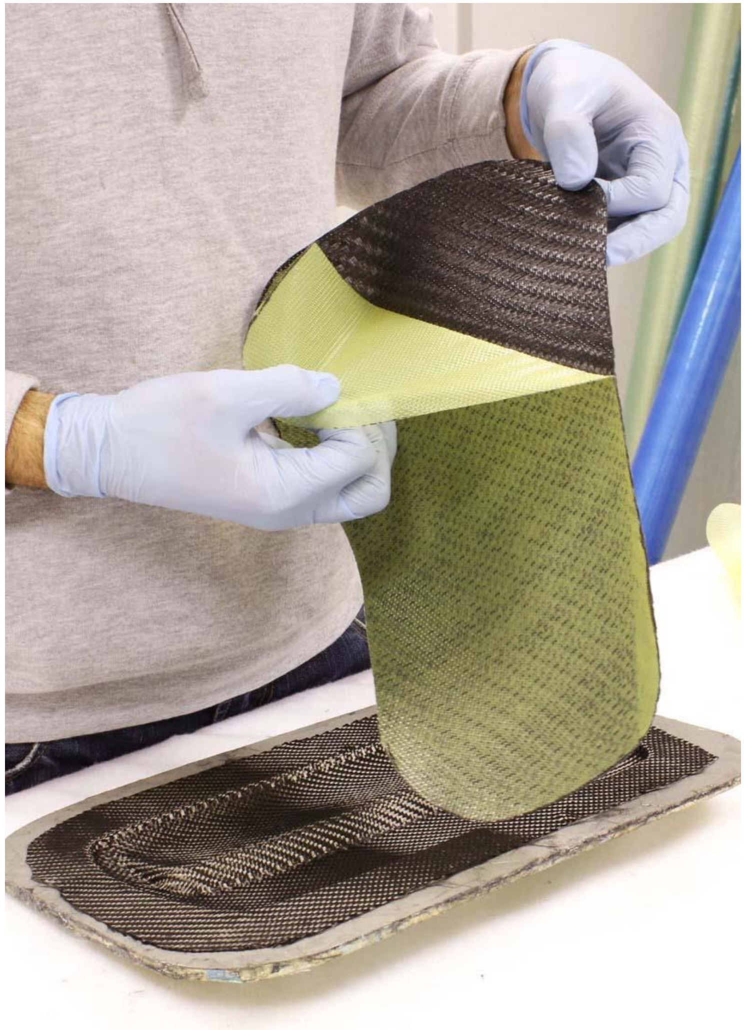
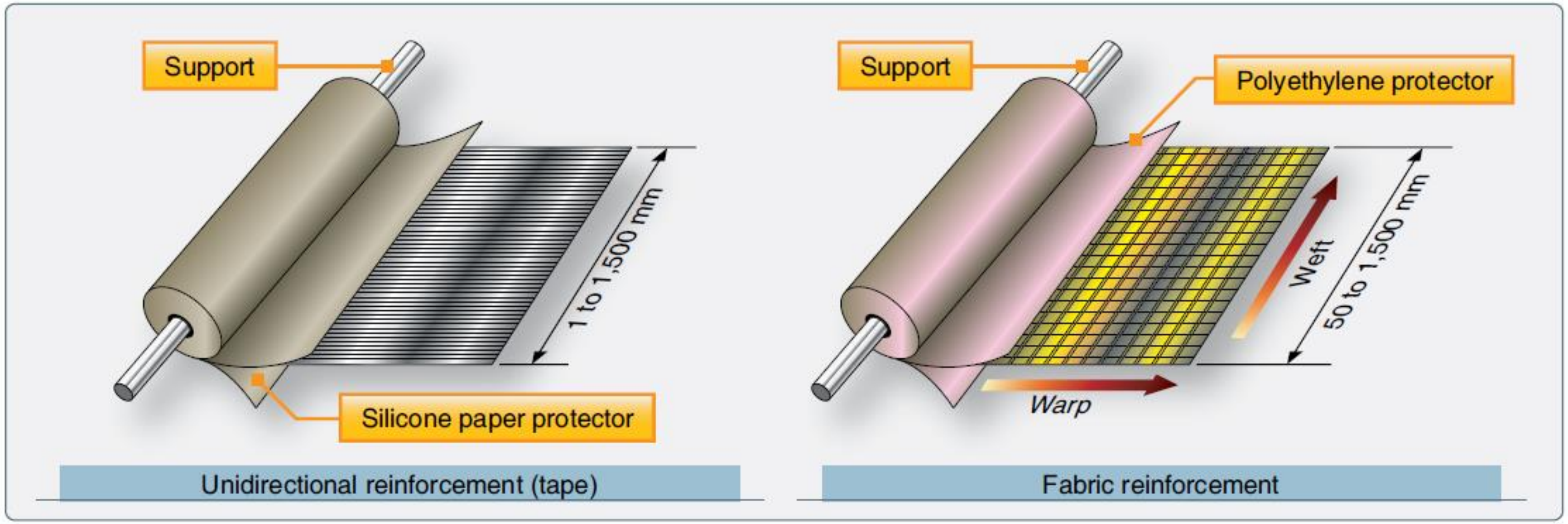


Fig 9-7: Example of Carbon Fiber Pre-pregs

Source: http://www.easycomposites.co.uk/Images/Guides/beginner-pre-preg/Large/what-is-pre-preg.jpg



Example of *Tape and fabric prepreg materials*

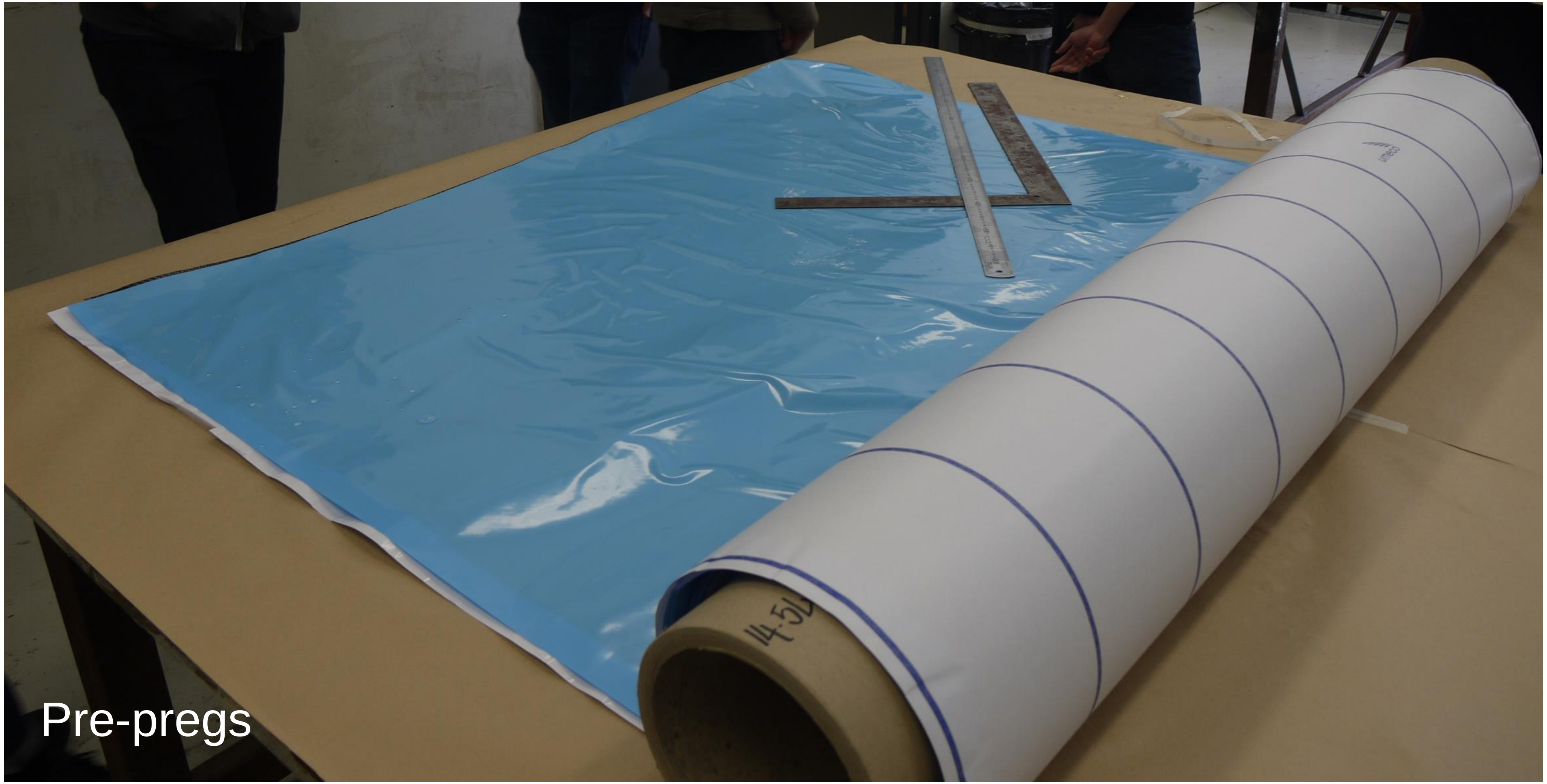
Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

- **Pre-pregs VS Wet lay-ups**

*Note: You will learn more about wet lay-ups in Chapter 11.

Matrix Materials: Preimpregnated Products (Pre-pregs)

- **Pre-pregs VS Wet lay-ups**



Quiz

Click the **Quiz** button to edit this object

Welcome to the quiz (Matrix Materials)

Click the "Start Quiz" button to proceed

Start Quiz

Adhesives: Film Adhesives*

- **Film adhesive**:
  - Thin films supported on a release paper and stored under refrigerated conditions (-18°C, or 0°F).
  - Are frequently supported by fibers that serve to improve handling of the films prior to cure, control adhesive flow during bonding, and assist in bondline thickness control.
  - Have a separator film or backing paper which is applied to keep the material from sticking to itself.
  - Tacky at room temperature and above, similar to pre-pregs.
  - Prior to application, to eliminate entrapped air pockets between the aircraft surface and adhesive film by pricking bubbles or "porcupine" rolling over the adhesive.
  - Used as a high strength adhesive to bond honeycomb to facesheet, precured detail parts and pre-pregs to original structures.*

Adhesives: Film Adhesives*

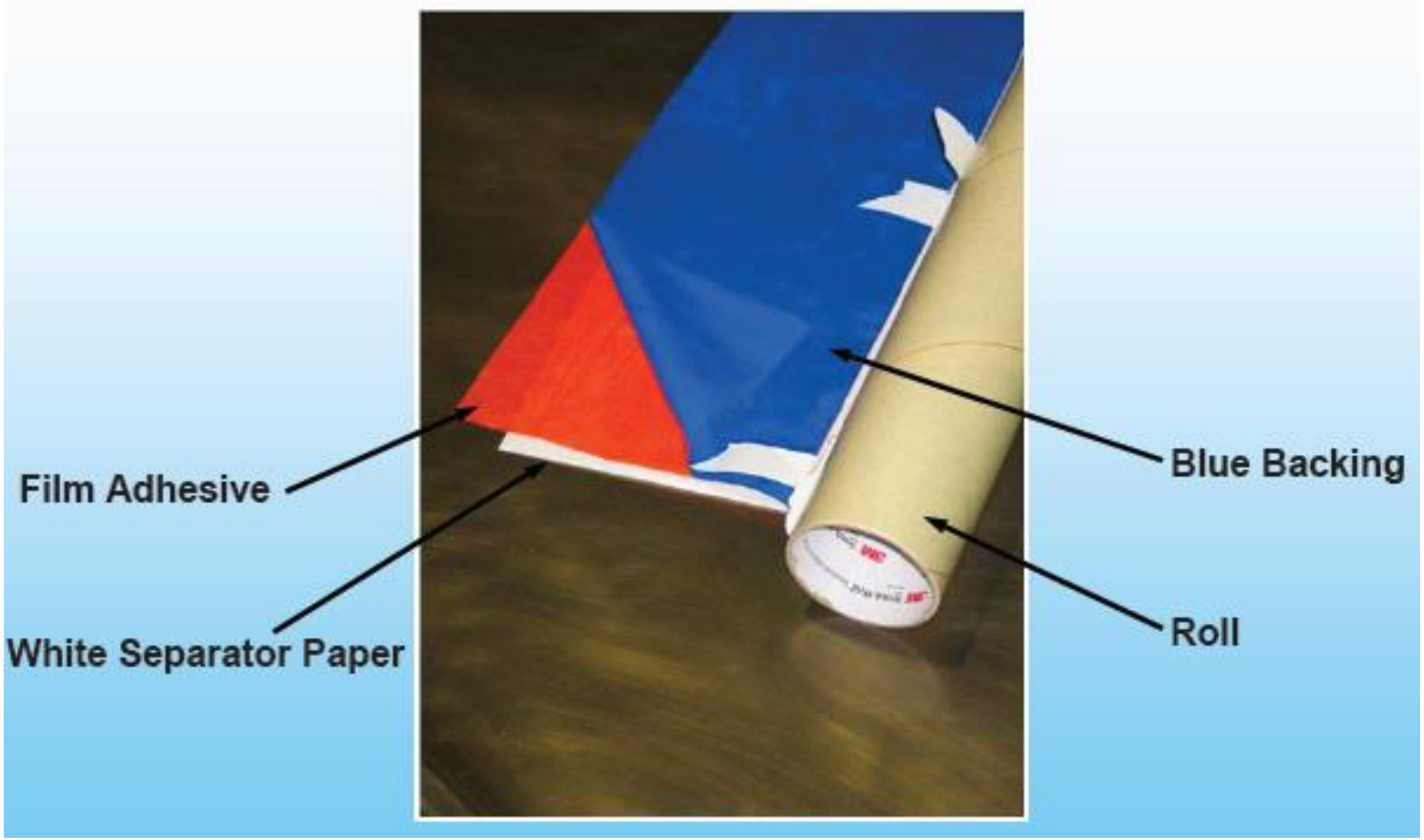


Fig 9-8: Example of Film Adhesives
Source: Extracted from Boeing 787 Composite Repair for Engineers (476)

Adhesives: Paste Adhesives

- **Paste Adhesive**:
  - Two components systems requiring careful weighing and thorough mixing to ensure strength is not compromised.
  - Have the advantage of lower temperature cure cycles and are easier to store and ship.
  - Disadvantage of lower strength, poor bond-line thickness control, and higher overall repair weight when compared to film adhesives.
  - Application of paste adhesives can be accomplished by brush, by spreading with a grooved tool, or by extrusion from cartridges or sealed containers using compressed air.
  - Used in the repair of composite parts as filler materials, for bonding repair core sections in place, and for bonding repair patches.

Adhesives: Paste Adhesives

Two-part paste adhesive

(2016). *Faa.gov*. Retrieved 15 September 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/media/FAA-H-8083-32-AMT-Powerplant-Vol-1.pdf

Adhesives: Paste Adhesives

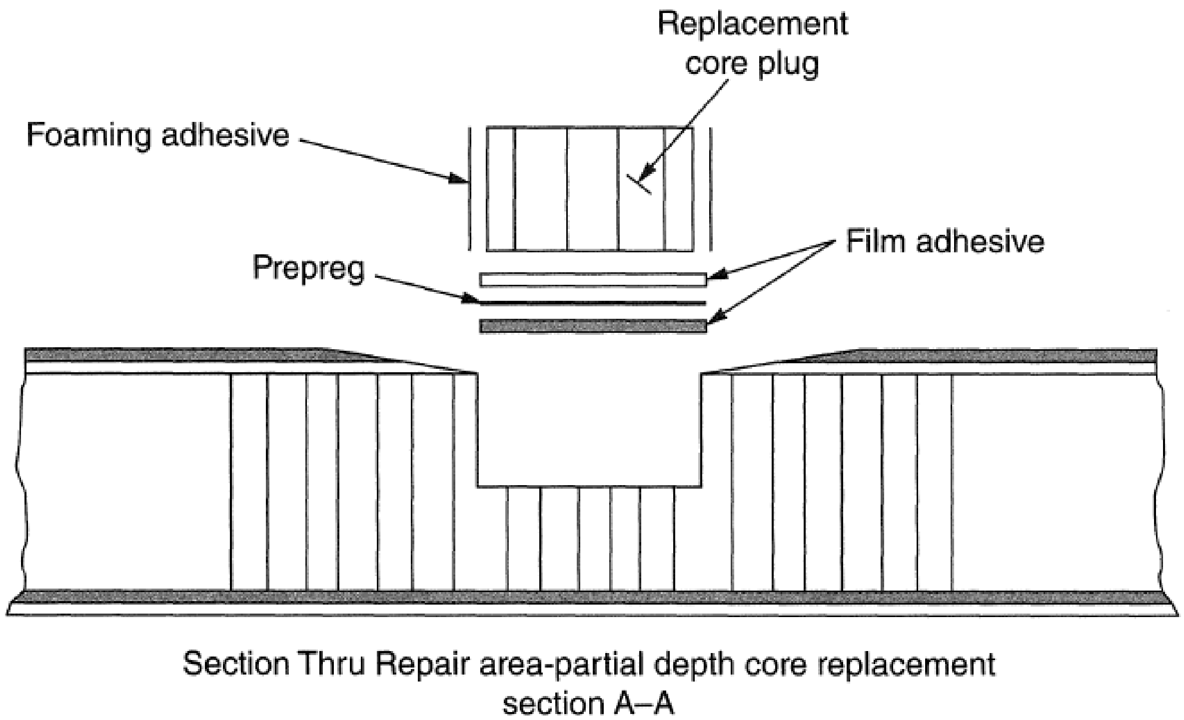


Figure 11-49 Replacement of damaged core

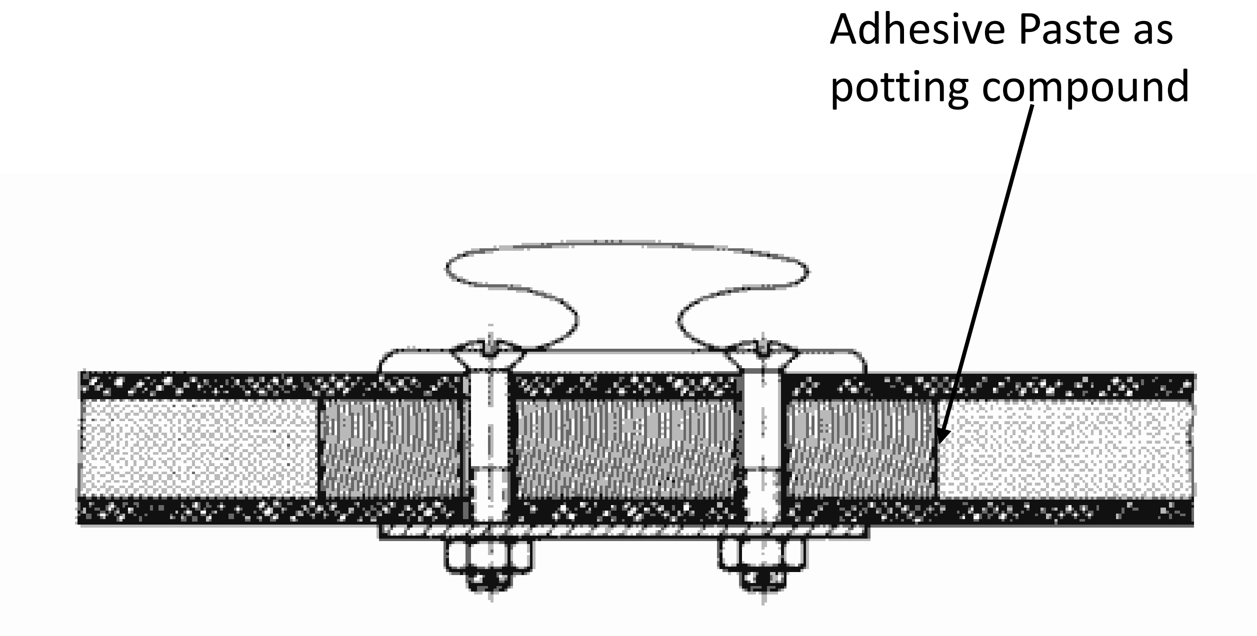


Fig 9-9: Example of adhesive paste as a potting compound
Source: https://boatbuildercentral.com/howto/foam10.php

Adhesives: Foaming Adhesives

- **Foaming adhesive**:
  - Consist of a thin unsupported epoxy film containing a blowing agent. During the rise to cure temperature, an inert gas is liberated causing an expansion or foaming action in the film.
  - The expansion must be performed under positive pressure to prevent over-expansion and reduced strength.
  - Used as a lightweight splice material for splicing honeycomb core repair sections.
  - Have the disadvantage of requiring a high-temperature cure cycle and must be shipped and stored at or below 0°F.
  - Sensitive to moisture, temperature, and contamination in the uncured state and must be handled and stored properly to prevent degradation.

Adhesives: Foaming Adhesives

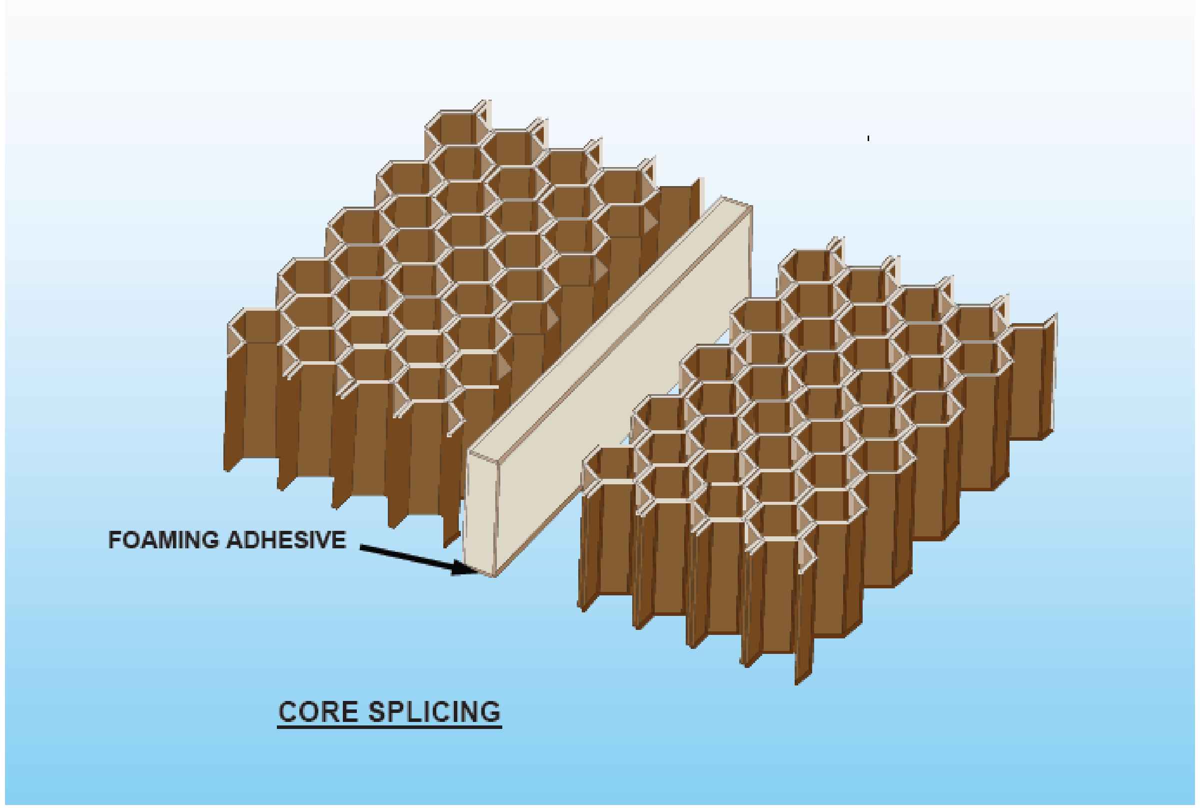
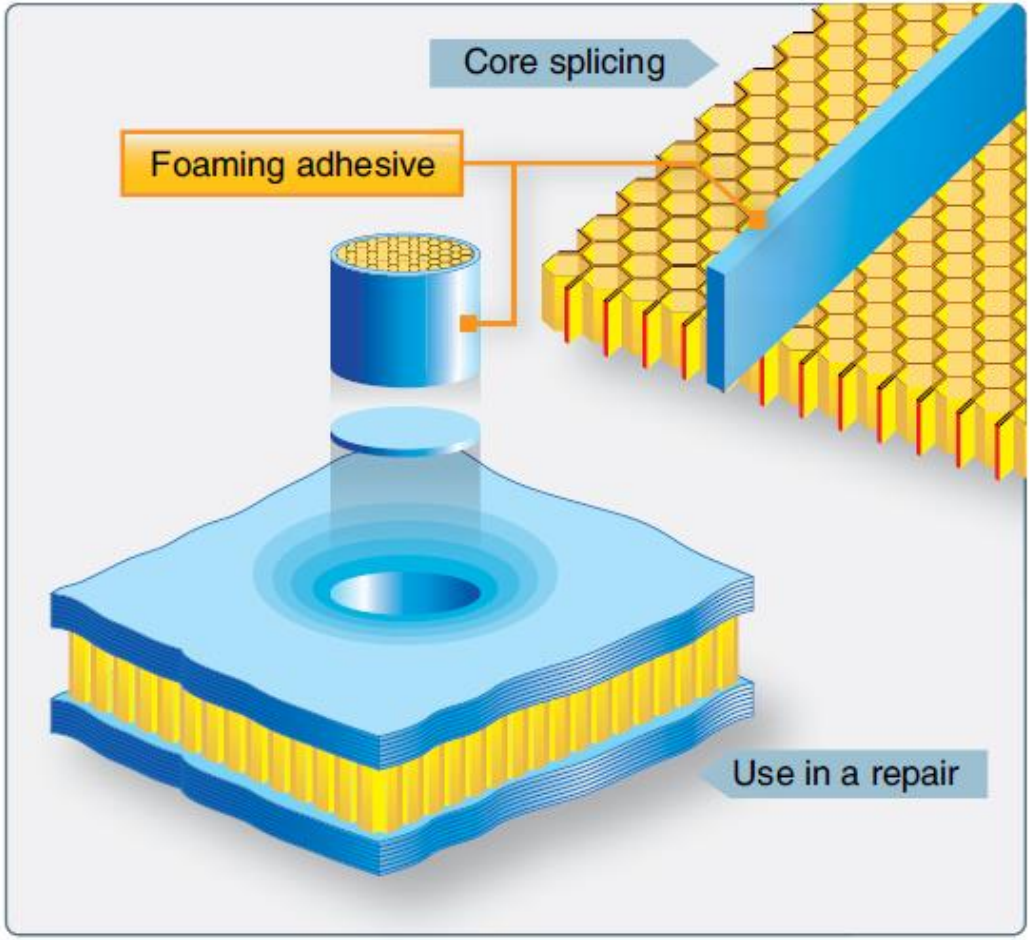


Fig 9-10: Example of Foaming Adhesives as adhesive between the new core and existing core
Source: Extracted from Boeing 787 Composite Repair for Engineers (476)



Example of Foaming Adhesives as adhesive between the new core and existing core
Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Quiz

Click the **Quiz** button to edit this object

Welcome to the quiz (Adhesive)

Click the "Start Quiz" button to proceed

Start Quiz

III. Sandwich Structures

Sandwich Structures

- A sandwich structure is a panel in which two face sheets (laminates) are bonded to a core.
- The core supports the face sheets against buckling and resists out of plane shearing.
- The core must have high shear strength and compression stiffness.
- Honeycomb core has an advantage of having a high ratio of stiffness-to-weight compared to other materials or construction.
- Most honeycomb sandwich structure are anisotropic.

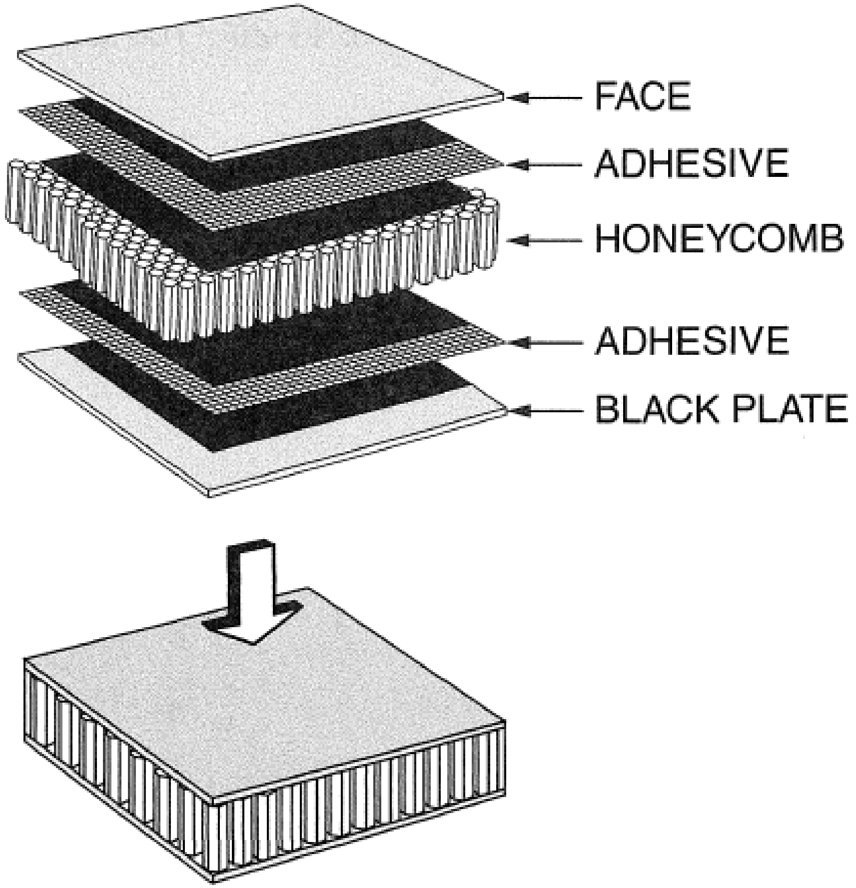
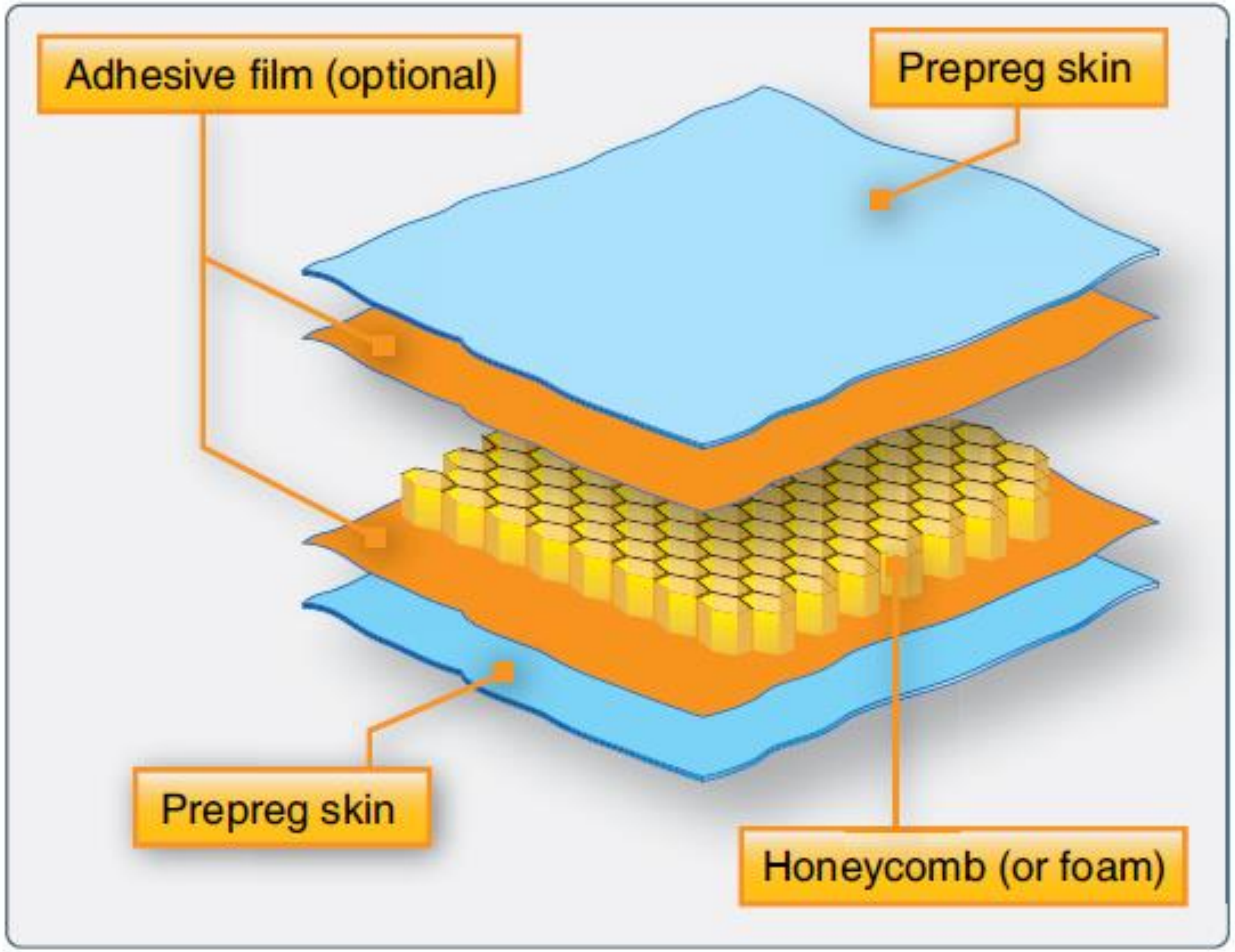


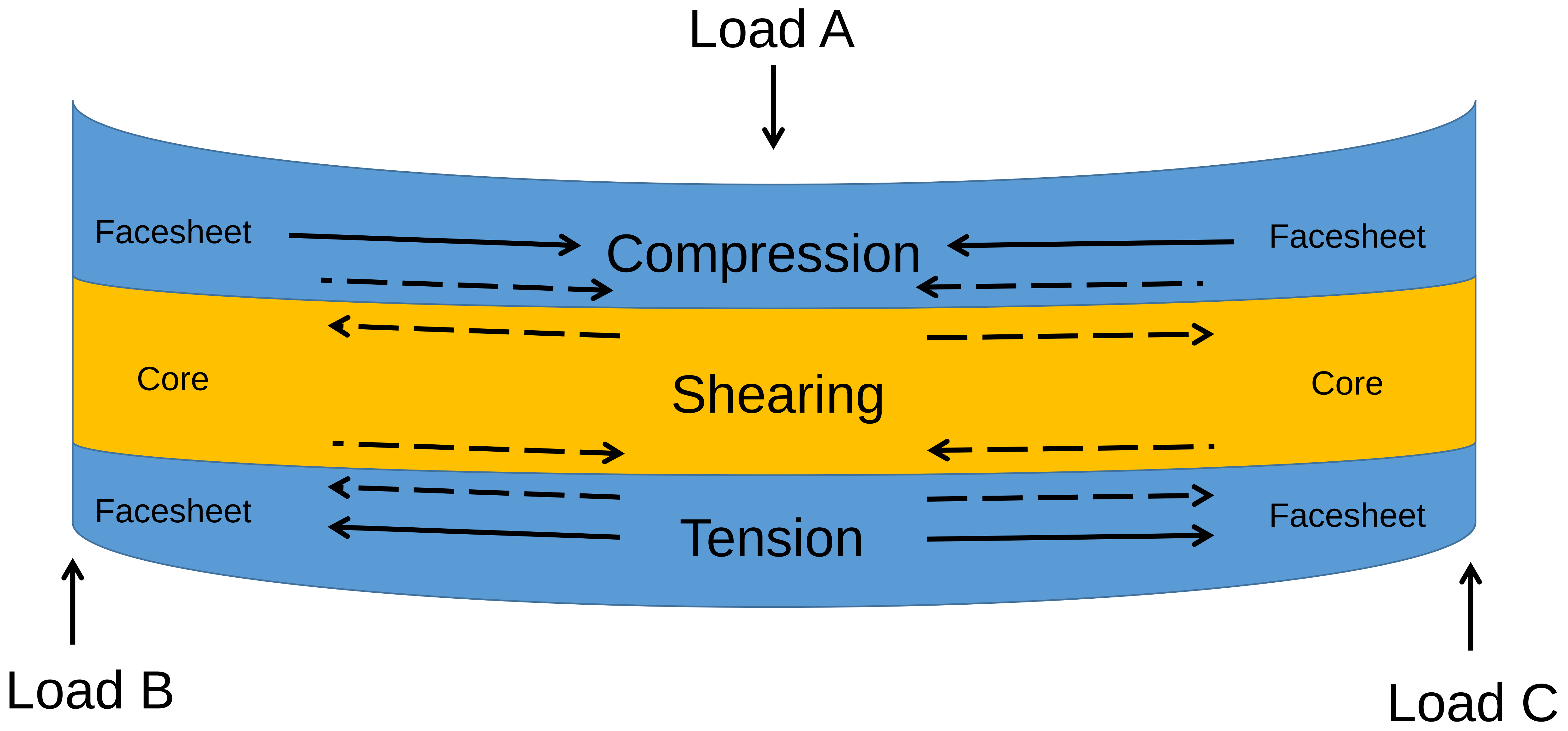
FIGURE 11-19 Honeycomb sandwich construction.

Sandwich Structures



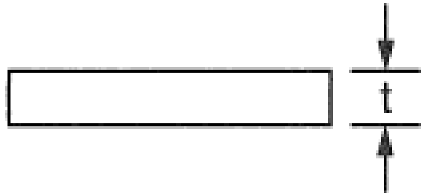
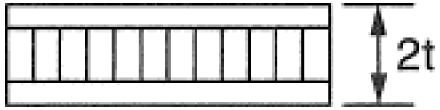
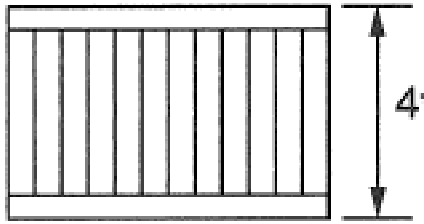
Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Sandwich Structures*



An example on how shearing stress (force) will take place under loads

Sandwich Structures: Properties

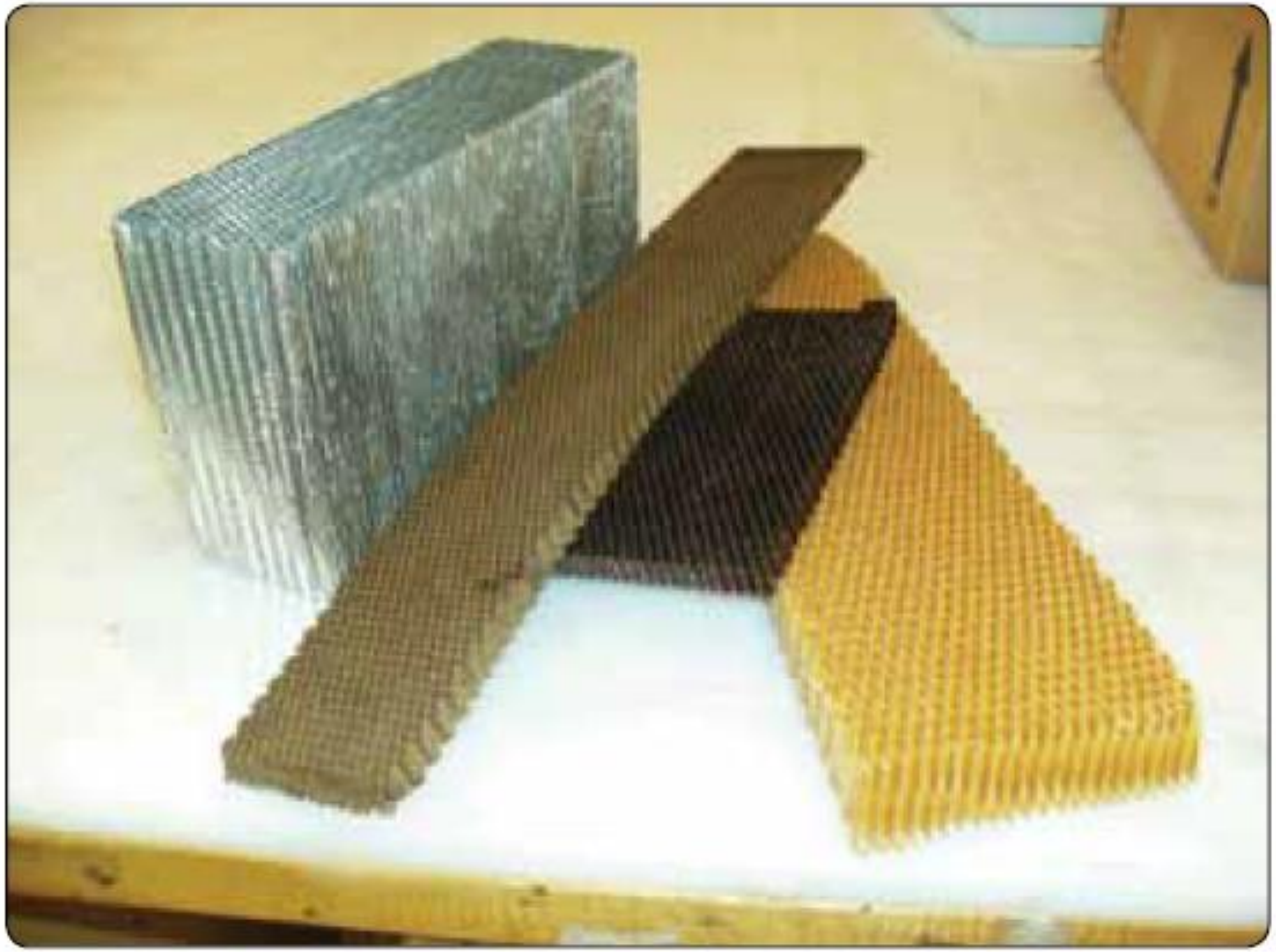
	Solid Metal Sheet	Sandwich Construction	Thicker Sandwich
			
Relative Stiffness	100	700 7 times more rigid	3700 37 times more rigid
Relative Strength	100	350 3.5 times as strong	925 9.25 times as strong
Relative Weight	100	103 3% increase in weight	106 6% increase in weight

A striking example of how honeycomb stiffens a structure without materially increasing its weight.

Sandwich Structures: Facing Materials & Core Materials

- **Facing Materials**:
  - Typically used are aluminum sheet or titanium sheet and fiberglass, Kelvar or carbon fiber laminates.
  - Face sheets of many laminate components are typically very thin, sometimes 3-4 plies for each face sheet.
- **Cores Materials**:
  - Typical core materials are Aluminium and Nomex (Paper impregnated with phenolic resin).

Sandwich Structures: Honeycomb*



Honeycomb Core Materials

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Sandwich Structures: Honeycomb*

- **Honeycomb:**
  - Metallic and nonmetallic honeycomb materials are used in combination with carbon fiber, fiberglass, and Kevlar face sheets.
  - Nonmetallic materials used are Fiberglass-reinforced honeycombs, Nomex, and Korex.
  - Hexagonal cells are the most common type and used for flat parts.
  - Overexpanded core decreases in shear strength in ribbon direction but increases shear strength in the width direction when compared to hexagonal honeycomb.*
  - Flexicore provides good formability for parts with tight contours without buckling the cell walls.

Sandwich Structures: Honeycomb*

- **Honeycomb:**
  - Ribbon Direction is the running direction of the honeycomb core. Typically, the cells are most easily to collapse in the direction perpendicular to ribbon direction.
  - Honeycomb core is available with different cell sizes/densities. Small sizes / higher densities provide better support for sandwich face sheets and is stronger and stiffer.*

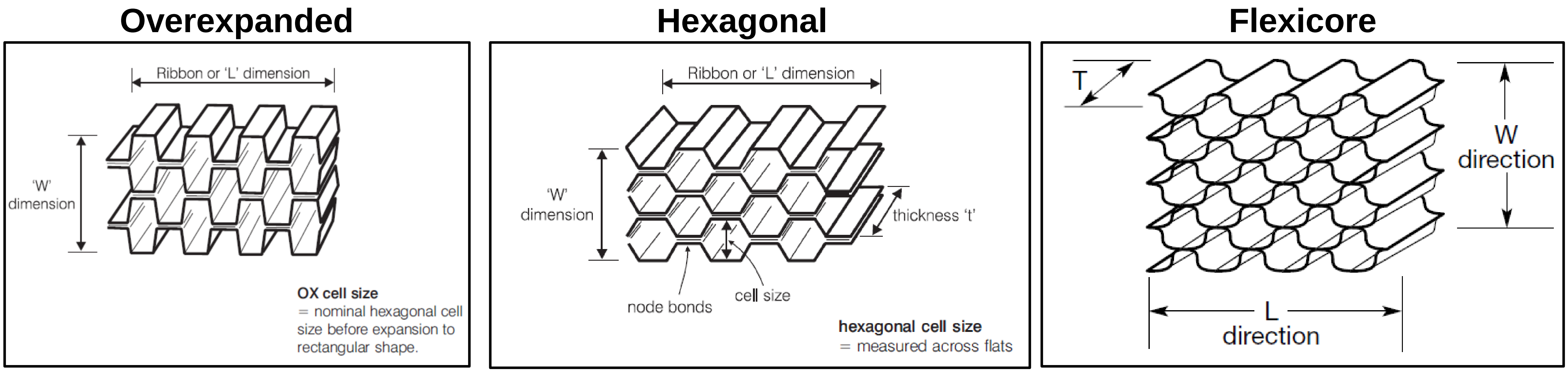


Figure 9-11: Examples of honeycomb cores (Overexpanded, Hexagonal and Flexicore)
Source: http://www.hexcel.com/resources/technology-manuals

Sandwich Structures: Foam & Balsa Wood

- **Foam**:
  - Are not commonly used on commercial-type aircraft.
  - Include a wide variety of liquid plastic materials that are filled with chemically released or mechanically mixed gas bubbles to produce rigid forms.
  - Tends to have lower mechanical properties than honeycomb.
- **Balsa Wood**:
  - A natural wood product with elongated closed cells.
  - Available in a variety of grades that correlate to the structural, cosmetic, and physical characteristics.
  - Density of balsa is less than one-half of the density of conventional wood products but has a considerably higher density than the other types of structural cores.
  - Modern aircraft floor panels, galley structures, and wardrobes may be fabricated from sandwich panels of balsa core faced with fiberglass or aluminium.

Description of Laminated Structure*

- Honeycomb sandwich has the following disadvantages compared to laminated structures:
  - Low impact damage resistance.
  - Prone to water intrusion.
  - Difficult to implement bolted repair. Typically, the core has to be potted with resin before installation of fasteners.*
  - Shapes/Contours of panels is dependent on the type and thickness of core used.*

Description of Laminated Structure

- Laminate structures are used for primary structures like wings and fuselage because:
  - More damage tolerant.
  - Easier to repair.
  - Easier to manufacture with automated manufacturing techniques.
  - Co-curing can be done with two or more laminated structures/components to eliminate secondary bonding processes or the use of fasteners.

Quiz

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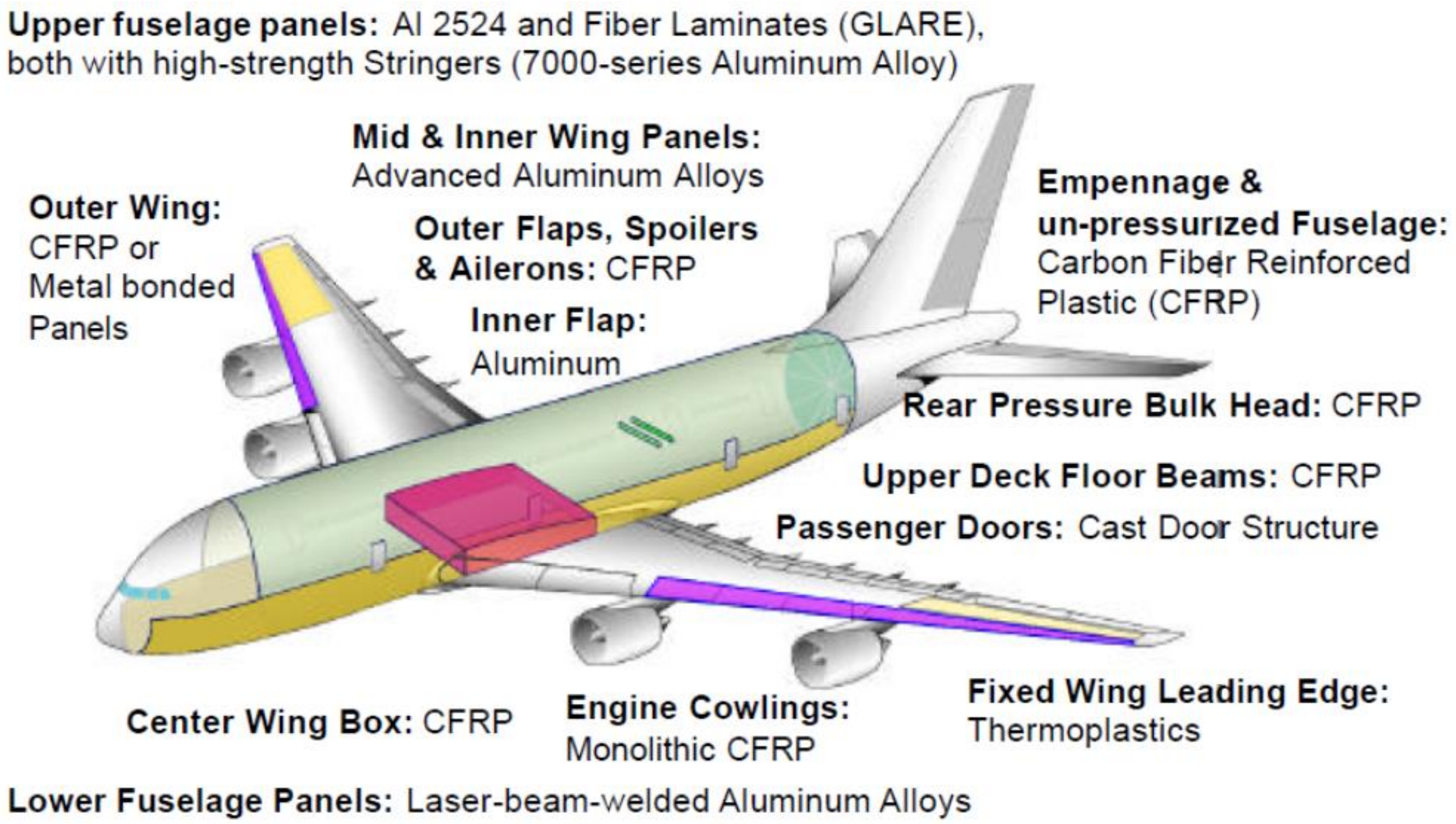
Welcome to the quiz (Sandwich Structure)

Click the "Start Quiz" button to proceed

Start Quiz

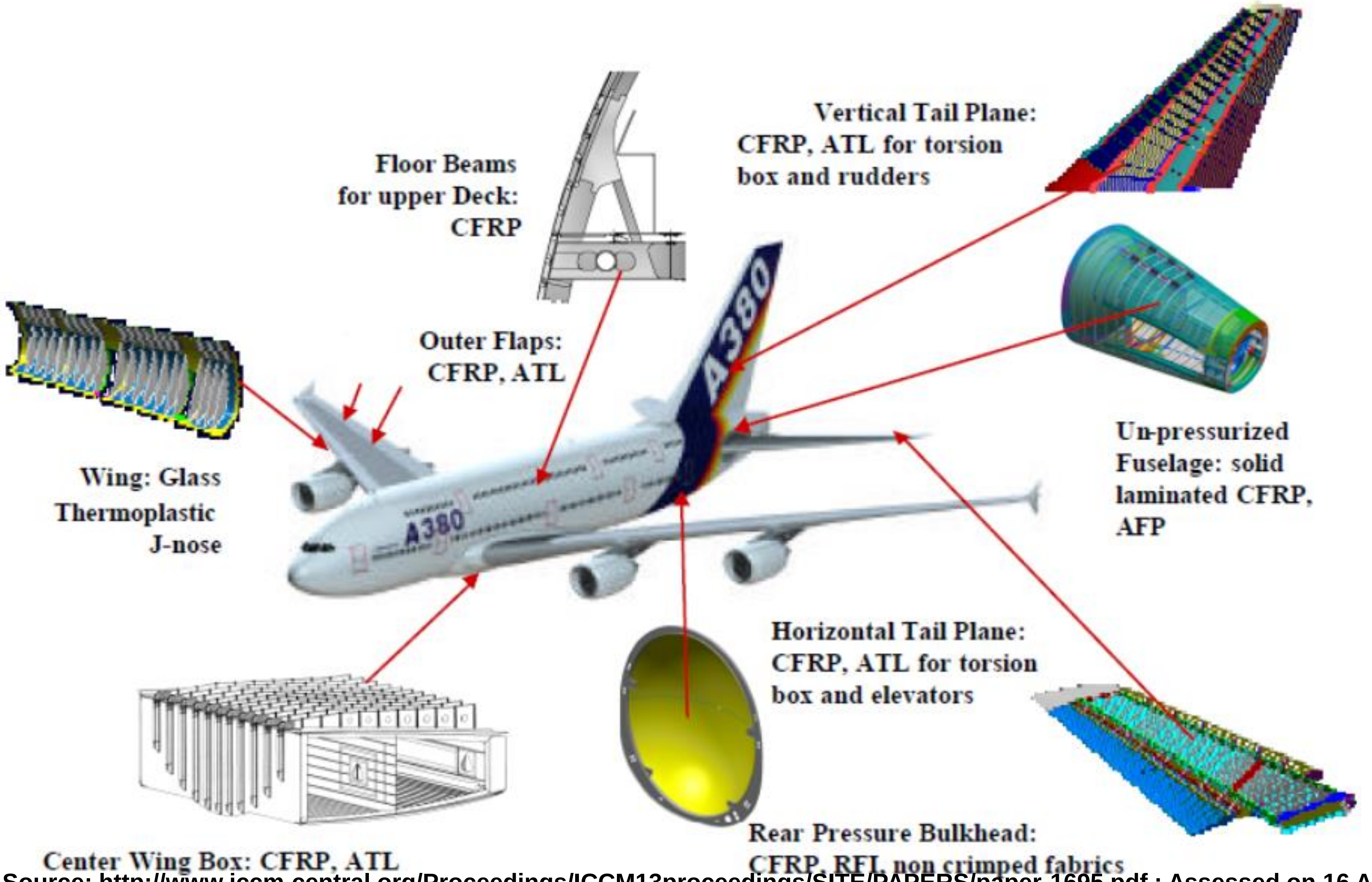
III. Sandwich Structures

Composite Materials used on A380



Source: http://www.iccm-central.org/Proceedings/ICCM13proceedings/SITE/PAPERS/paper-1695.pdf ; Assessed on 16 Apr 2015

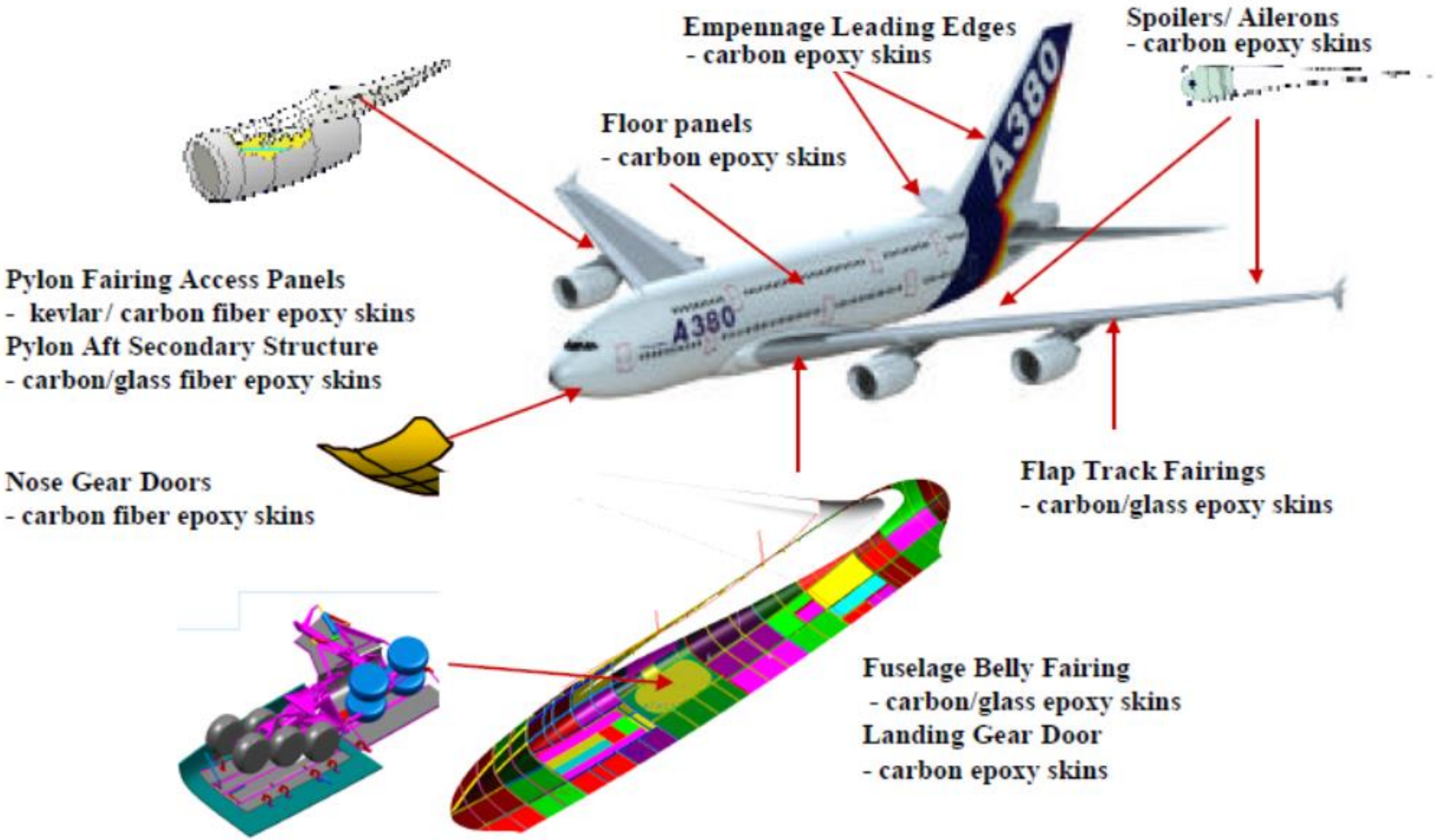
Composite Materials used on A380



Source: http://www.iccm-central.org/Proceedings/ICCM13proceedings/SITE/PAPERS/paper-1695.pdf ; Assessed on 16 Apr 2015

III. Sandwich Structures

Composite Materials used on A380



Source: http://www.iccm-central.org/Proceedings/ICCM13proceedings/SITE/PAPERS/paper-1695.pdf ; Assessed on 16 Apr 2015

Ⅳ. Manufacturing and In-Service Damage

Introduction

- Damage to composite may affect only a few surface plies over a large area or a depth of many plies in a smaller area.
- Failure to use protective devices during maintenance may cause damages ranging from small surface scratches to more severe defects that would include punctures.
- Once damage is identified, the extent of damage must be determined and then it can be classified to determine the disposition of the defective part.
- Improper damage removal and repair techniques can impart greater damage than originally sustained to the composite material.

Manufacturing Defects

- Manufacturing damage includes anomalies such as porosity, microcracking, and delaminations resulting from processing discrepancies.
- Inadvertent (non-process) damage can occur in detail parts or components during assembly or transport or during operation.
- Resin rich if too much resin is used: Undesirable but for non-structural applications, this is not necessarily bad but it will add weight.
- Resin starved if too much resin is bled off during the curing process or if not enough resin is applied during the wet lay-up process. Resin-starved areas are indicated by fibers that show to the surface.
- 60:40 fiber-to-resin ratio (by volume after curing) is considered optimum.
- Damage can occur within the composite material and structural configuration and the extent of damage controls repeated load life and residual strength and is critical to damage tolerance.

Fiber Breakage & Matrix Imperfection

- **Fiber Breakage**:
  - Can be critical if the structure is designed to be fiber dominant.
  - Fiber failure is typically limited to a zone near the point of impact.
- **Matrix Imperfection**:
  - Usually occur on the matrix-fiber interface, or in the matrix parallel to the fibers.
  - Accumulation of matrix cracks (a.k.a. micro-cracks) can cause degradation of matrix-dominated properties like significantly reduction of properties dependent on the resin or the fiber/resin interface, such as interlaminar shear and compression strength.
  - Slight reduction of properties if affected composite is fiber-dominated.
  - For high-temperature resins, micro-cracking can have a very negative effect on properties.
  - All the matrix defects may develop into delaminations, which are more critical damage.

Delamination and Debonds & Combinations of Damages

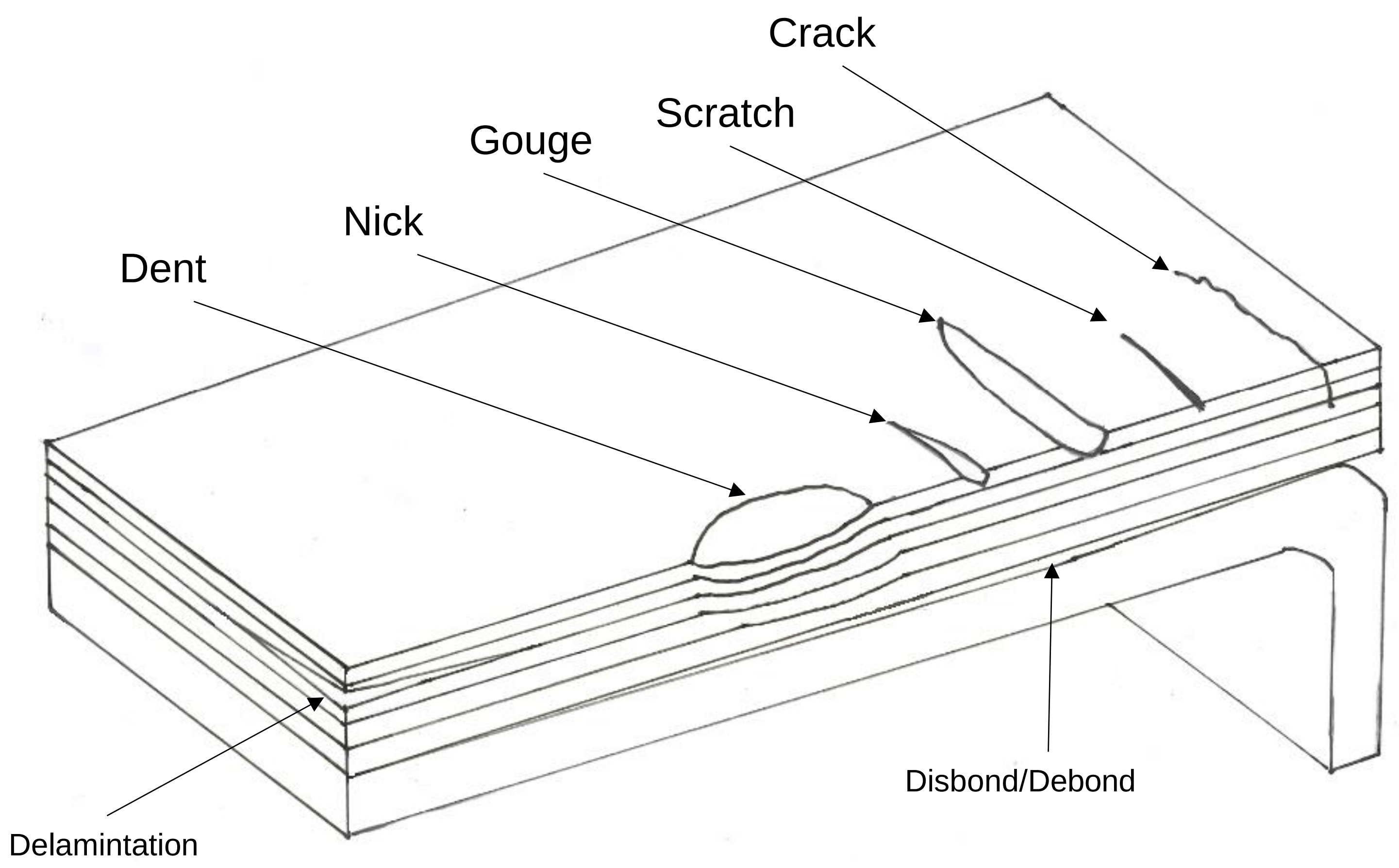


Fig 9-12: Example of defects found on a composite

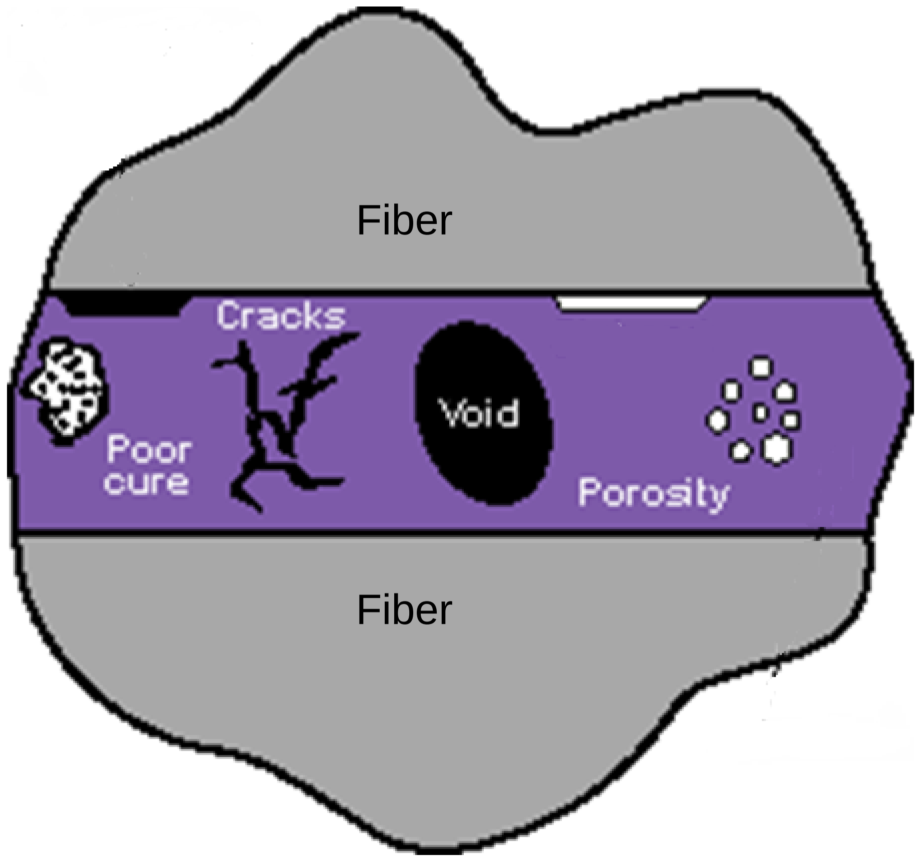


Fig 9-13: Example of defects found in the matrix of a composite
Source: http://www.twi-global.com/technical-knowledge/faqs/process-faqs/faq-what-is-the-difference-between-debonding-and-delamination-in-adhesive-joints-coatings-and-composites-and-other-defects-found/

- **Delamination and Debonds (Disbonds)**
  - Delaminations form on the interface between the layers in the laminate.
  - Debonds or disbonds can form along the bondline between two elements.
  - Delaminations or debonds can grow when subjected to repeated loading and can cause catastrophic failure when the laminate is loaded in compression under certain conditions.
- **Combinations of Damages**
  - High-energy impacts by large objects (i.e., turbine blades) may lead to broken elements and failed attachments. May include significant fiber failure, matrix cracking, delamination, broken fasteners, and debonded elements.
  - Damage caused by low-energy impact is more contained, but may also include a combination of broken fibers, matrix cracks, and multiple delaminations.

Delamination and Debonds & Combinations of Damages

Figure 9-14a: Example of combinations of damage on a sandwich structure (Radome) after bird-strike

Delamination and Debonds & Combinations of Damages

Figure 9-14b: Example of combinations of damage on a sandwich structure after bird-strike
Source: http://www.airliners.net/aviation-forums/tech_ops/read.main/184925/

In-service Defects

- **Flawed Fastener Holes**
  - Improper hole drilling, poor fastener installation, and missing fasteners may occur in manufacturing.
  - Hole elongation can occur due to repeated load cycling in service.
  - Typically, composite structures have low bearing strength. Flawed fastener holes need to be repaired once found.

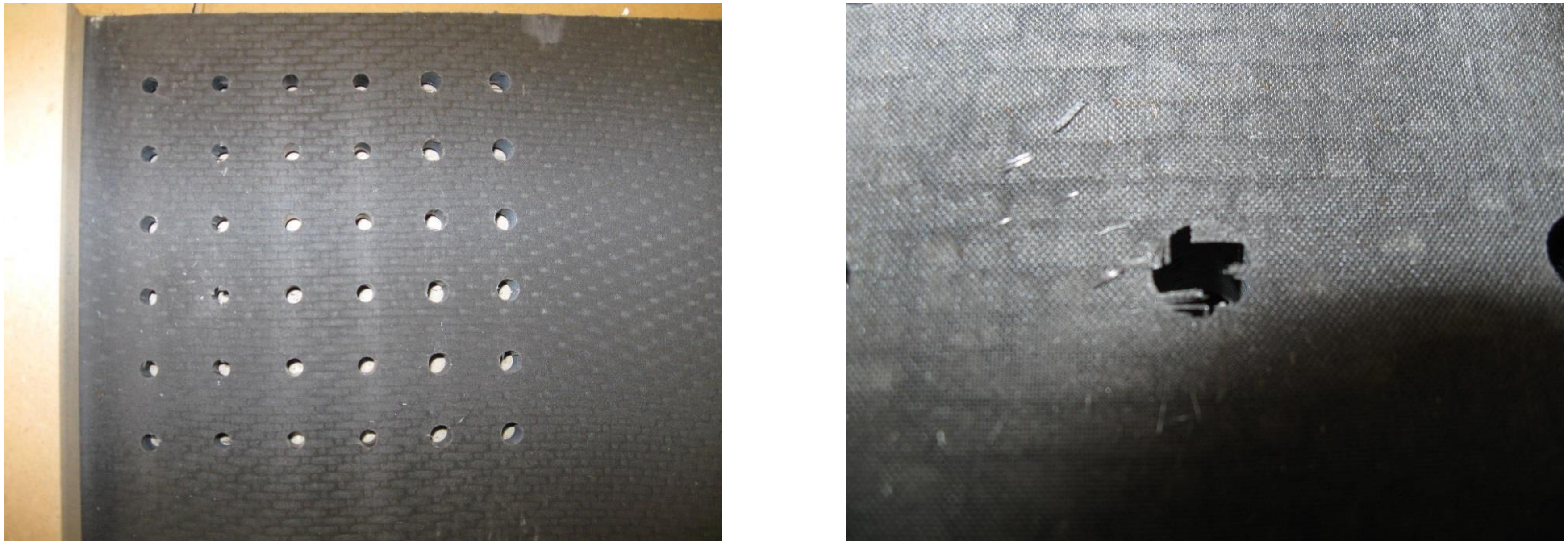
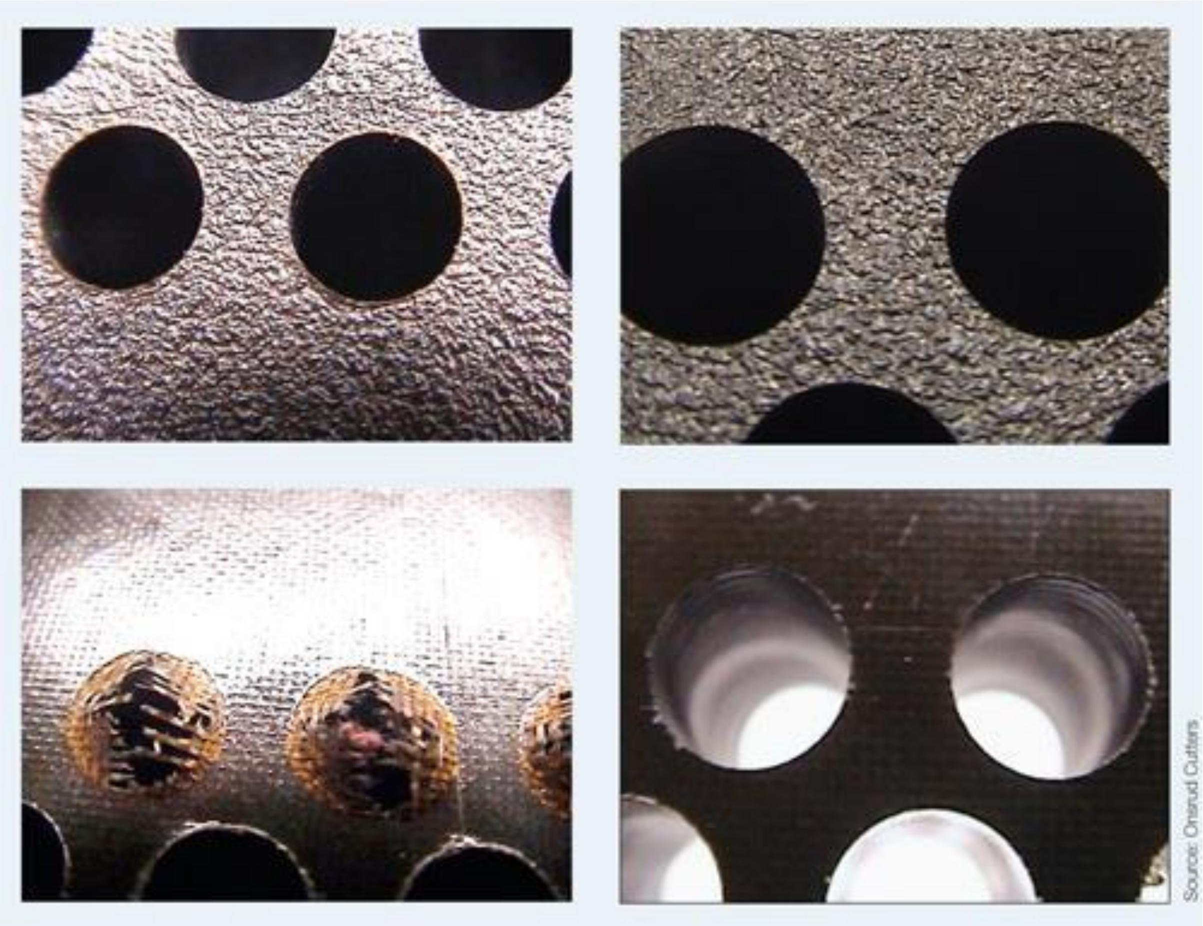


Figure 9-15: Example of drilling in composite laminate (left) and example of fiber breakage on the exit (right)
Source: http://www.mmsonline.com/articles/how-to-machine-composites-part-4----drilling-composites

Official (Closed) - Non Sensitive

In-service Defects

- **Flawed Fastener Holes**



Drilling a clean hole is especially challenging with composites. A worn or dull tool can produce holes with frayed edges and delamination. The photos contrast unacceptable and acceptable entry (top left and right) and unacceptable and acceptable exit holes (bottom left and right). Source: Onsrud Cutter

Source: Machining carbon composites: Risky business : CompositesWorld . (2016).Compositesworld.com. Retrieved 26 April 2016, from http://www.compositesworld.com/articles/machining-carbon-composites-risky-business

In-service Defects*

- **Low Resistant to Impact**:
  - Structures have adequate stiffness and strength, but low resistant to loads that the composites are not designed for. For example, impact load from dropping of heavy tools, etc.
- **Liquid Ingression**:
  - Exposure to water and Skydrol (Hydraulic fluid) can take place due to impact or manufacturing defects when undetected.
  - Water create additional damage in repaired parts when cured above boiling point. Removal of water before repair is cured is recommended.
  - Freeze-Thaw Cycle*: Water ingression will expand in volume as ice during flight and melt when on ground. This cycle repeats every flight and may lead to severe delamination or debond.

In-service Defects

- **Liquid Ingression**:
  - Complete removal of Skydrol is almost impossible. Removal of contaminated core and adhesive is recommended before repair.

- **Erosion**:
  - Erosion capabilities of composite materials are less than aluminium so their application in leading edge surfaces has been generally avoided.
  - Erosion coatings are available to provide protection but the durability and maintainability of erosion coatings are less than ideal.
  - Edges of doors or panels can erode if they are exposed to the air stream attributed to improper design or installation/fit-up.

In-service Defects*

- **Corrosion:**
- Metal structures in contact or in the vicinity of composite parts may show corrosion damage due to:
  - Inappropriate choice of aluminium alloy (For example lightning protection fibers).
  - Damaged corrosion sealant of metal parts during assembly or at splices.*
  - Insufficient sealant, and/or lack of glass fabric isolation plies at the interfaces of spars, ribs, and fittings.*
- **UV radiation:**
  - Composite structures need to be protected by paint to prevent the effects of UV light.

V. Nondestructive Inspection of Composite

V. Nondestructive Inspections of Composites

- Damage to composite components is not always visible to the naked eye and the extent of damage is best determined for structural components by suitable nondestructive test (NDT) methods also called nondestructive inspection (NDI).
- Composites are generally more difficult to inspect than metals because laminated structures contain multiple ply orientations with numerous ply drop-offs.
- It is important that technicians conducting NDI be trained and certified in the method they are using as NDI ranges from simple visual inspections to very sophisticated automated systems with extensive data handling capabilities.

V. Nondestructive Inspections of Composites

TABLE 11-1 Comparison of NDI Methods for Fault Detection

Method of Inspection	Type of Defect							
	Disbond	*Delamination*	*Dent*	*Crack*	*Hole*	*Water ingestion*	*Overheat and burns*	*Lightning strike*
Visual	X*	X*	X	X	X		X	X
X-ray	X	X		X		X		
Ultrasonic TTU	X	X						
Ultrasonic pulse ECHO		X				X		
Ultrasonic bondtester	X	X						
Tap test	X†	X†						
Infrared thermography	X‡	X‡				X		
Dye penetrant								
Eddy current				X§				
Shearography	X‡	X‡		X§				

Note:
*For defects that open to the surface.
†For thin structure (3 plies or less).
‡The procedures for this type of inspection are being developed.
§This procedure is not recommended.

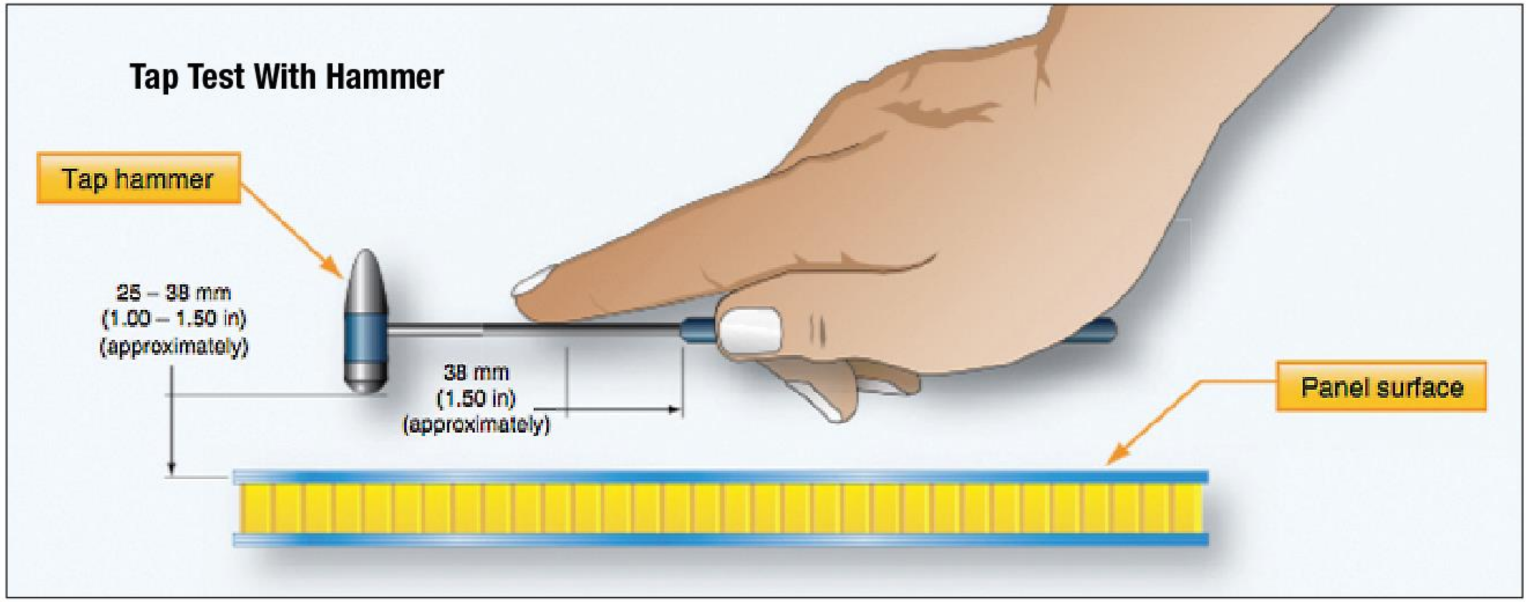
Visual Inspection

- **Visual inspection:**
  - Is often used to inspect metallic and composite aircraft.
  - Large areas can be inspected quickly and if damage is suspected more detailed inspection methods can be used to verify the damage.
  - Flashlights, magnifying glasses, mirrors, and boroscopes are employed as aids to magnify defects which otherwise might not be seen easily and to allow visual inspection of areas that are not readily accessible.
  - Resin starvation, resin richness, wrinkles, ply bridging, discoloration (due to overheating, lightning strike, etc.), impact damage by any cause, foreign matter, and blisters are some of the discrepancies readily discernible by a visual inspection.
  - Visual inspection cannot find internal flaws in the composite, such as delaminations, disbonds, and matrix crazing.

Audible Sonic Testing (Coin Tapping)*

- **Audible Sonic Testing (Coin Tapping/Tap Test)**:
  - Consists of lightly tapping the surface of the part with a coin or other suitable object.
  - Acoustic response of the damaged area is compared with that of a known good area.
  - A dull sound is a good indication that some delamination or disbond exists, although a clear, sharp tapping sound does not necessarily ensure the absence of damage, especially in thick panels.
  - The acoustic response of a good part can also vary dramatically with changes in geometry.
  - If results are questionable, the technician can compare the results to a tap test of other like parts which are known to be good or use another inspection method.
  - The automated tap test (a.k.a woodpecker) uses a solenoid instead of a hammer and this type of device will give a more consistent reading than the manual tap test
  - Tap test inspection loses it reliability for face sheet with 4 plies or more.*

Audible Sonic Testing (Coin Tapping)

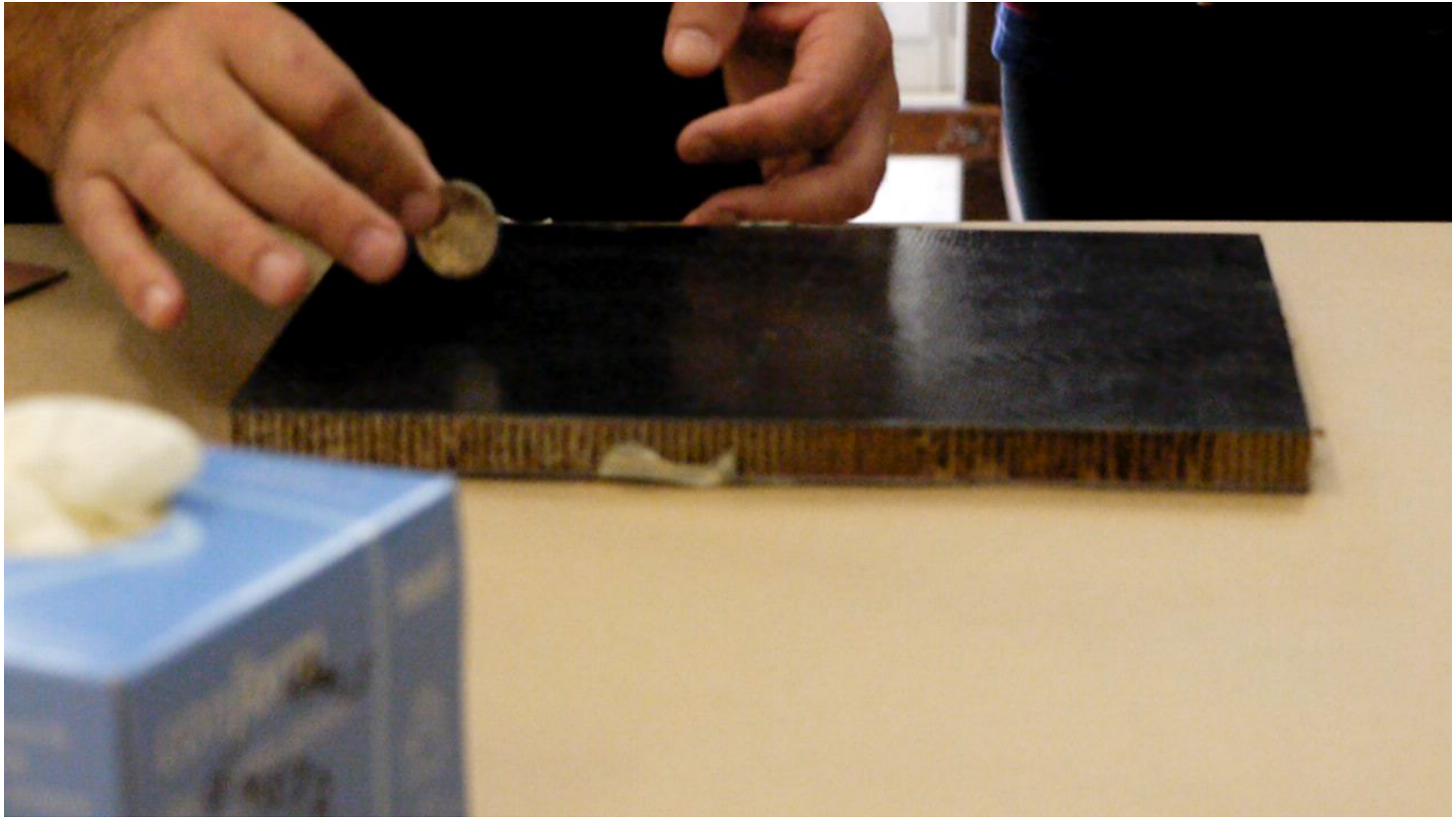


Source: http://amumagazine.com/?p=3154

Source: http://www.jrtech.co.uk/products/ndt-and-inspection

Audible Sonic Testing (Coin Tapping)

A video of how tap test is done



Ultrasonic Inspection*

- **Ultrasonic Inspection**:
  - A very useful tool for the detection of internal delaminations, voids, or inconsistencies in composite components not otherwise discernible using visual or tap methodology.
  - There are many ultrasonic techniques but each technique transmits a high frequency sound wave (ultrasonic) into the part to be inspected.
  - The route and echo of the ultrasonic wave is evaluated to determine presence and extent of any defects.
  - Transmission, pulse echo, phase array, and bond tester are types of ultrasonic equipment for inspection.
  - For sandwich structures, pulse echo is not preferred due to the change in materials (face sheet to core to face sheet).*

Ultrasonic Inspection

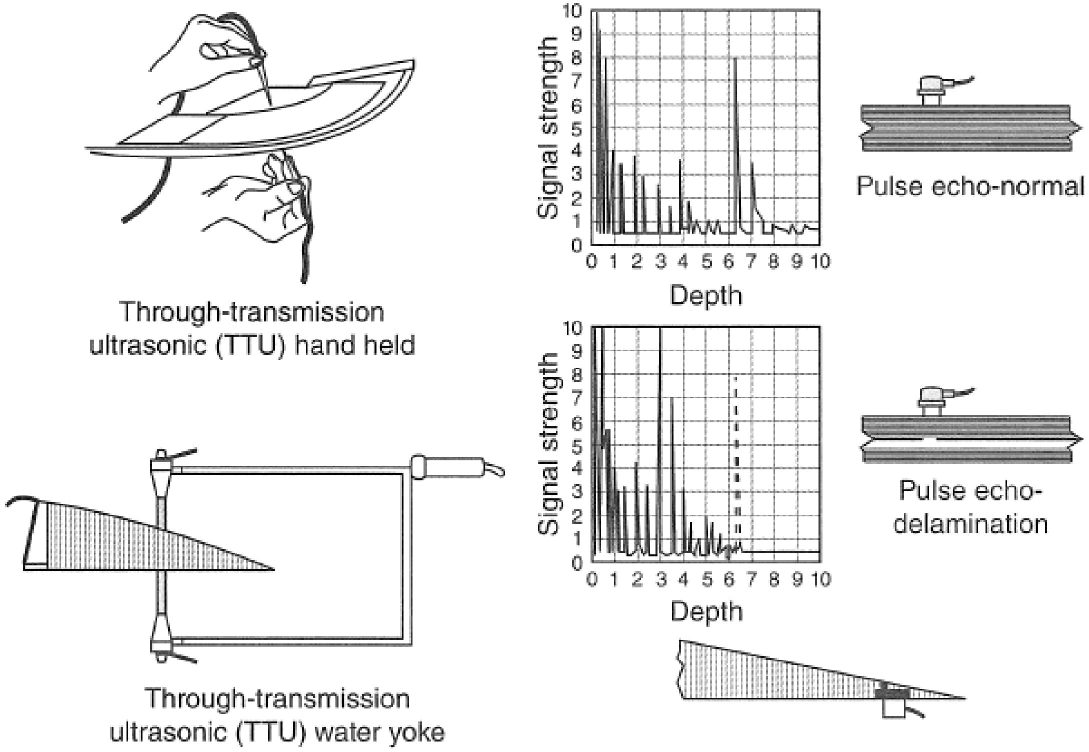
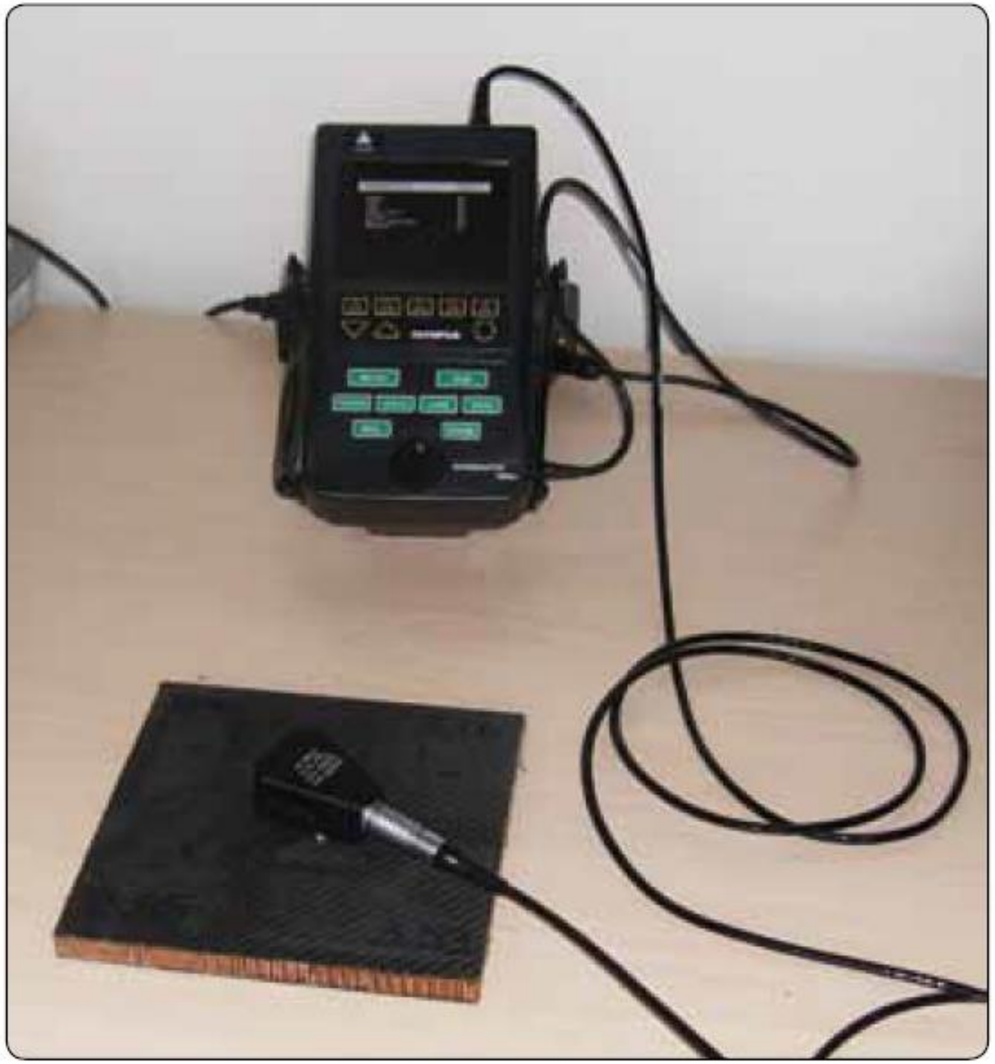


FIGURE 11-27 Through transmission and pulse echo ultrasonics.

NDI Equipment

Extracted from Boeing Aero QTR 4 2014

Ultrasonic Inspection

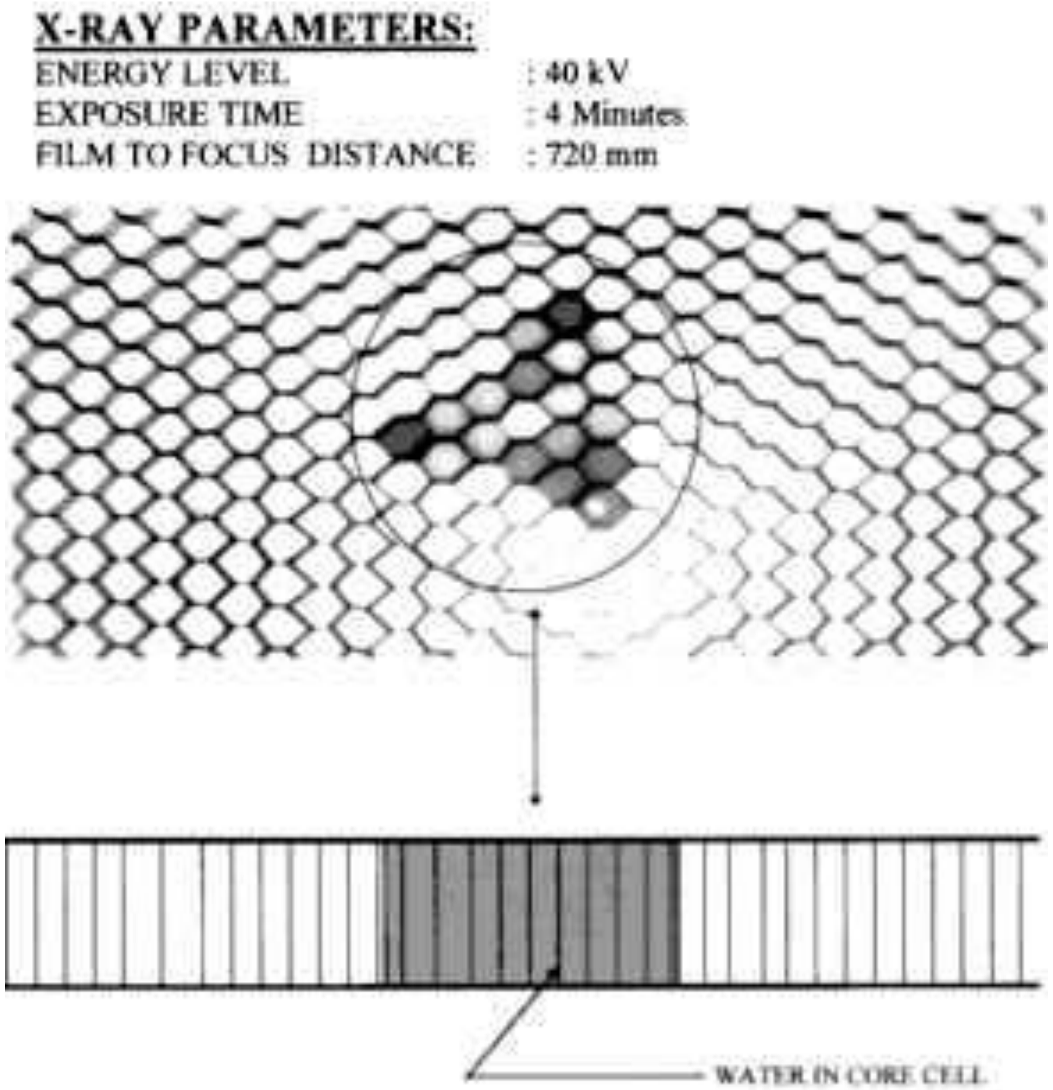


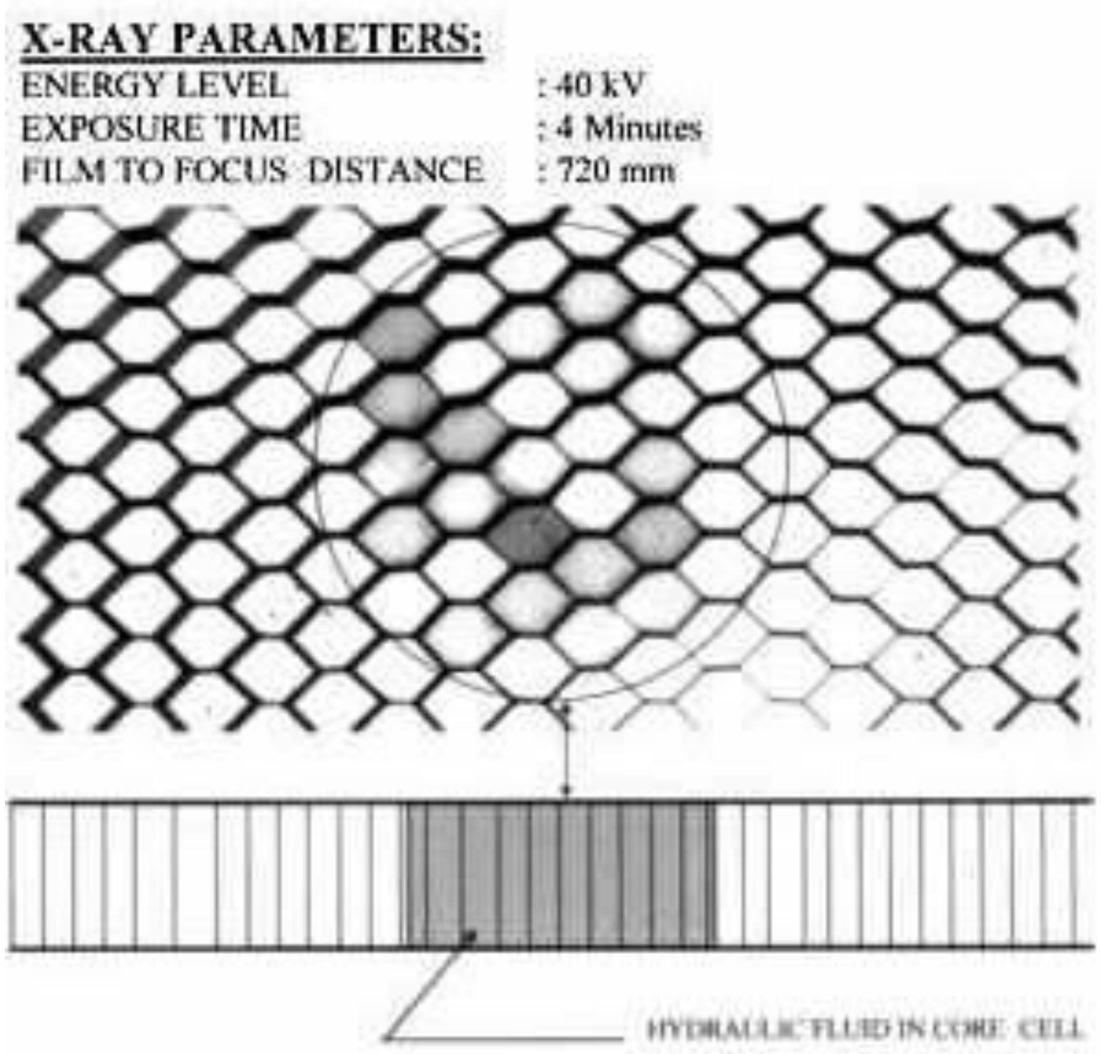
Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Radiography

- **Radiography (X-ray)**:
  - Passing X-rays through the part or assembly being tested while recording the absorption of the rays onto a film sensitive to X-rays.
  - Variations in material thickness or density and the presence of flaws affect the intensity of the transmitted radiation.
  - Voids and other discontinuities attenuate the radiation to different degrees and are revealed as contrasting light and dark areas (shadows) on the recorded image.
  - Defects less dense than surrounding material (such as voids in a composite laminate) absorb less radiation and are shown on X-ray film as darker areas as compared to nearby images.
  - Defects more dense than surrounding areas absorb more radiation (such as water in honeycomb sandwich assemblies) and are shown as lighter areas on X-ray film when compared to nearby images.

Radiography



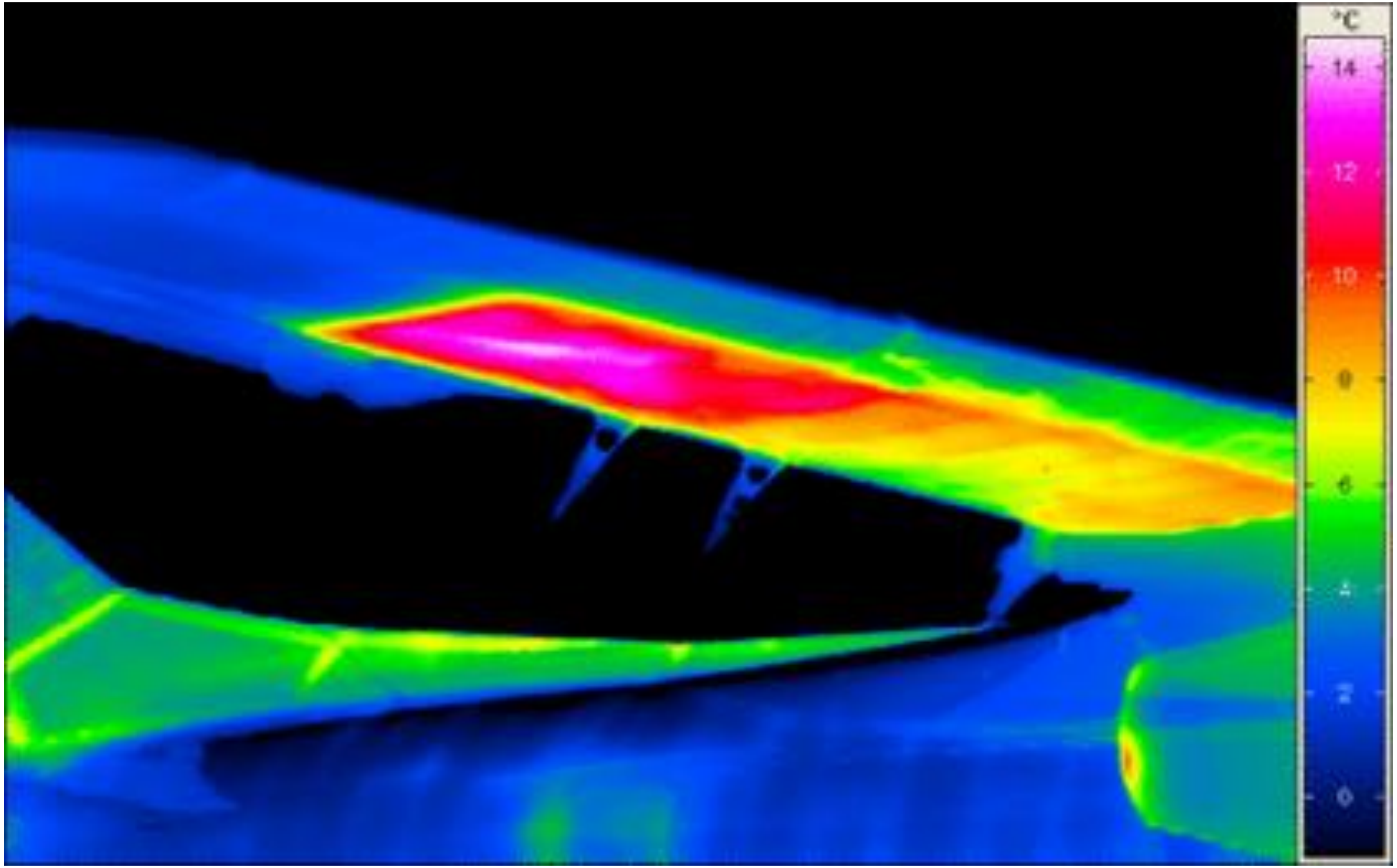


http://www.ndt.net/article/dgzfp02/papers/p36/p36.htm

Thermography*

- **Thermography**:
  - All thermographic techniques rely on differentials in thermal conductivity between normal, defect-free areas and those having a defect.
  - Defect-free areas conduct heat more efficiently than areas with defects, the amount of heat that is either absorbed or reflected indicates the quality of the bond.
  - Defects that affect the thermal properties include disbonds, cracks, impact damage, panel thinning, and water ingress into composite materials and honeycomb core.
  - Normally, a heat source is used to elevate the temperature of the part being examined while observing the surface heating effects using an infrared camera.*
  - Thermography can also be done when the aircraft just landed as the aircraft is still cold from flight.*
  - Thermal methods are most effective for thin laminates or for defects near the surface.

Thermography*



Source: http://www.infratec-infrared.com/thermography/application-area/process-optimisation/aviation.html

Quiz

Click the **Quiz** button to edit this object

Welcome to the quiz.
(Manufacturing and In-Service Damage)

Click the "Start Quiz" button to proceed

Start Quiz

VI. Damage Assessment

Damage Categories

- **Permanent repair:**
  - Restores the strength of the original structure like stiffness, static strength up to ultimate load, durability for the remaining life of the component.
  - Satisfies original part damage tolerance requirements, and restores functionality of aircraft systems.
  - Other criteria applicable in repair situations are to minimize aerodynamic contour changes, minimize weight penalty, minimize load path changes, and be compatible with aircraft operations schedule.

Damage Categories

- **Interim repairs:**
  - Repair design criteria for temporary/interim repairs can be less demanding but borders permanent repair if repair is to be embodied for a considerable time.
  - Permanent repair is preferred by aircraft owners and original equipment manufacturers.
  - Temporary repairs may damage parent structure, necessitating a more extensive permanent repair or part scrapping.
  - All temporary repairs have to be approved before the aircraft can be restored to operational status.
- **Cosmetic repair:**
  - Does not affect the structural integrity of the component or aircraft.
  - Carried out to protect and decorate the surface and will not involve the use of reinforcing materials.

Quiz

Click the **Quiz** button to edit this object

Welcome to the quiz
(Nondestructive Inspection of Composite)

Click the "Start Quiz" button to proceed

Start Quiz

VII. Composite Repairs

- Fibres are laid in layers using a resin system (matrix) and, therefore, composites manufactured this way are known as laminates.
- A sandwich structure is a panel in which two face sheets (laminates) are bonded to a core.

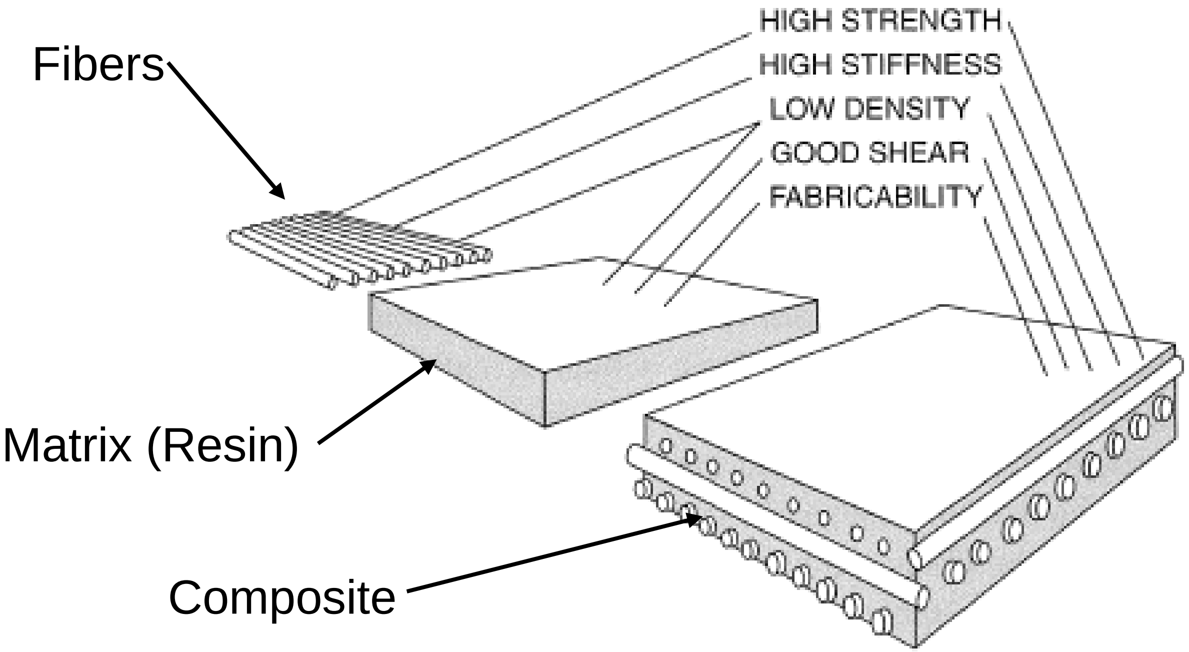


FIGURE 11-2 Solid laminate using reinforced fibers.

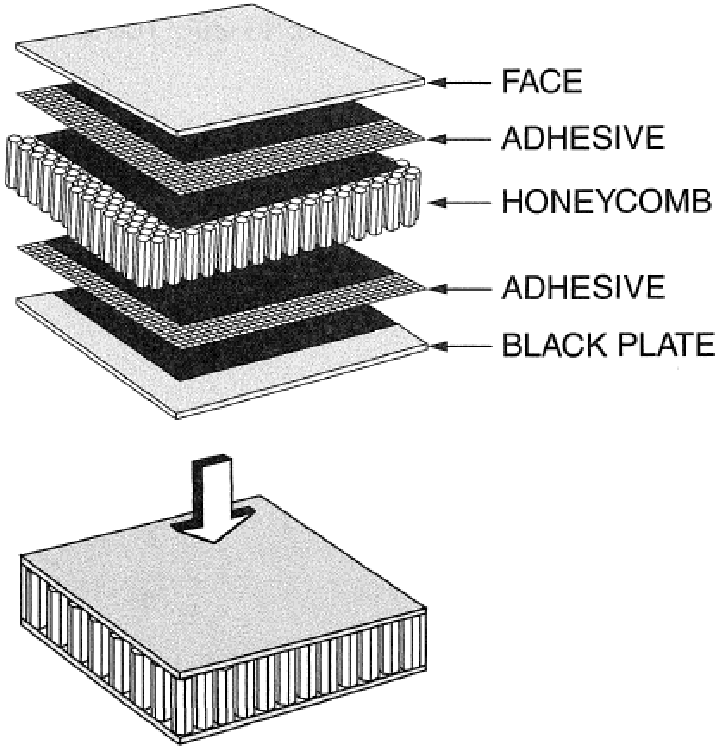
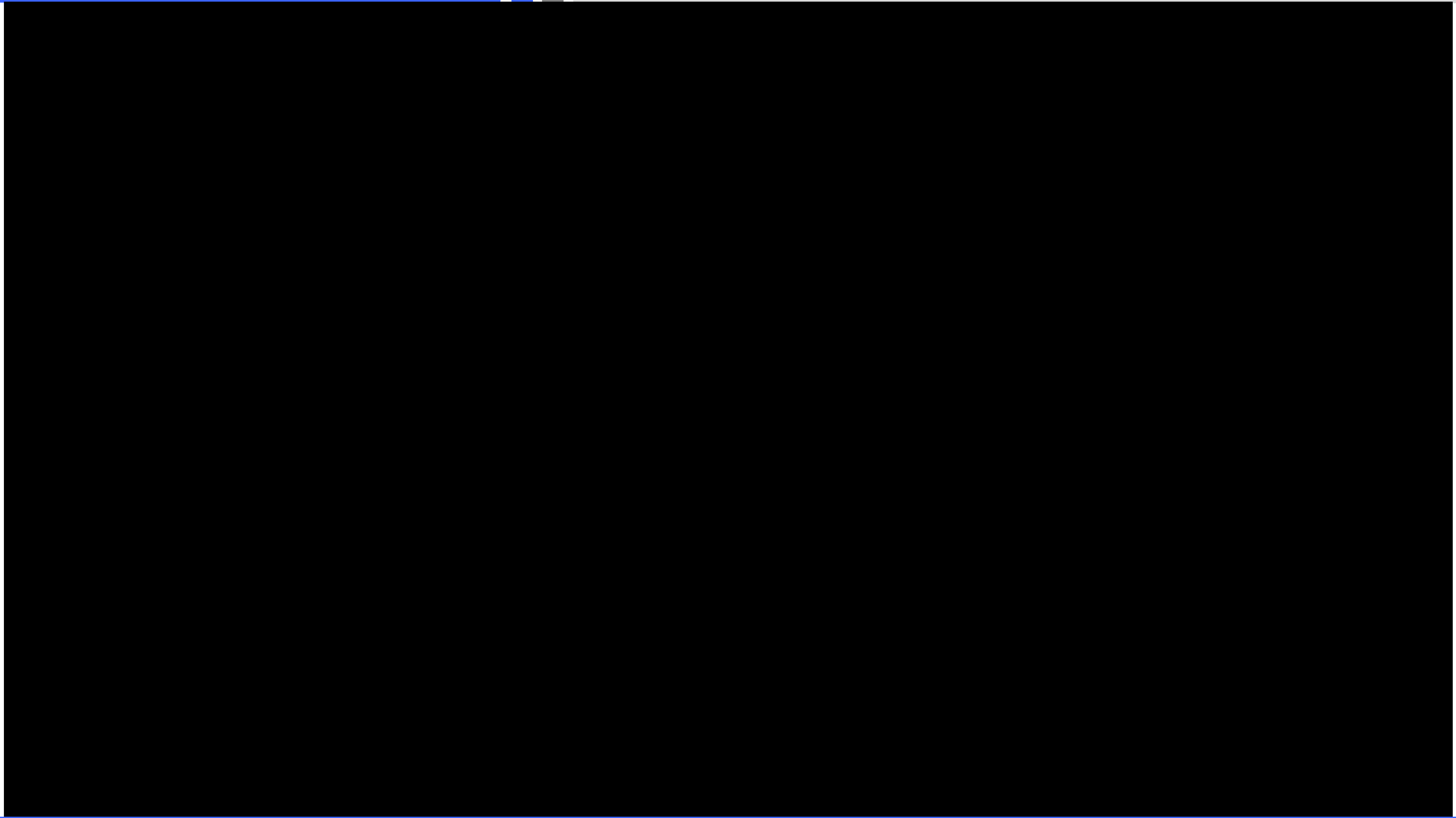


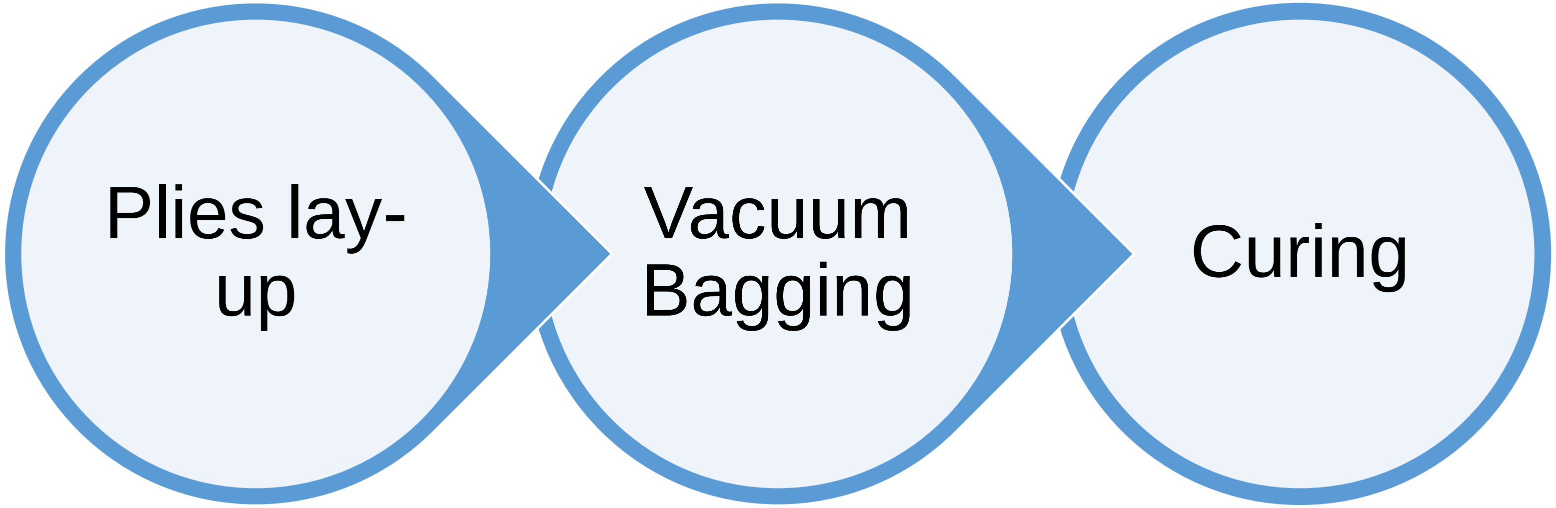
FIGURE 11-19 Honeycomb sandwich construction.

Recap*



Recap*

Stages of Composite Fabrication



- Vaccum bagging is the most common method used to apply pressure during the curing of the composite.
- Curing is a term which refers to the toughening or hardening of a polymer material by cross-linking of polymer chains, brought about by electron beams, heat or chemical additives.

Stages of Composite Repairs

Damage removal → Core Replace-ment if required → Repair Plies lay-up → Vacuum Bagging → Curing

- Composite bonded repairs are typically done in-situ on the aircraft using portable curing equipment on-site or the damaged panel can be removed to be repaired in workshop.
- The most common way to apply pressure to a composite repair is to use a vacuum bag technique.
- The most common way to apply heat to a composite repair is to use heat bonder, thermocouple and heat blanket.

Vacuum Bag Materials

Figure 7-31. *Hand tools for laminating.*

Figure 7-32. *Air tools used for composite repair.*

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Vacuum Bag Materials

- **Caul Plate**:
  - To provide a more uniform heated area and a smoother finish.
- **Release Agents**:
  - A.k.a. mold release.
  - To apply on the surface where it is in contact with some form of adhesion so the part can be release after cure.
- **Peel ply**:
  - Used to create a surface suitable for secondary bonding.
  - Uncoated peel ply can be difficult to remove after cure.

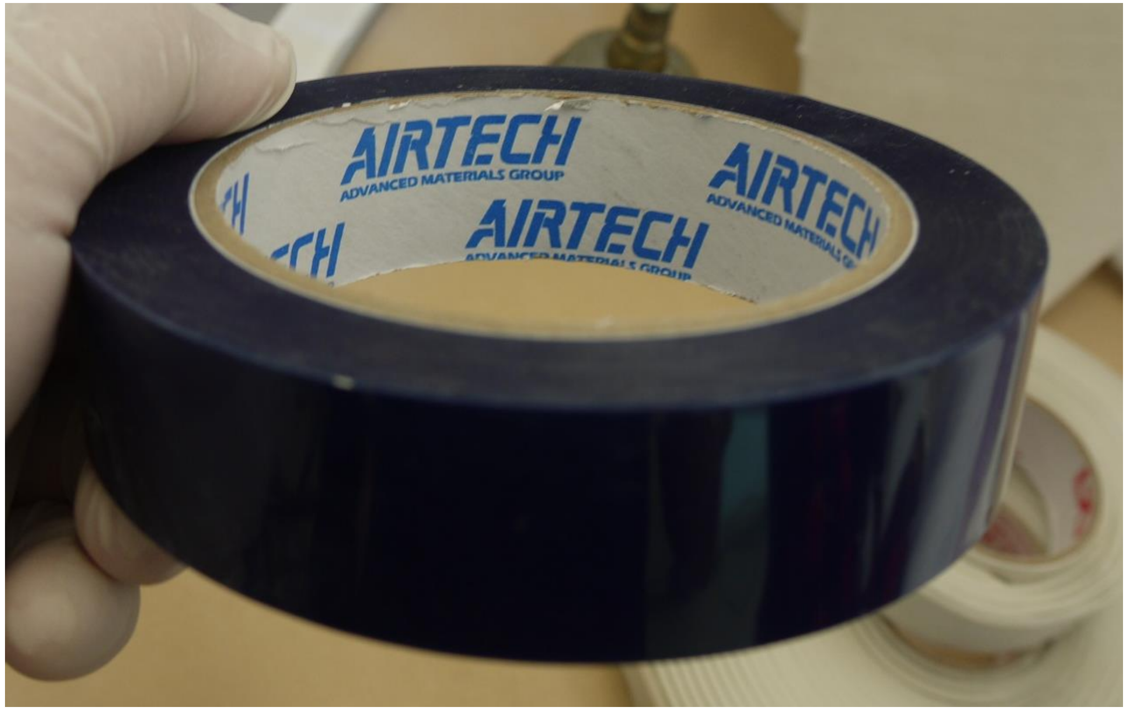
Vacuum Bag Materials

- **Heat-resistant Sealant Tape**:
  - A.k.a. sticky tape.
  - Used to seal the vacuum bag to the surface and has a temperature rating for different cure temperature.
- **Perforated Release Film**:
  - Used to allow air and volatiles to escape out of the repair area and prevents the bleeder ply to stick to the repair.
  - An alternative for perforated release film is porous Teflon-coated fiberglass.
- **Solid Release Film**:
  - Used to ensure that the repair patch will not stick to a working surface and also to prevent resin to damage (touch) the heat blanket or caul plate.

Vacuum Bag Materials

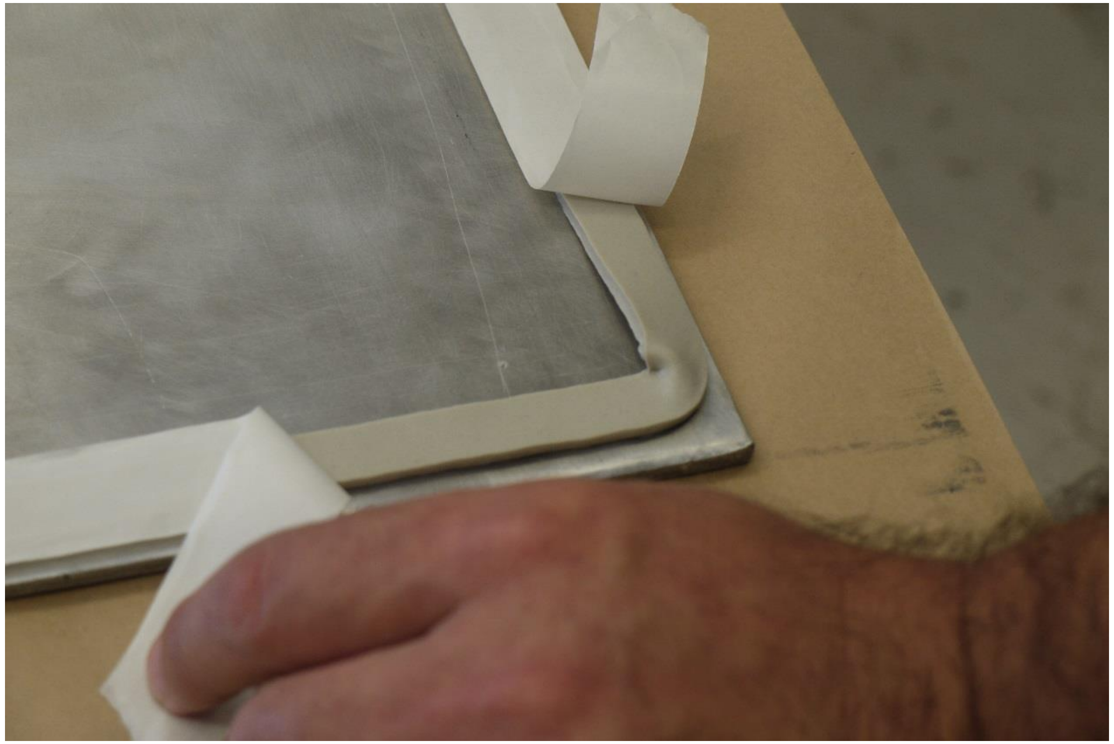
- **Bleeder ply**:
  - To bleed off excess resin and remove air and volatiles trapped in the repair area.
- **Breather material**:
  - To create a path for air to escape out of the vacuum bag. Placed on top of the heater blanket or the repair if used.
- **Vacuum bag film**:
  - To seal off the repair from atmosphere to ensure vacuum can take place.
  - It has different temperature rating for different cure temperature.
  - In order to bridge in corners, pleats are to be made in the bag.

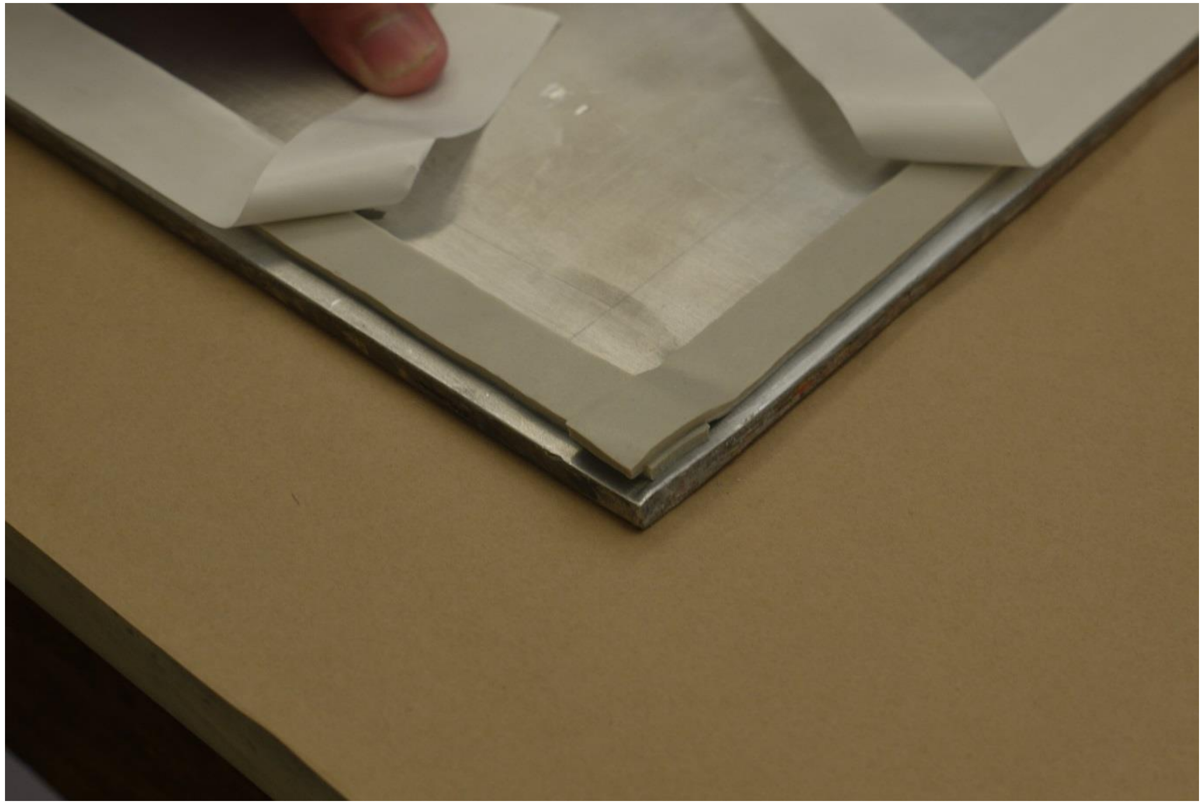
Vacuum Bag Materials



Masking tape (Flash tape) used to hold the vacuum bag materials in place.
Different colours indicated different temperature range

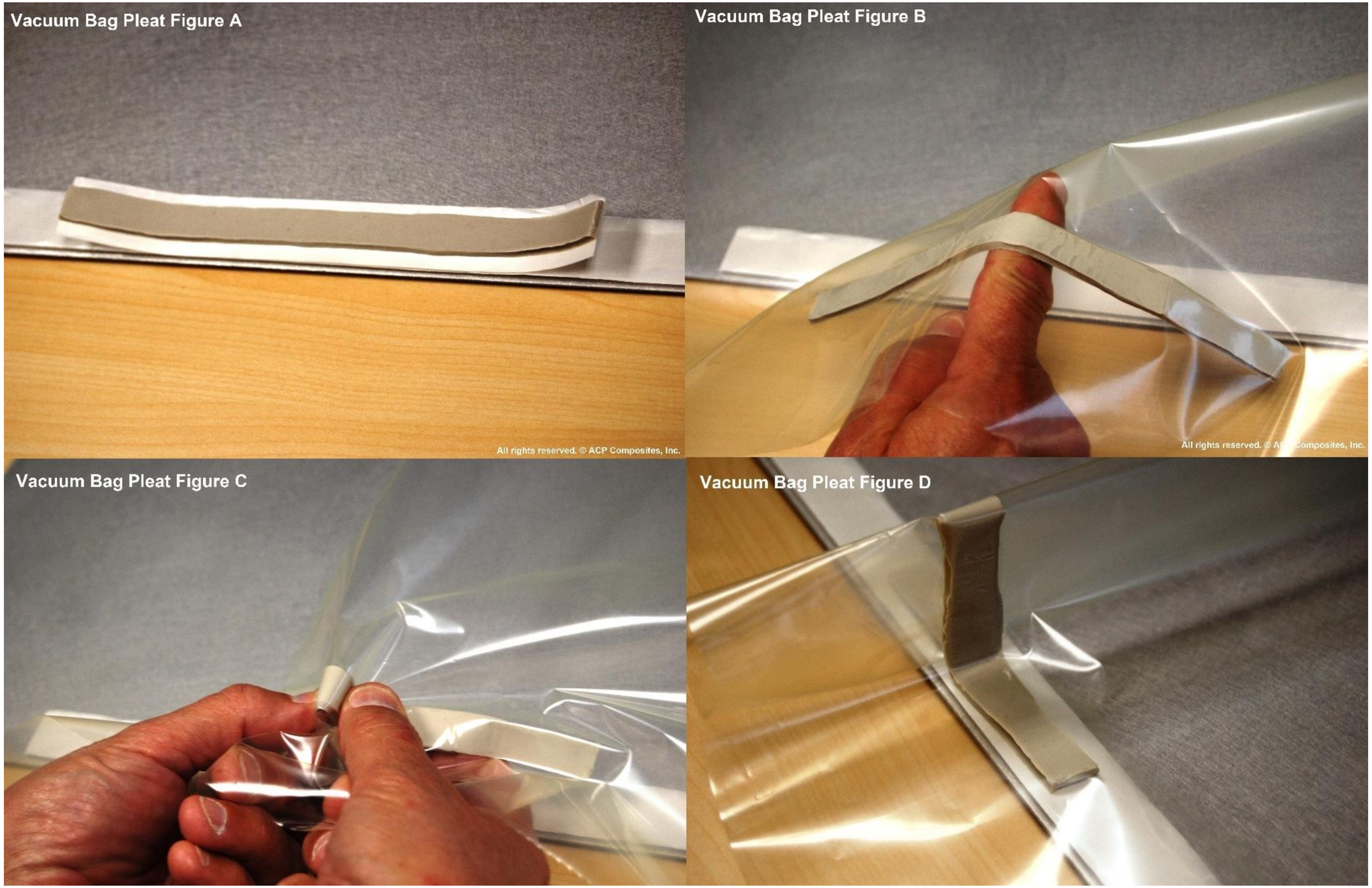
Vacuum Bag Materials





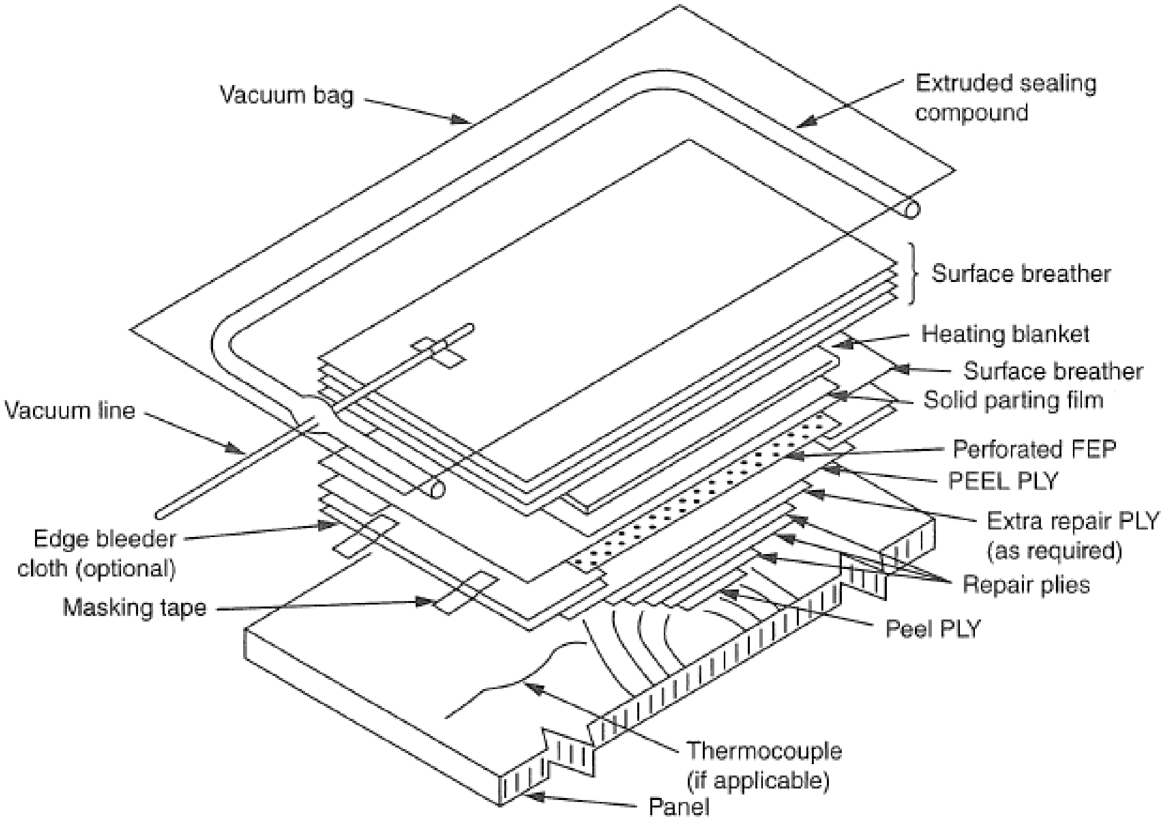
Sealant Taping at corners

Vacuum Bag Materials

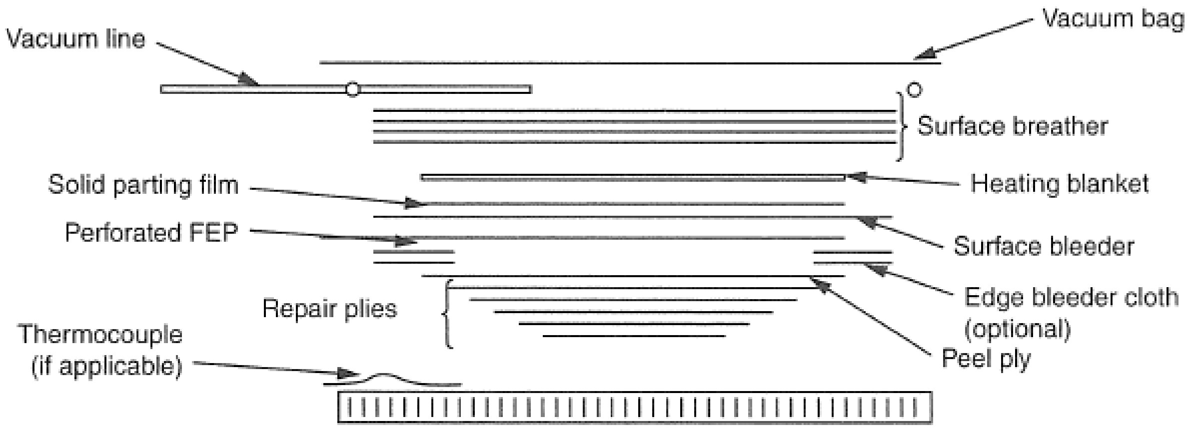


Source http://www.acpsales.com/Vacuum-Bagging-Pleats.html

Vacuum Bag Materials



Cutaway view of
bagging sequence for skin PLY repair (Wet layup)



Section thru LAYUP for skin PLY repair
(Wet layup)

- **Vacuum Equipment:**
  - A vacuum pump is used to evacuate the air and volatiles from the vacuum bag so that atmospheric pressure will consolidate the plies.
  - A vacuum port is installed in the vacuum bag; the bottom part is placed in the vacuum bag on top of the breather and the bagging film is cut above the bottom part and the top part of the vacuum port is installed with a quarter turn.
- **Vacuum bag technique:**
  - The repair patch is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible film placed over it.
  - Both single-side vacuum bag and envelope bag techniques are used depending on the application.
  - After the vacuum bag is installed, a leak check must be performed to check if the vacuum bag holds vacuum. Repair manual will specify the minimum required vacuum and the maximum allowable leakage per minute.

Single-Side Vacuum Bagging, Envelope Bagging & Alternate Pressure Application

- **Single-side vacuum Bagging**:
  - The vacuum bag is applied with sealant tape (tacky tape) to one side of the damaged aircraft part and a vacuum is applied.
- **Envelope Bagging**:
  - A process in which the part to be repaired is completely enclosed in a vacuum bag or the bag is wrapped around the end of the component to obtain an adequate seal.
- Alternate Pressure Application are
  - **C clamps**: Used when there is no room for a vacuum bag or the vacuum bag will crush the part. Pressure distribution pads or spring clamps are used to prevent damage.
  - **Shot Bags and Weights**: Limited due to the low pressure that can be applied and the fact that the part or repair needs to be horizontal.
  - **Shrink tape**: Heat is applied to the tape with a heat gun to make the tape shrink.

Single-Side Vacuum Bagging, Envelope Bagging & Alternate Pressure Application

Figure 7-35. *Bagging materials.*

Figure 7-36. *Bagging of complex part.*

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Bleeding Techniques

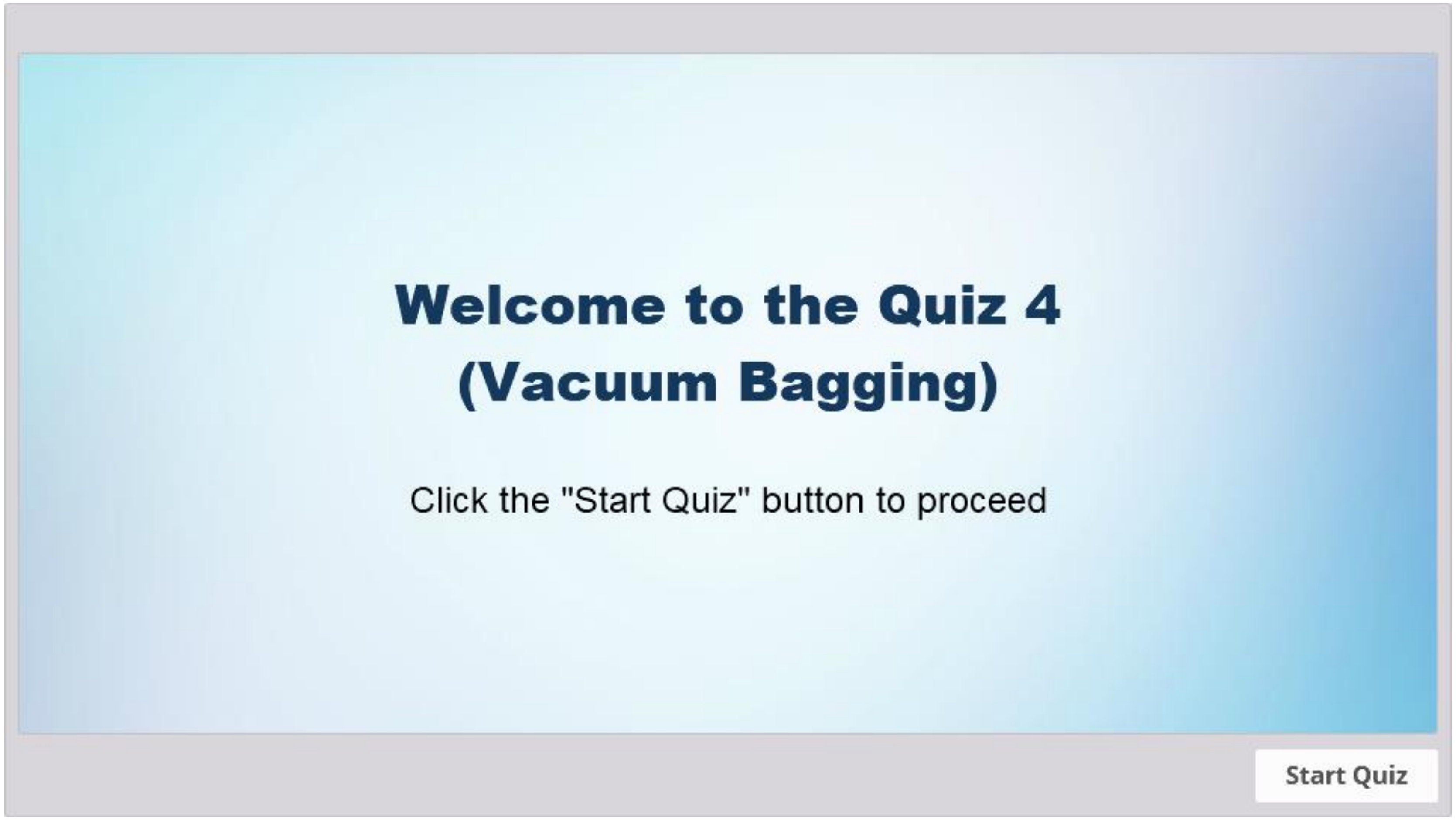
- Wet lay-ups tend to be resin rich and it is easy to introduce air during the mixing of the resin and impregnating of the repair plies.
- The amount of bleeding can be controlled through the use of bleeding cloth in the vacuum bag.
- Some pre-pregs are made with a higher percentage of resin than is required and need to be bled during the curing process.
- Most pre-pregs, however, are net resin pre-pregs and has just enough resin and no further bleeding is required.

Bleeding Techniques

- **Vertical Bleed Out**:
  - The traditional way of bleed out using a vacuum bag technique is to use a perforated release film and a breather material / bleeder ply on top of the repair.
  - Holes in the release film allow air to breathe and resin to bleed off over the repair area.
  - If the lay-up is thick, the top plies tend to be resin starved and the bottom plies, resin rich.
- **No Bleed Out**:
  - Pre-pregs with 32 to 35 percent resin content are typically no-bleed systems as they contain exactly the amount of resin needed in the cured laminate.
  - Bleed out of these pre-pregs will result in a resin-starved repair or part.
  - A sheet of solid release film (no holes) will be placed on top of the pre-preg during the vacuum bagging to prevent bleed out.

Quiz

Click the **Quiz** button to edit this object



Hand Lay-up

- **Hand Lay-up:**
  - Technician uses knives and scissors to cut the material (wet lay-up or pre-preg) to size and stacks the plies in the repair area.
  - Hand lay-up greatly depends on the skill of the technician who performs the repair.
  - Common mistakes are inclusion of backing materials and incorrect ply orientation.

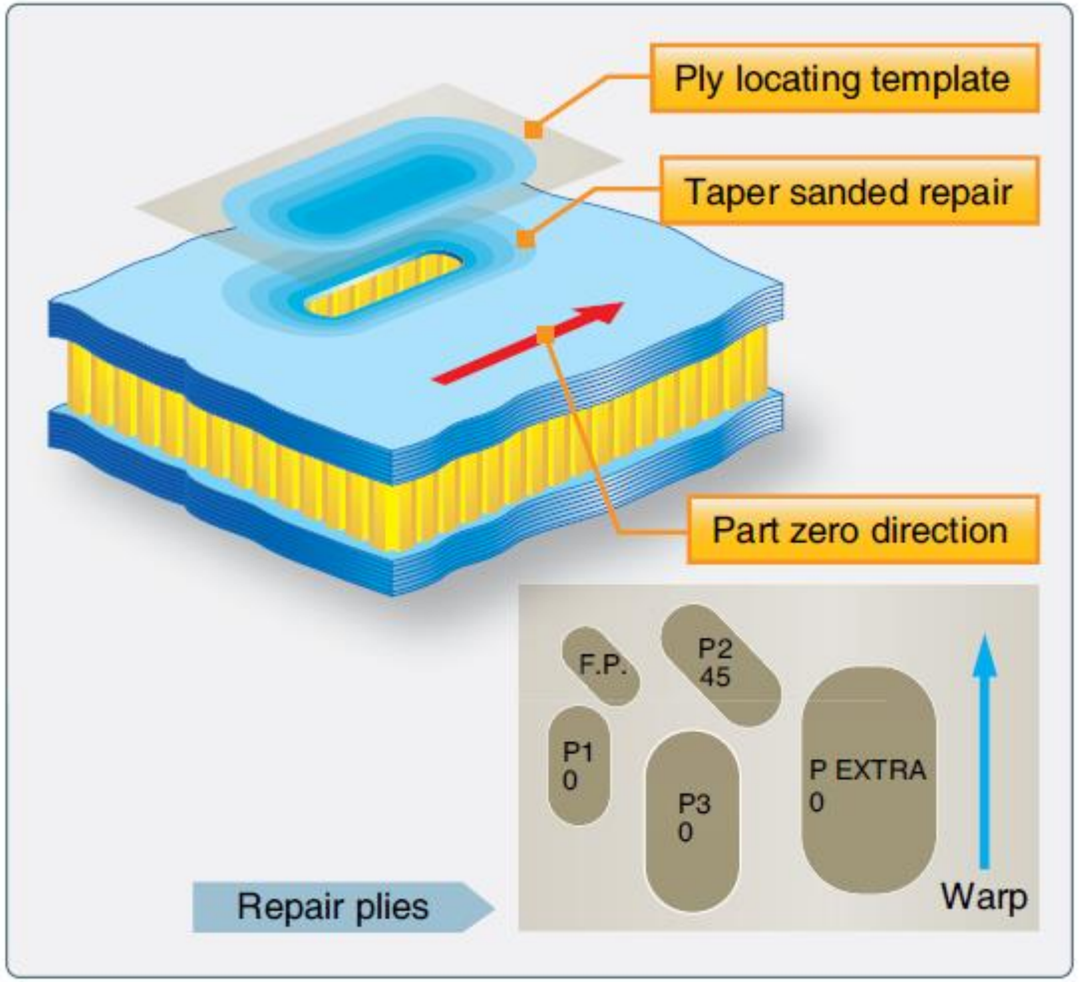


Figure 7-46. *Repair layup process.*

Source: (201https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf
6). *Faa.gov*. Retrieved 26 April 2016, from

Wet Lay-ups

- **Wet lay-up process**:
  - Wet lay-up repairs are often used with fiberglass for non-structural applications but other fiber materials can be used for wet lay-up too.
  - A dry fabric is placed on a piece of release film and impregnated with a resin system (typically a two-part resin system).
  - **Advantages**: Wet lay-up cure at room temperature is easy to accomplish and the separate materials can be stored at room temperature for a long time.
  - **Disadvantages**: Room temperature wet lay-up will not restore the strength and durability of the original structure, which was cured at a much higher temperature in the first place.

Pre-preg

- **Pre-preg**:
  - Most primary structures of transport composite aircraft are made from pre-preg materials that are cured at elevated temperatures.
  - Has the same properties as the original structure to repair these primary structures.
  - A dry fiber material that is impregnated with a resin during the pre-preg manufacturing process. Resin system is in the B stage of cure and needs to be stored in a freezer below 0°F to prevent further curing.
  - Repairs consists of four or less plies can be made in one stack-up.
  - If more plies are required for thicker laminate, structures consolidation is needed to remove air trapped between each pre-preg layer.
  - The trapped air can be removed by vacuum bagging and some manufacturers use the double vacuum debulk concept to create thick ply stack-ups.

Storage and Handling of Pre-pregs, Film Adhesives, and Foam Adhesives:

- Stage B material require freezer storage.
- Storage life and maximum time out of the freezer, prior to use, are specified by material manufacturers.
- The maximum time allowed for storing pre-preg material at low temperature is called the storage life which is typically 6 months to a year.
- The maximum time allowed for material at room temperature before the material cures is called the mechanical life.

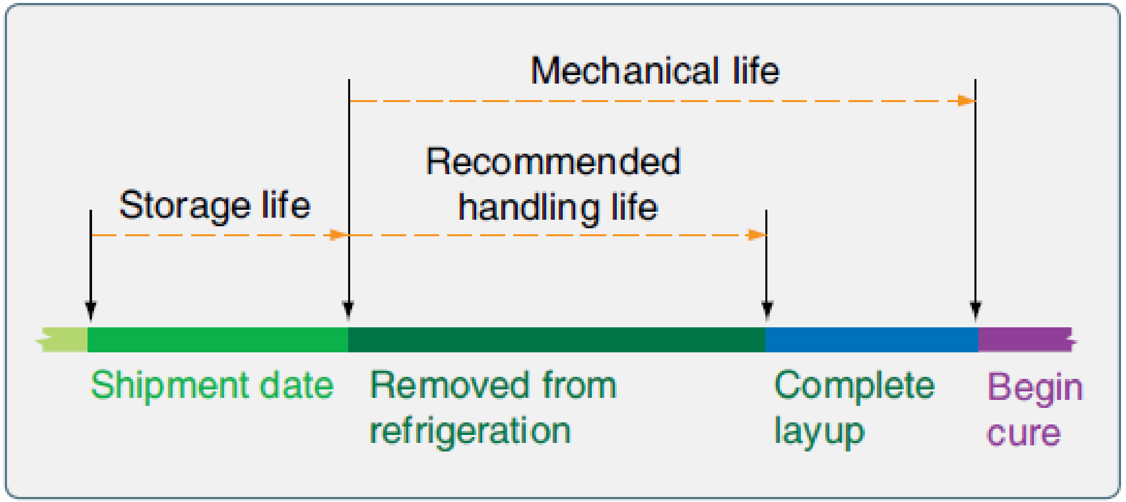


Figure 7-45. *Storage life for prepreg materials.*

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

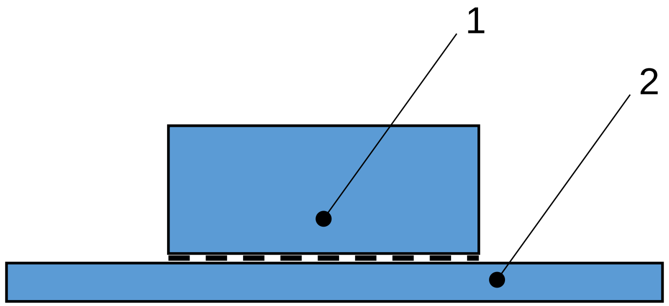
Mixing Resin*

- **Mixing Resins:**
  - Resin systems used for aircraft repair are usually two-part epoxy systems.
  - Failure to use the correct mixing ratio for A and B components will result in incorrect curing.
  - Resin must be mixed slowly to prevent air entering and fully for at least 3 min.
  - After mixing the resin, there is also a pot life for the resin.*

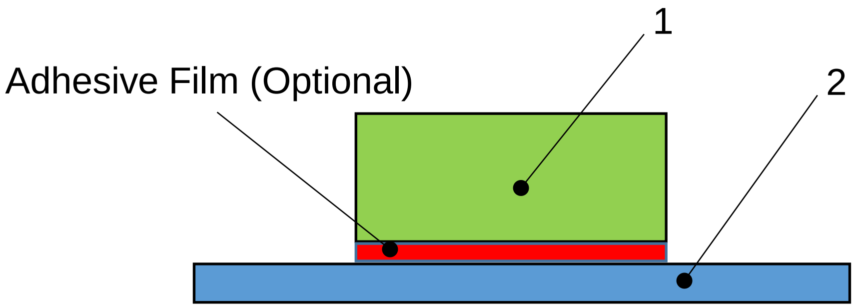
Bonding Processes

- **Co-curing**
  - Is a process wherein two parts are simultaneously cured together.
  - Advantages are excellent fit between bonded components and guaranteed surface cleanliness.
- **Secondary bonding**
  - Ultilises pre-cured composite detail parts and uses a layer of adhesive to bond two pre-cured composite part together.
  - Pre-cured parts usually have peel ply cured onto the secondary bonding surfaces.
- **Co-bonding**
  - One of the detail parts is pre-cured, with the mating part being cured simultaneously with the adhesive.
  - Film adhesive is often used to improve peel strength.

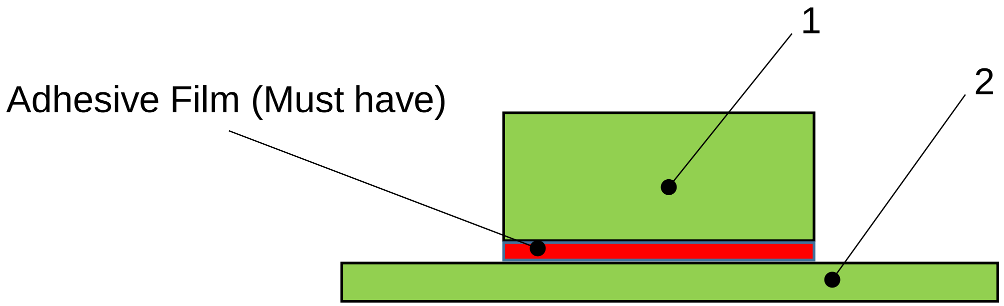
Bonding Processes



<u>Co-curing</u>:
Component 1 uncured (blue).
Component 2 uncured (blue).
Components cured together effectively creating a single part.



<u>Co-bonding</u>:
Component 1 pre-cured (green).
Component 2 uncured (blue).
Components bonded together during cure cycle of component 2.



<u>Secondary bonding</u>:
Component 1 pre-cured (green).
Component 2 pre-cured (green).
Components bonded together with a separate bonding operation.

Impregnation Process

- **Fabric Impregnation with a Brush or Squeegee**

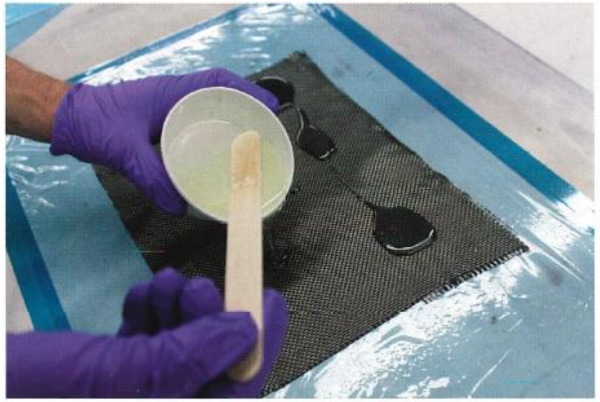


FIGURE 11-37a Apply resin on top of the fabric.

FIGURE 11-37b Use squeegee to impregnate the fabric.

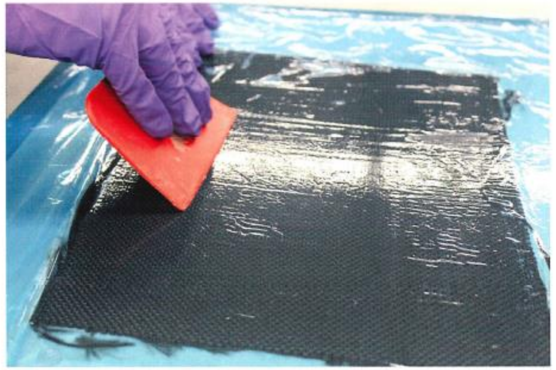


FIGURE 11-37c Place release film on top of wet fabric and use squeegee to remove air from the lay-up.

- **Fabric Impregnation using a Vacuum Bag**

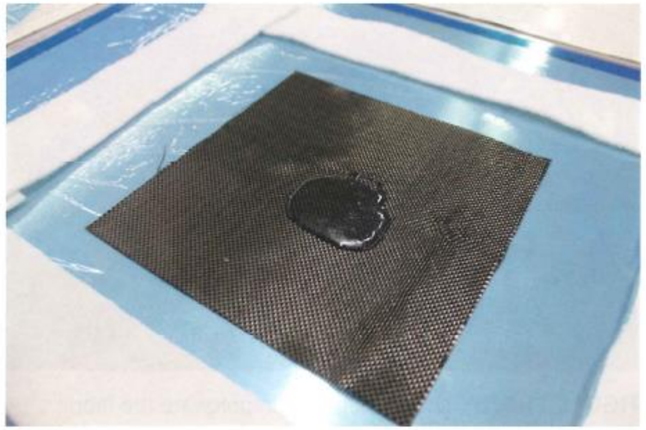


FIGURE 11-38a Pour resin in the middle of the fabric.

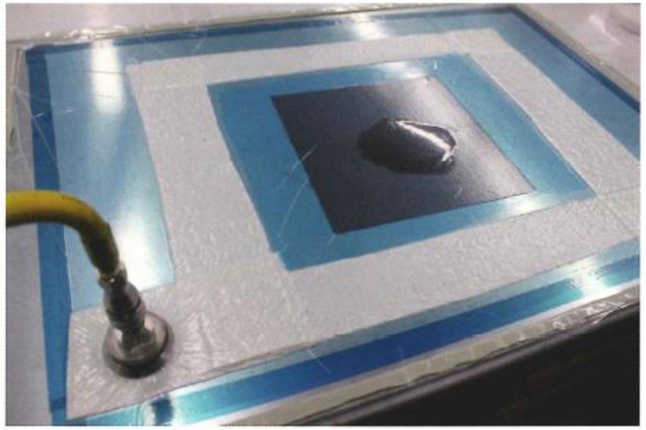


FIGURE 11-38b Apply release film and vacuum bag.

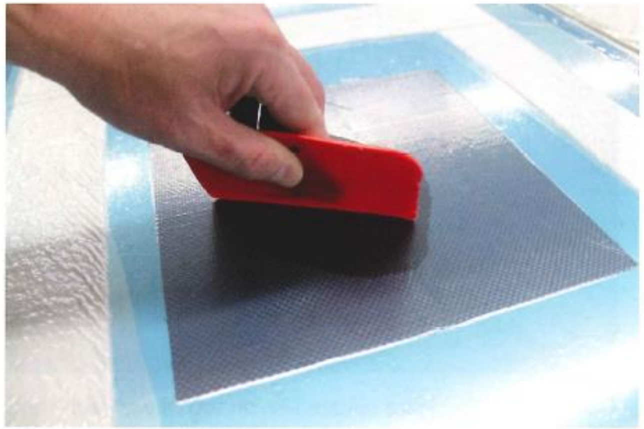


FIGURE 11-38c Use squeegee to impregnate the fabric.

Curing of Composite Material*

- **Room Temperature Curing**:
  - Will not restore the strength or durability of the original elevated temperature cure components
  - Takes a long time for epoxy resins to cure at room temperature.
- **Elevated Temperature Curing**:
  - Some resins cure at elevated temperature to achieve their optimum mechanical properties.
  - Heating equipment are required to ensure the correct cure take place.
  - Always consult the repair manual or technical data sheet for the correct cure recipe.

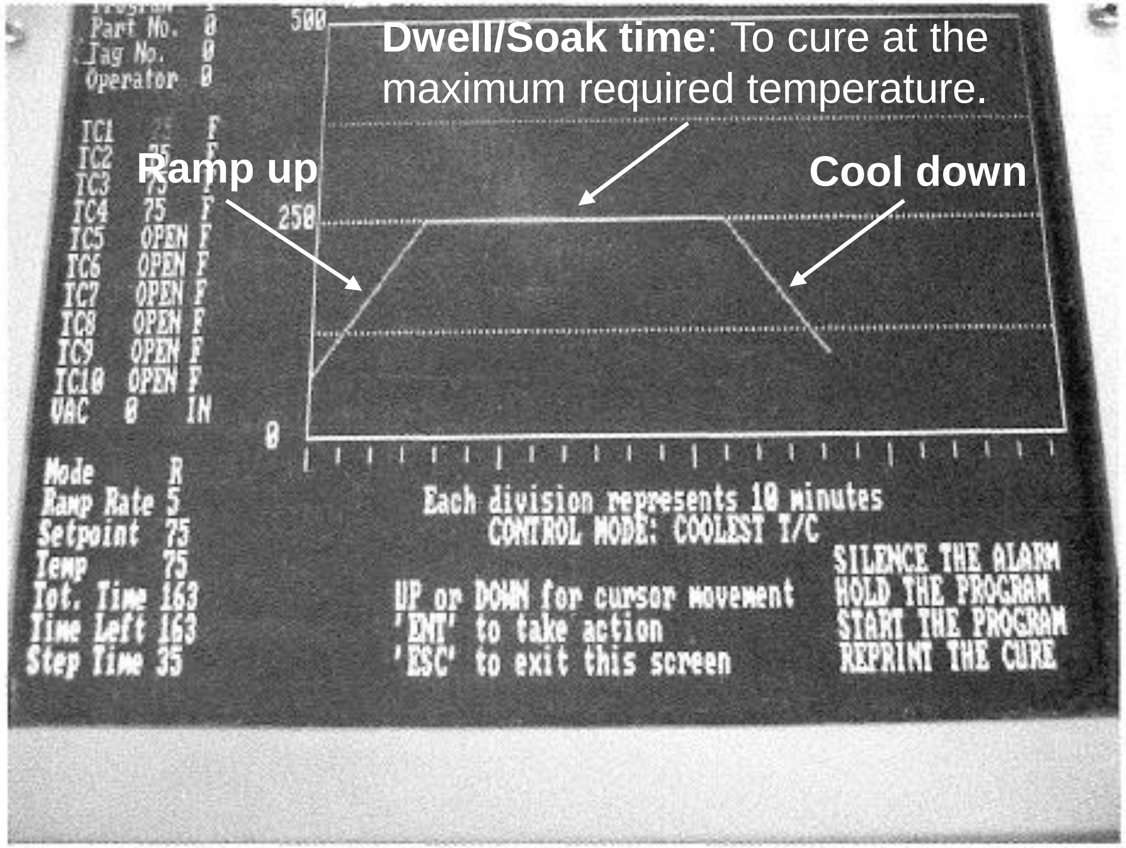


FIGURE 11-39 Heat bonder showing programmed cure recipe.

Curing of Composite Material*

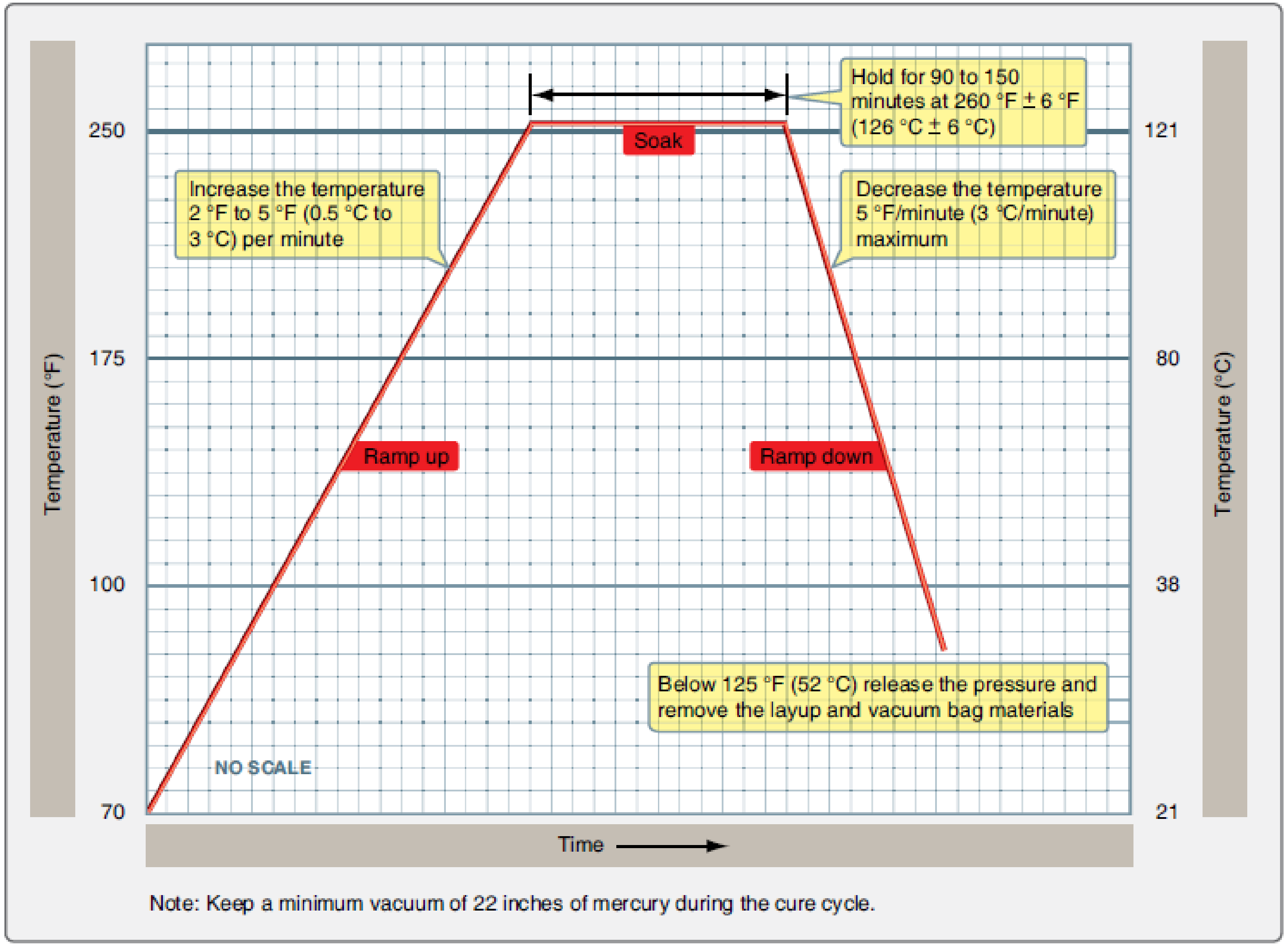


Figure 7-62. *Curing the repair.*

Source: (2016). *Faa.gov*. Retrieved 26 April 2016, from https://www.faa.gov/regulations_policies/handbooks_manuals/aircraft/amt_airframe_handbook/media/amt_airframe_vol1.pdf

Curing Equipment

- **Heat Bonder**:
  - A programmable temperature control unit that can control the heating equipment.
- **Thermocouples:**
  - Temperature measuring device, to be used as a feedback to the heat bonder.
- **Autoclave**:
  - Used by MROs or OEMs.
  - Able to be accurately controlled with temperature as high as 650°F and pressure up to 250 psi.
  - Nitrogen gas is used to pressurise the chamber during the cure.
- **Curing oven**:
  - Components typically enter the oven vacuum bagged. Able to control both vacuum and heat if its programmable.

Curing Equipment

- **Heat lamps**:
  - Used for curing composite repair below 150°F. Difficult to control the heat applied.
- **Hot Air Modules:**
  - Can be used to heat large areas of any contours and material up to 900°F.
  - Good for curing non-contact material like paints.
- **Heater Blankets**:
  - Come in all shapes and sizes.
  - Transfer heat from blanket to component via conduction.
  - Often used together with vacuum bagging to ensure adequate contact.
  - Most common used heating equipment on aircraft.

Curing Equipment

FIGURE 11-42 Autoclave.

FIGURE 11-41 Curing oven.

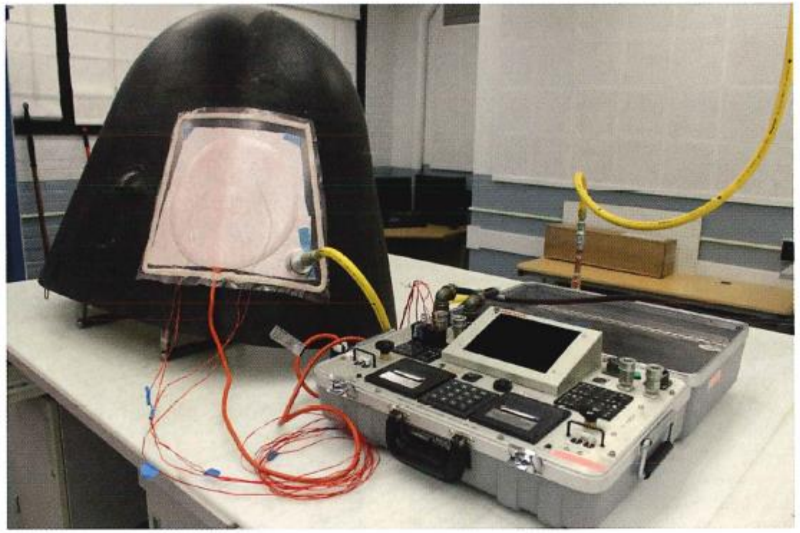


FIGURE 11-43 Heat bonder used for curing a repair to nose radome.

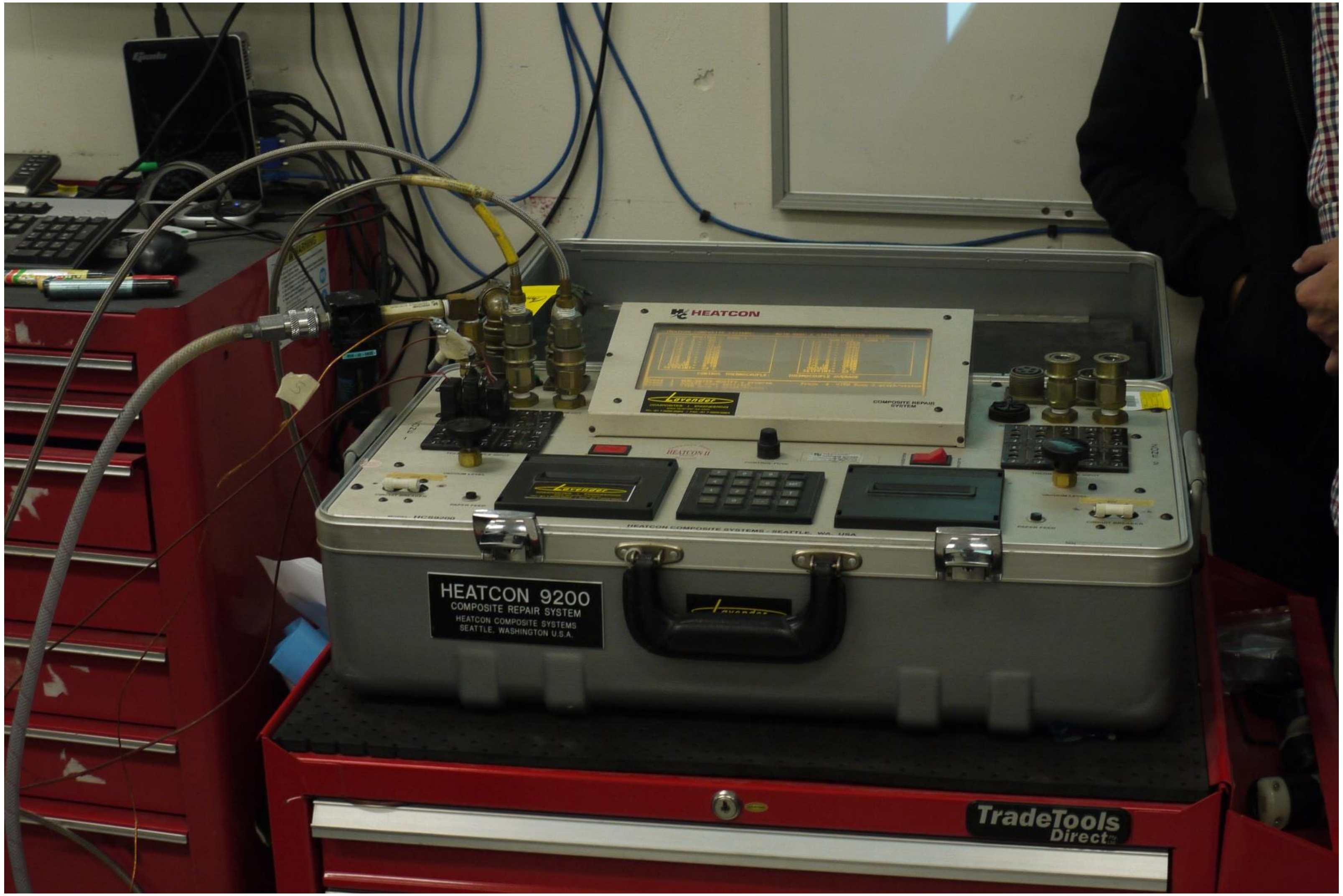
Curing Equipment

Heat Bonder

Curing Equipment

Heat Blanket

Curing Equipment



Heat Bonder fully connected.

Quiz

Click the **Quiz** button to edit this object

Bolted Repair VS Bonded Repair*

TABLE 11-2 Bonded versus Bolted Repair Matrix

Bonded versus Bolted Repair	Bolted	Bonded
Lightly loaded structures: laminate thickness less than 0.1 in		x
Highly loaded structures: laminate thickness between $\frac{1}{8}$ and 0.5 in	x	x
Highly loaded structures: laminate thickness larger than 0.5 in	x	
High peeling stresses	x	
Honeycomb structure		x
Dry surfaces	x	x
Wet and/or contaminated surfaces	x	
Disassembly required	x	
Restore unnotced strength		x

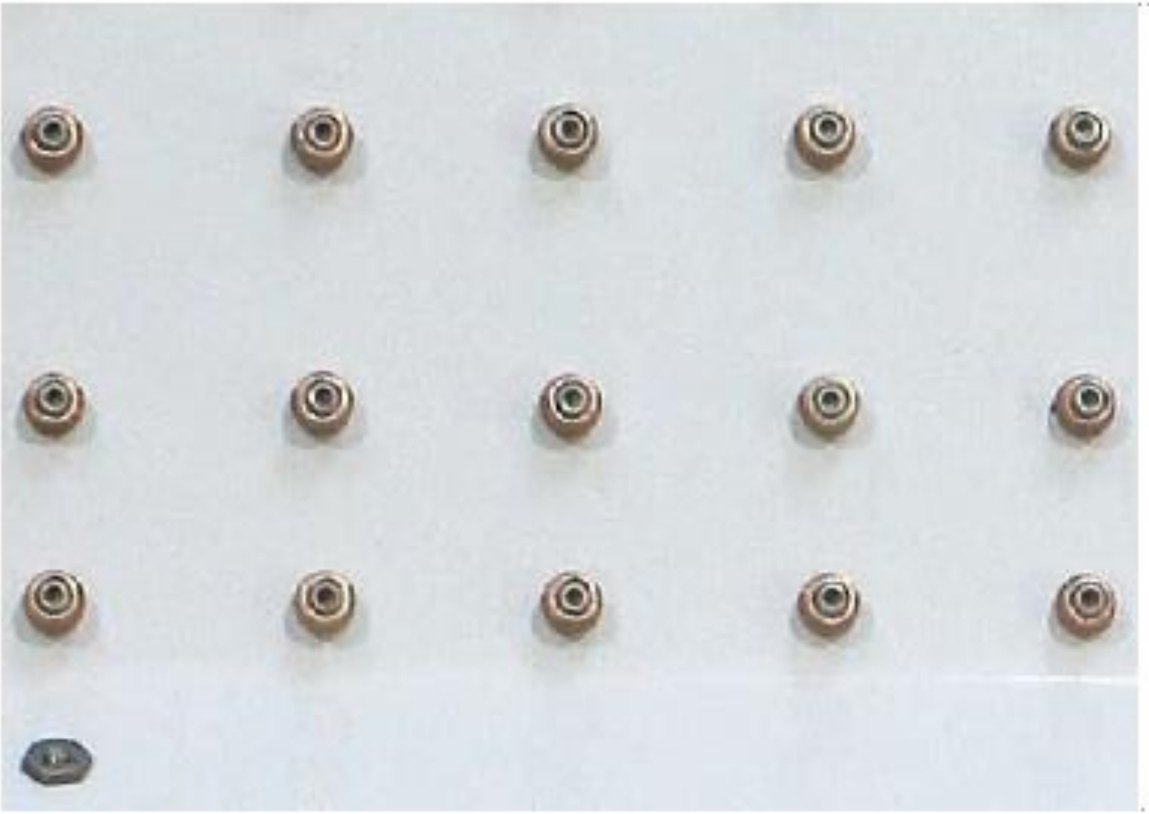
When to use Bonded Repair and Bolted Repair

Bolted Repair VS Bonded Repair*

Bolted Repair*	Bonded Repair*
Repair typically used on composite thicker than 0.125” to ensure sufficient bearing area for load transfer.	Thick composite repair requires several curing sessions or the use of autoclave.
Faster processing time.	Slower processing time.
Repair may be Category B (Require inspections).	Category A (Permanent Repair)
Repair material not sensitive.	Repair material is time, temperature, and moisture sensitive
Heavier repair (additional weight)	Lightweight.
Hole drilling must be done with care and caution. Introduction of new holes reduce cross-sectional area.	No protruding parts or additional fastener holes.
Have to pot the core for sandwich structure before fasteners can be installed.	More suitable for sandwich structure as there is no potential for moisture ingression and no bearing load on the thin face sheets.
No risk of heat damage due to curing.	Risk of heat damage on original structure and porosity forming during the curing of repair plies

Extracted from Boeing Aero QTR 4 2014

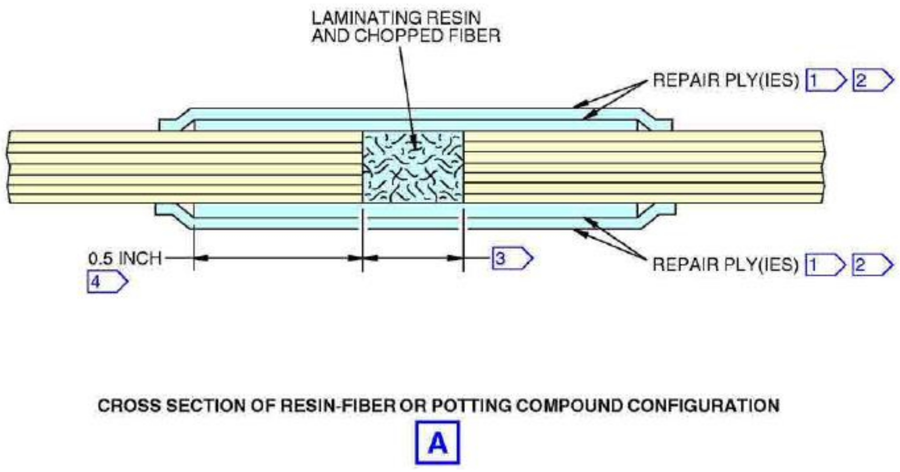
Bolted Repair VS Bonded Repair*



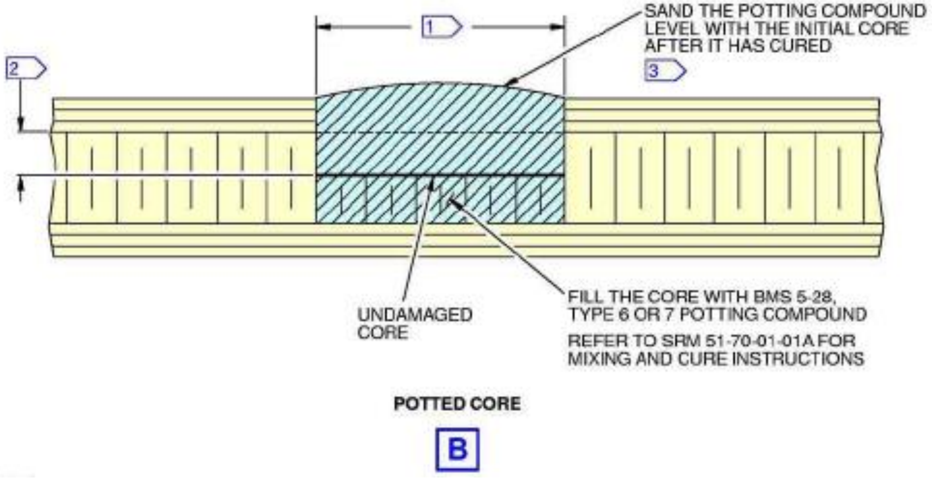
Bolted Repair (left) VS Bonded Repair (Right)
Extracted from Boeing Aero QTR 4 2014

Composite Honeycomb Sandwich Repairs: Minor Core Repair (Filler and Potting Repairs)

- Potting repair is done by filling up the damaged area with potting compound, it may be mixed with chopped fibers or materials.
- These repairs does not restore the full strength of the damaged area.
- Potting compound is used as filler for cosmetic repair to edges and skin panels.
- Potting compounds are also used in sandwich honeycomb panels as hard points for bolts and screws.
- Potting compound can be heavier than the original core and affect flight control balance.



Extracted from Boeing B787 SRM



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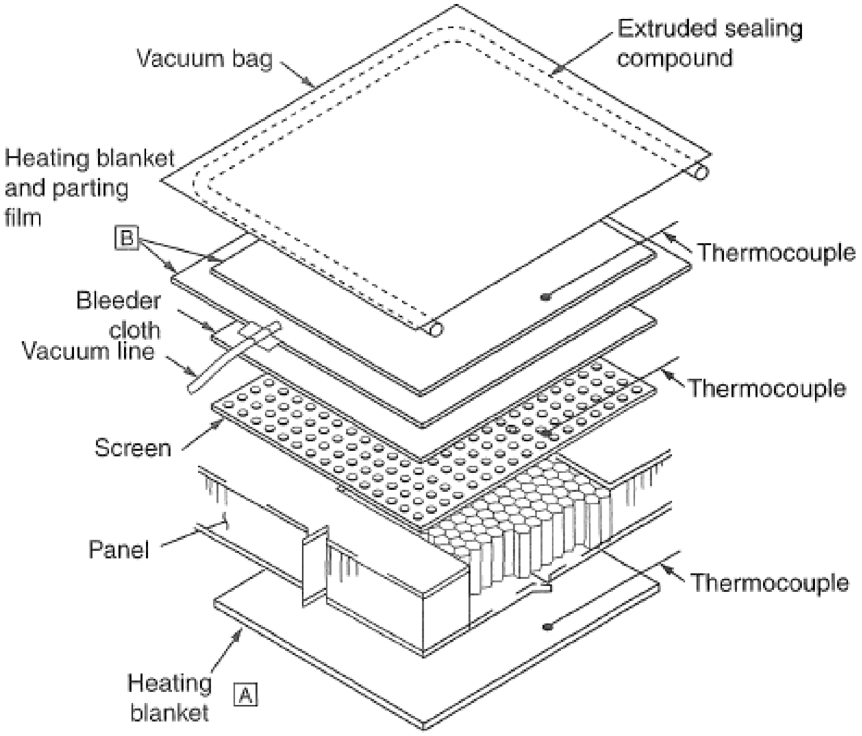
VII. Composite Repairs

Composite Honeycomb Sandwich Repairs: Bonded Scarf Repair Example of Honeycomb Sandwich Panel

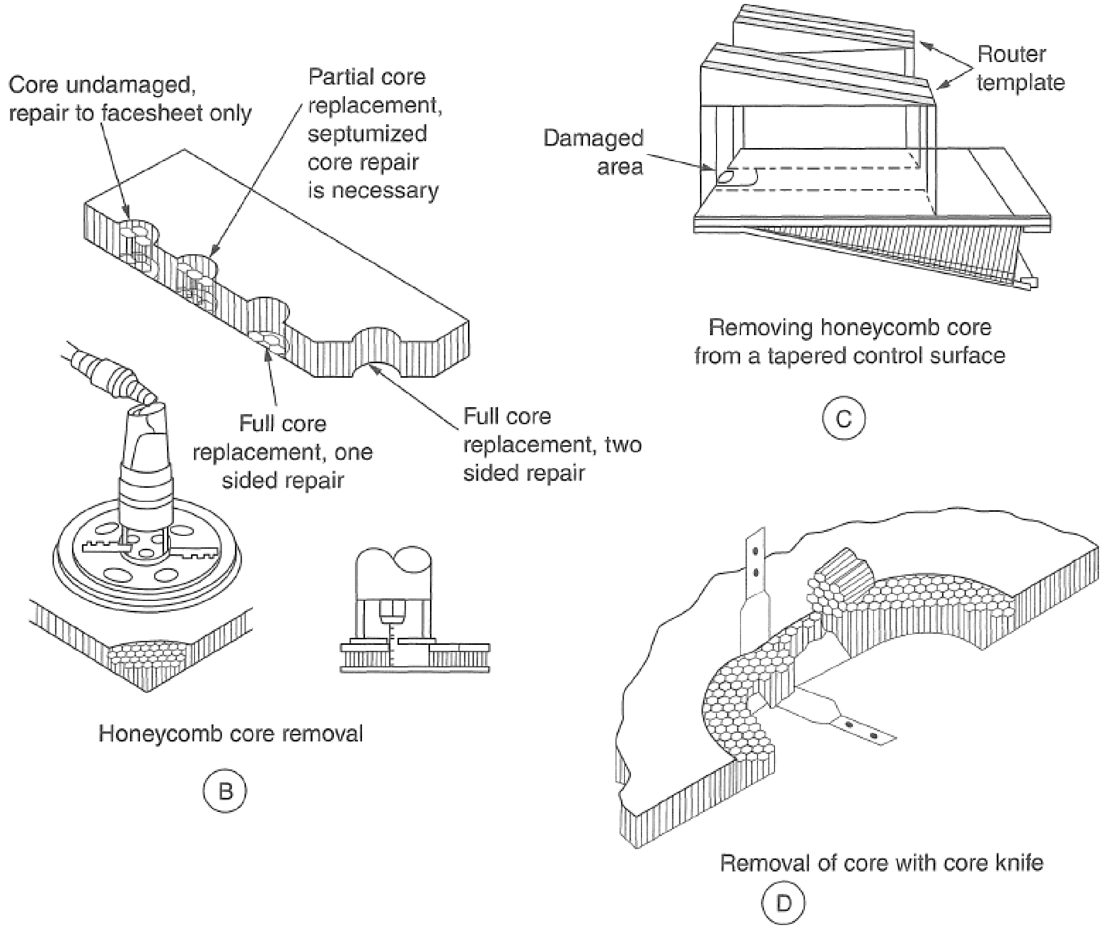


Notes

[A] Preferred location of heating blanket when opposite face is accessible.

[B] Alternate location of parting film and heating blanket when opposite side is in-accessible. This location may be used for an additional heating blanket to accelerate watch removal.

FIGURE 11-46 Water removal using vacuum bag procedure.



Instructions

1. Remove the face sheet and/or core with a power router, using a router template to protect the undamaged part of the face sheet. Refer to SRM 51–10–02 for instructions on the use of a router and template.

   Note: The router can be adjusted to remove: One of the face sheets only, a face sheet and part of the core, a face sheet and all of the core, or both the face sheets and core. See detail B.

2. If you are routing tapered part, you can use wedge shaped router templates. This will permit the router to cut the core material parallel with the lower surface. See detail C.
3. It is permitted to remove honeycomb core with a care knife. See detail D.

FIGURE 11-47 Removal of damaged laminate and core.

Steps taken for typical bonded repair*

1. **Evaluate the damage**: To determine extent of the damage using visual inspection and proper NDI inspection.
2. **Remove the damaged area**: Take care not to enlarge the repair area during damage removal.* #
3. **Taper sand a uniform taper around repair area:** To taper sand accordingly per the type of repair to be implemented.* #
4. **Dry and clean the repair area**: To ensure no moisture prior to the curing of the bonded repair. Cleanliness is important to ensure proper adhesion.* #
5. **Replace the damaged core**: To cut a replacement core to size and to be bonded to the repaired area prior to patch curing.* #

Note: The order of the steps is different from the textbook.

Steps taken for typical bonded repair*

6. **Prepare the repair plies and lay-up**: To prepare the repair plies per manual requirement and lay the repair plies per original orientation. Typically, 1-2 extra plies are added. Additional fiberglass ply may be applied for smoothing purposes.

7. **Vacuum bag the repair**: To apply pressure to the repair during curing.

8. **Cure the repair**: Ensure proper cure cycle is applied.

9. **Remove vacuum bag and inspect the repair**: To ensure structural integrity of the repair, NDI may be required depending on repair procedure chosen and size of repair.

Example of a bonded scarf repair of a honeycomb sandwich panel

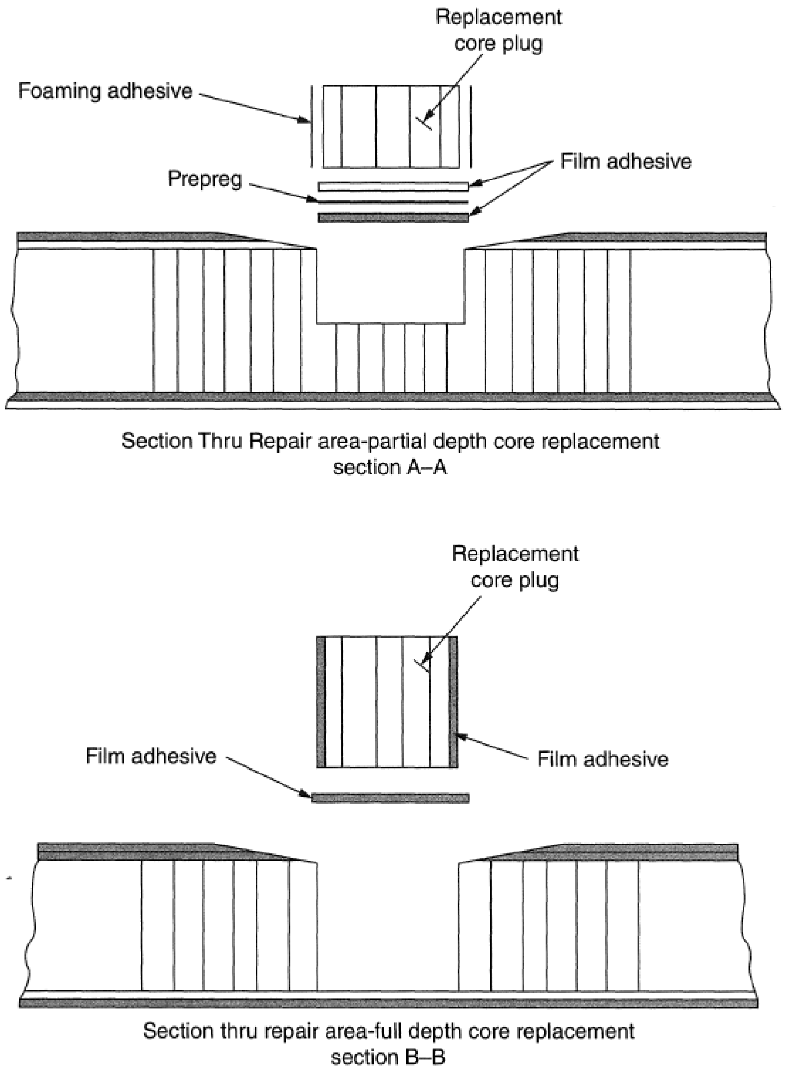


FIGURE 11-49 Replacement of damaged core.

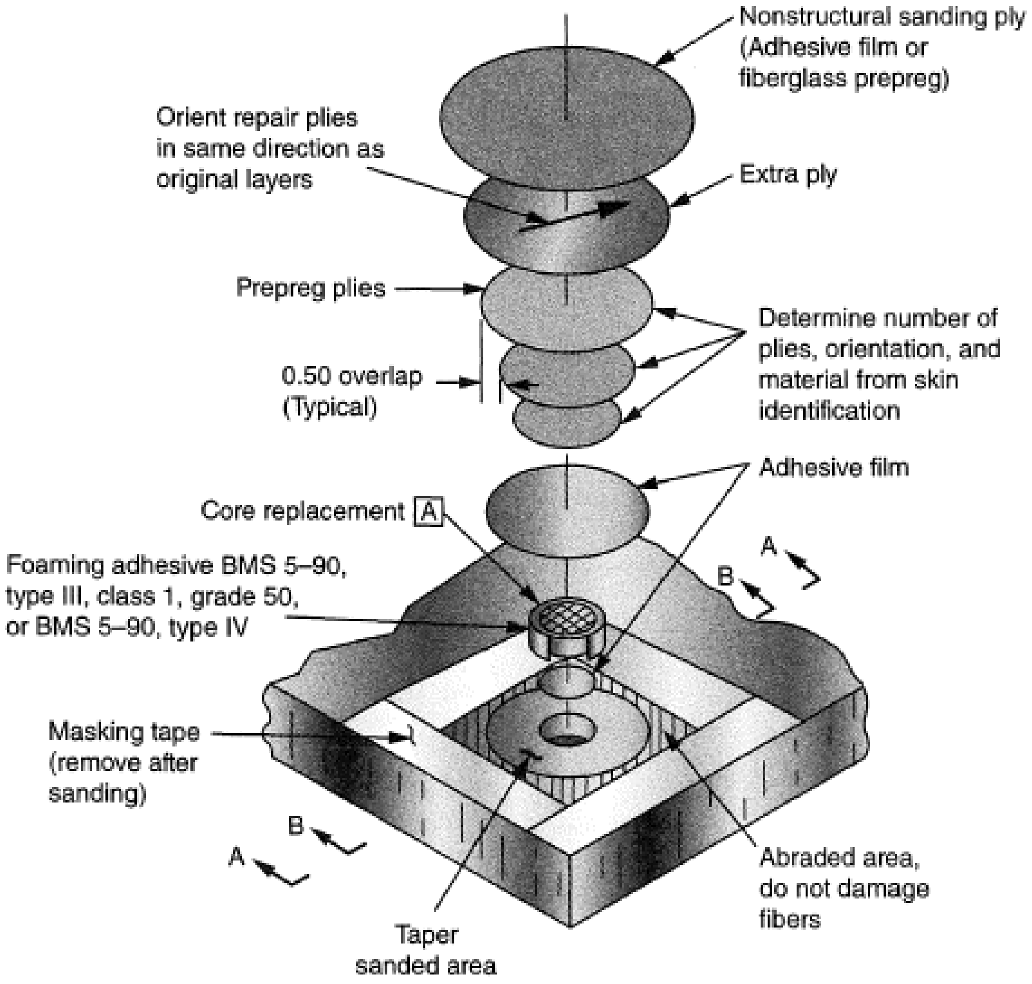


FIGURE 11-50 Repair ply placement.

Example of Honeycomb Ramp Corner Repair

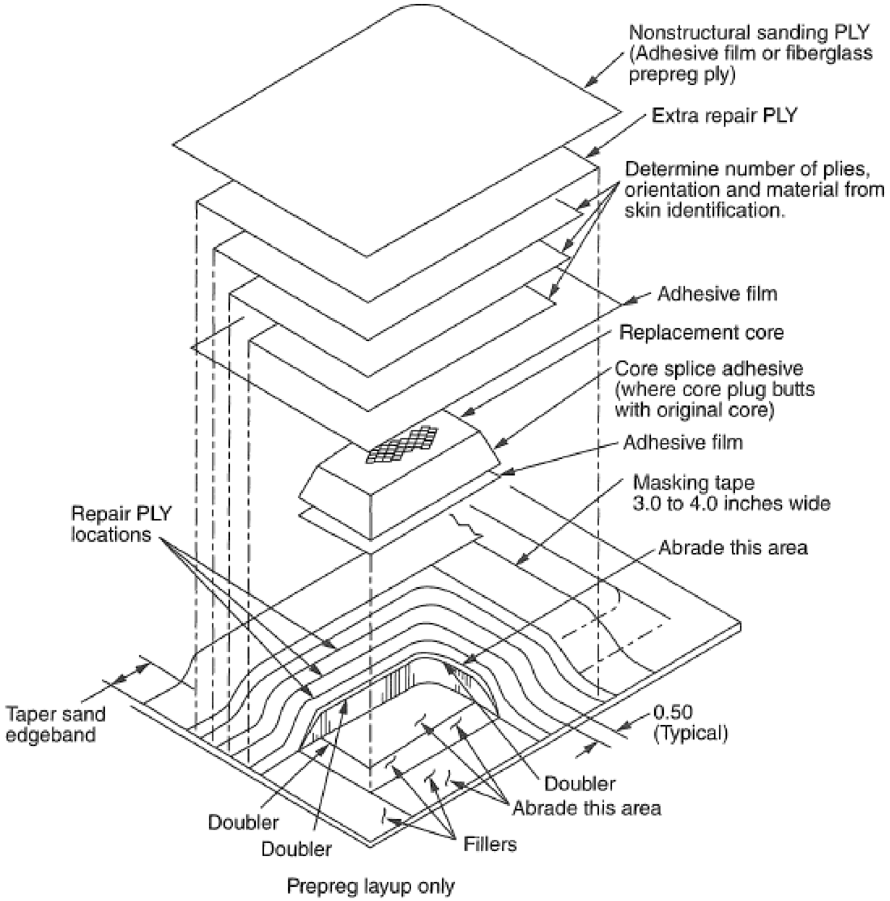


FIGURE 11-53 Repair of corner ramp.

Example of Sandwich structures with Foam Core Repair

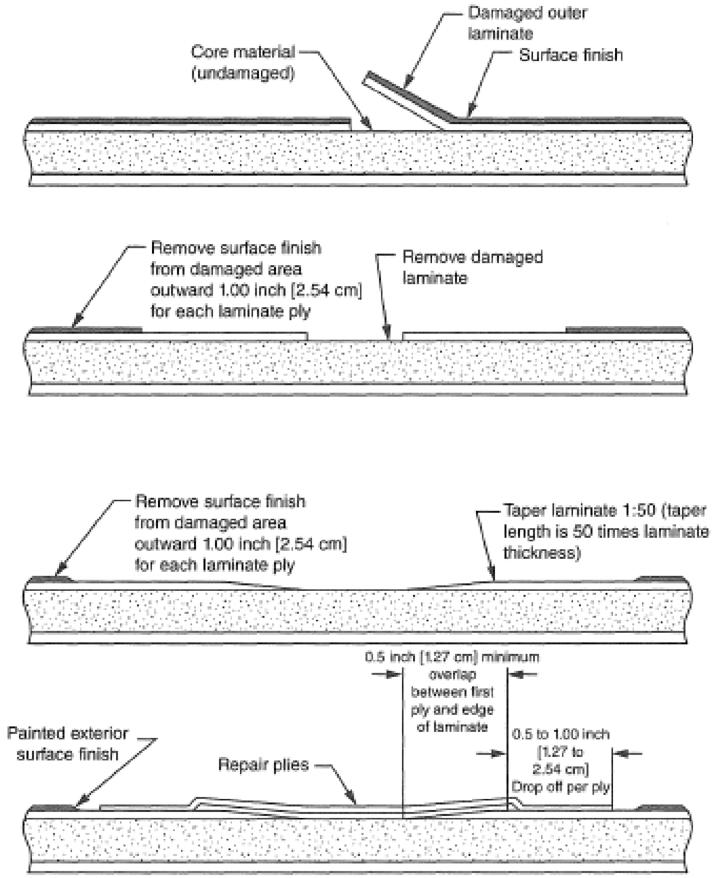


FIGURE 11-54 Repair scenarios for bonded foam cores.

Solid Laminates: Bonded Patch Repairs

- Solid laminate structures ranging from 4 plies to over 100 plies are used.
- Thinner laminate structures can effectively be repaired with a bonded repair.
- Thicker laminate structures are often repaired with a bolted repair.
- **Bonded Patch Repairs**:
  - Are often made with pre-preg materials but lower cure temperature using wet lay-up is feasible.
  - Similar to repairing face sheet of a sandwich structure.
  - Are often scarfed or stepped for a flush repair but external patch repairs could also be used.
  - Stepped repair has the advantage that the adhesive does not run down the bond-line.
  - Stepped repair is often avoided as it is considered very difficult if not impossible to achieve with manual tooling and non-flat surfaces.

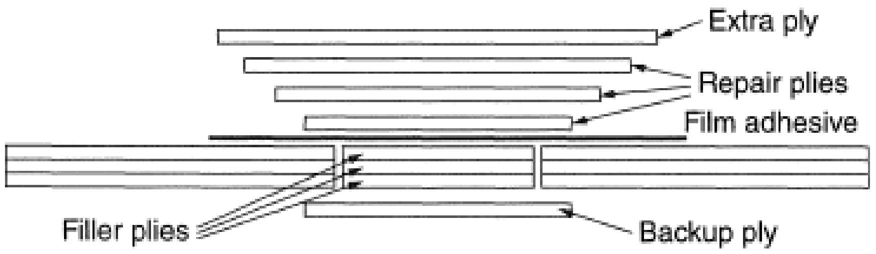


FIGURE 11-56 External patch repair for laminate structure.

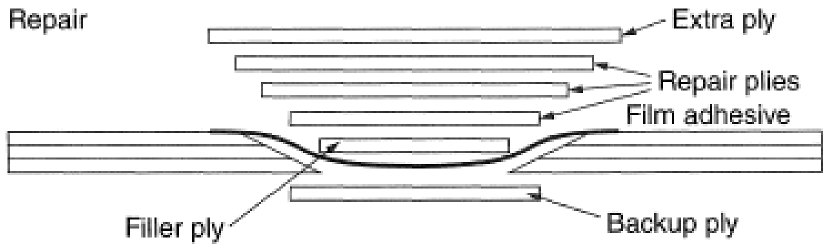


FIGURE 11-57 Scarf repair for laminate structure.

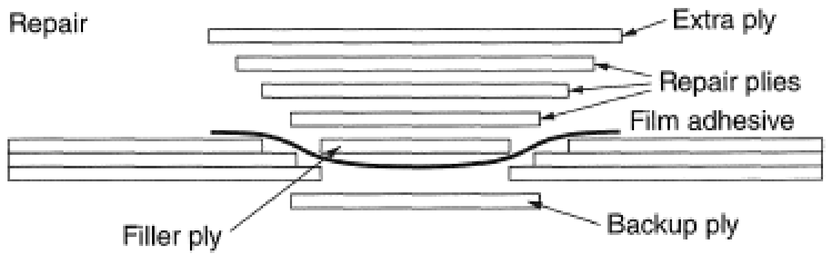
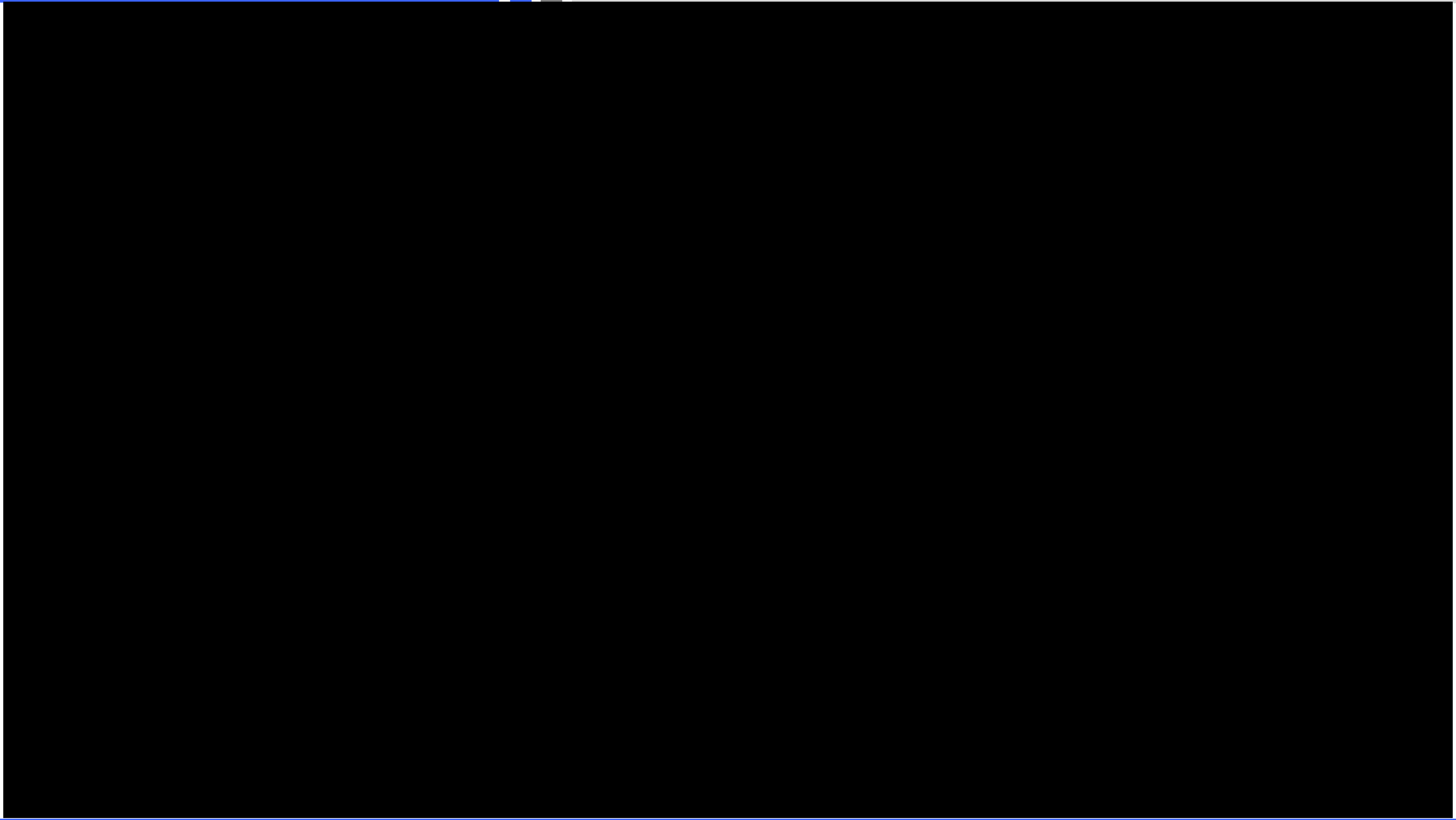


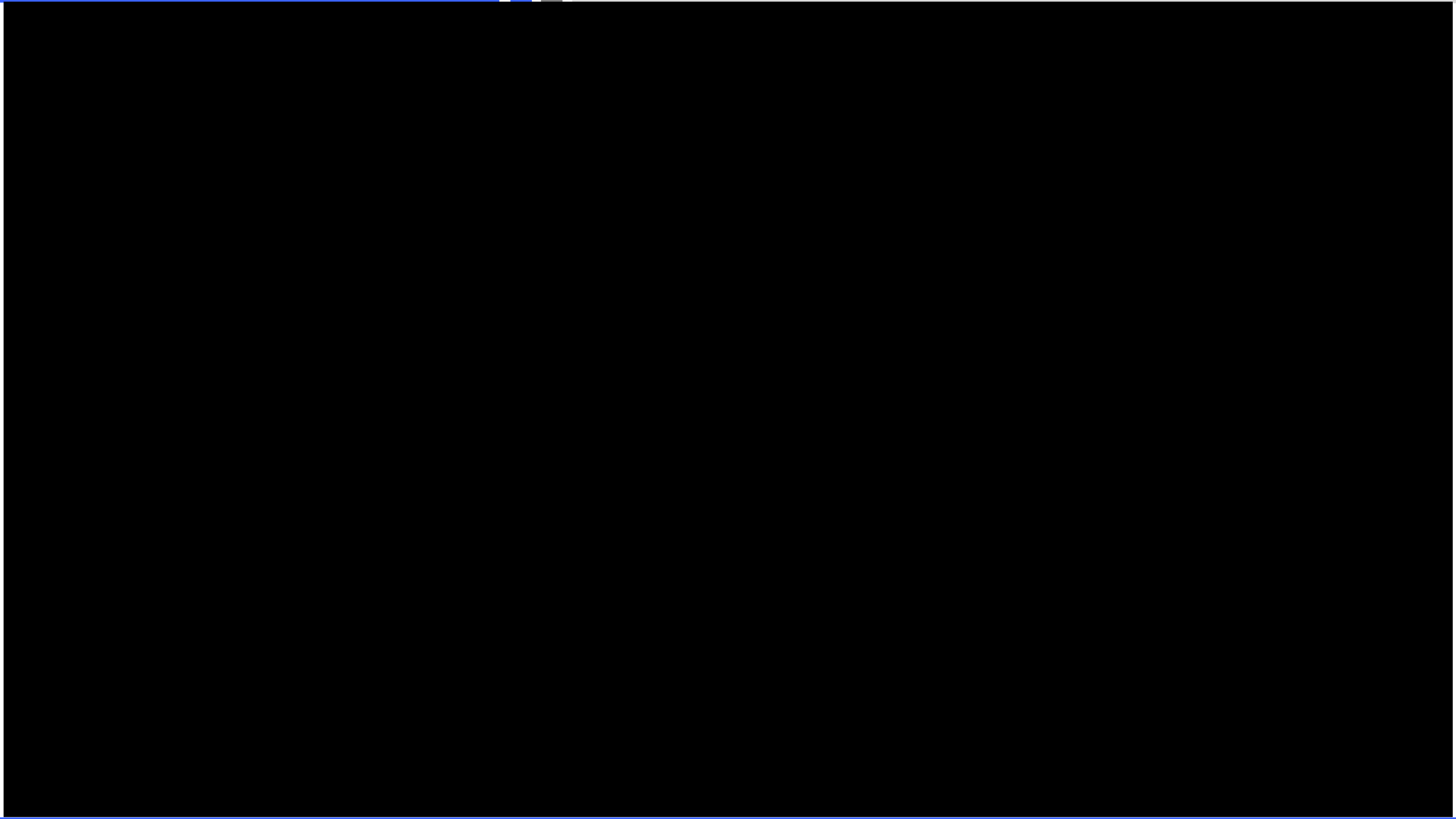
FIGURE 11-58 Stepped repair for laminate structure.

VII. Composite Repairs

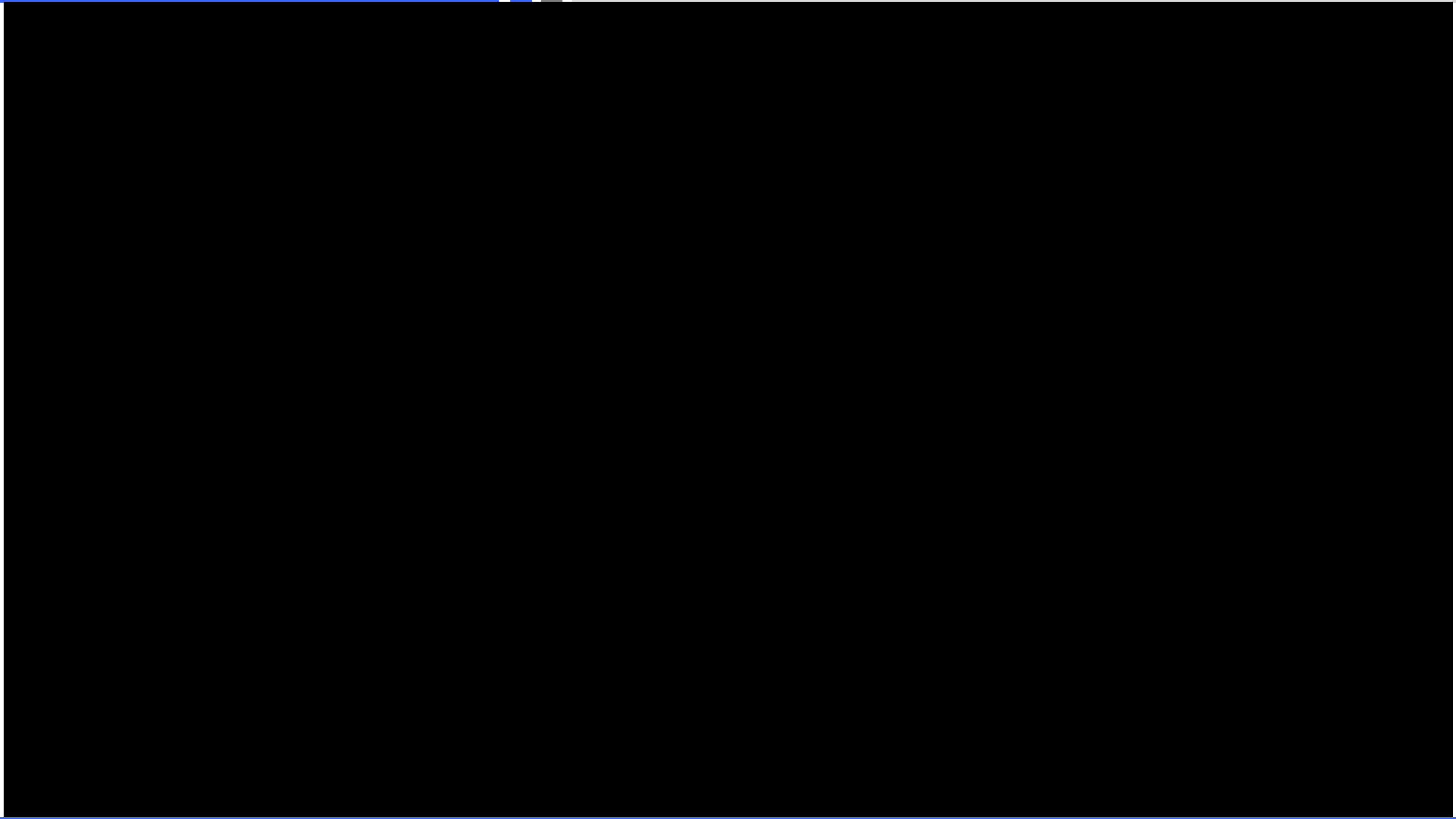
Solid Laminates: Bonded Patch Repairs



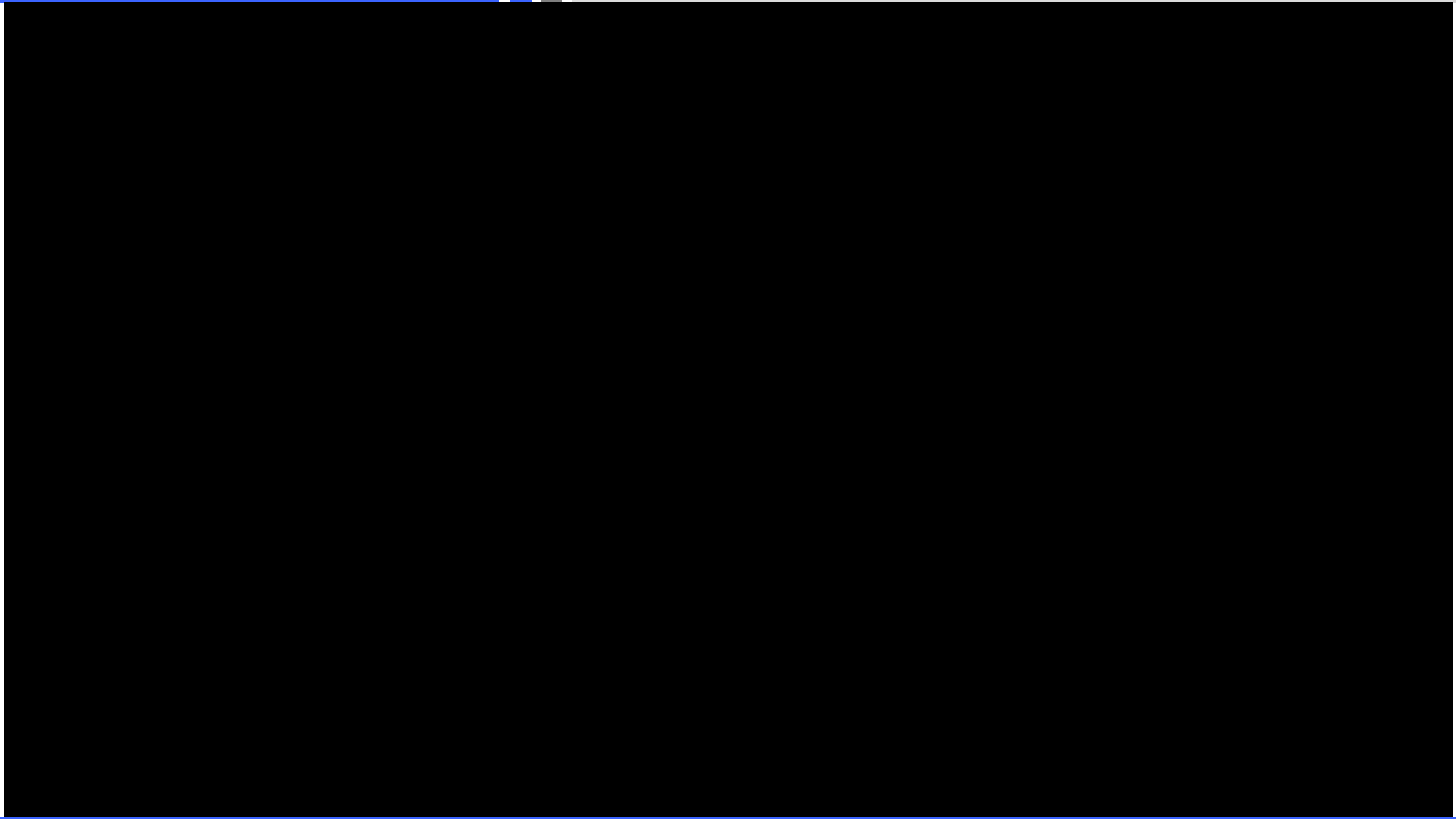
Solid Laminates: Bonded Patch Repairs



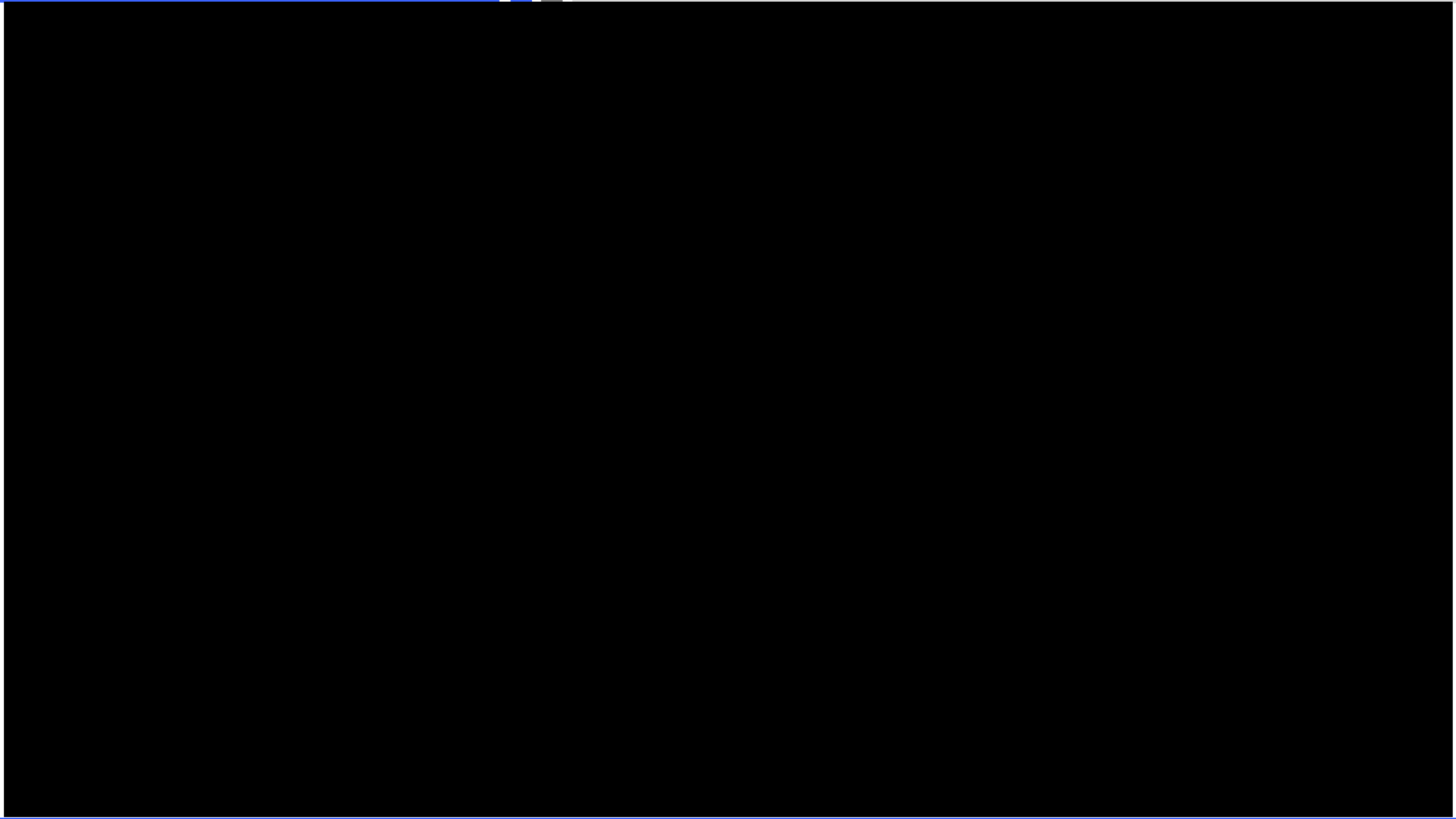
Solid Laminates: Bonded Patch Repairs



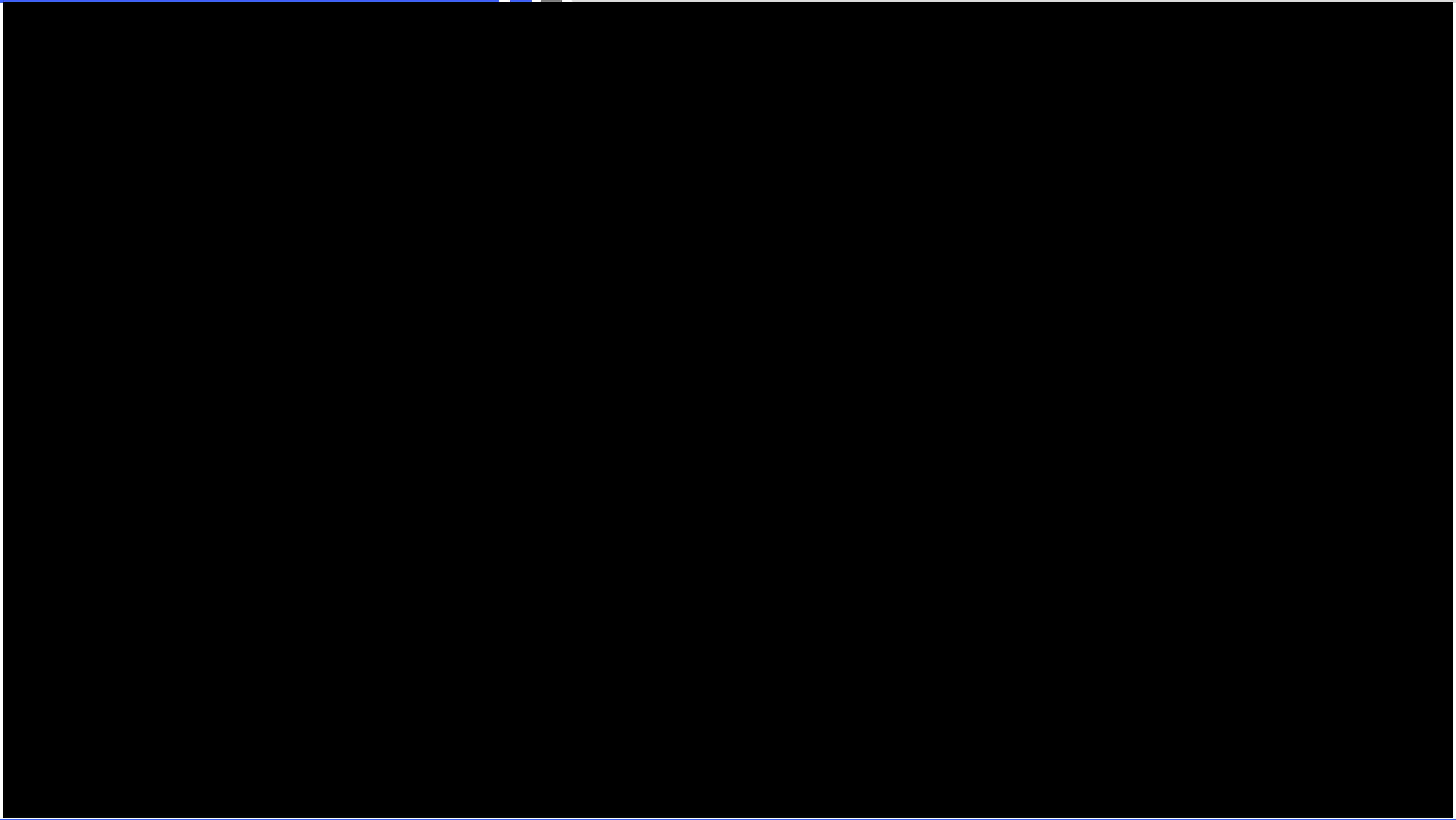
Solid Laminates: Bonded Patch Repairs



Solid Laminates: Bonded Patch Repairs



Solid Laminates: Bonded Patch Repairs



Solid Laminates: Resin Injection Repairs*

- Used to repair minor damage on laminate due to delamination.
- Disadvantages of the resin injection method are
  - The fibers are cut due to drilling holes.
  - Difficult to remove moisture from the damaged area and to achieve complete infusion of resin.
- Can be used on face-sheet or the core of sandwich structure for delamination and disbond.*

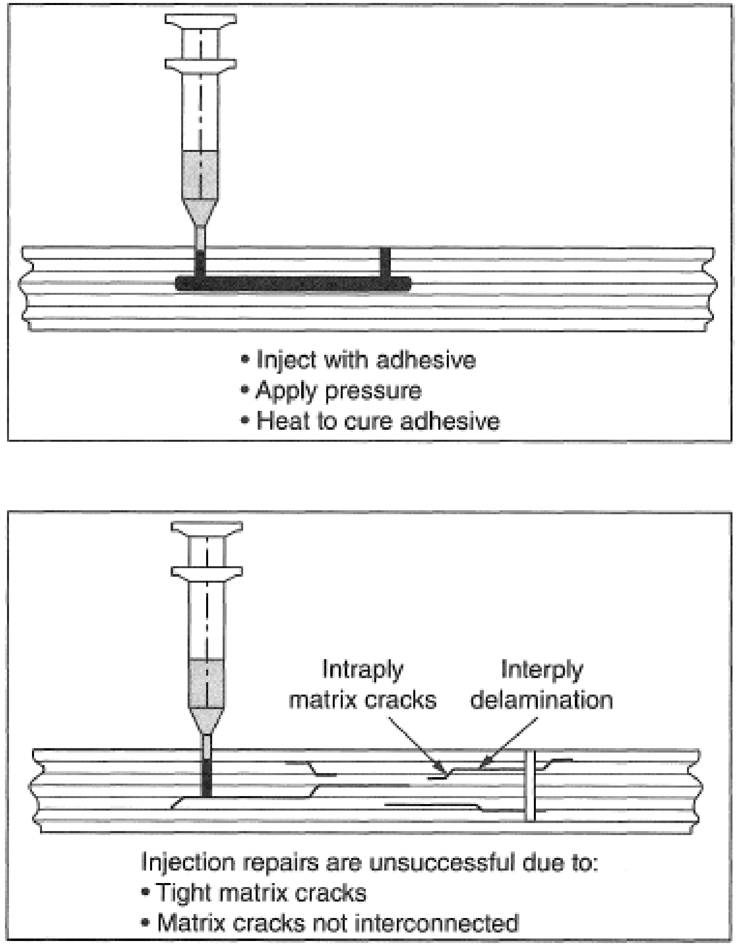
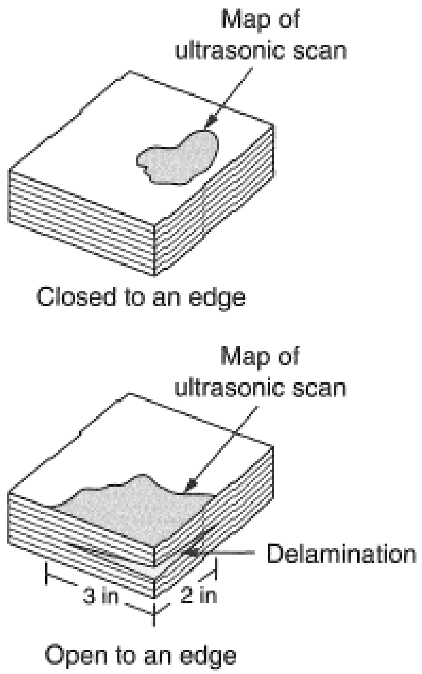
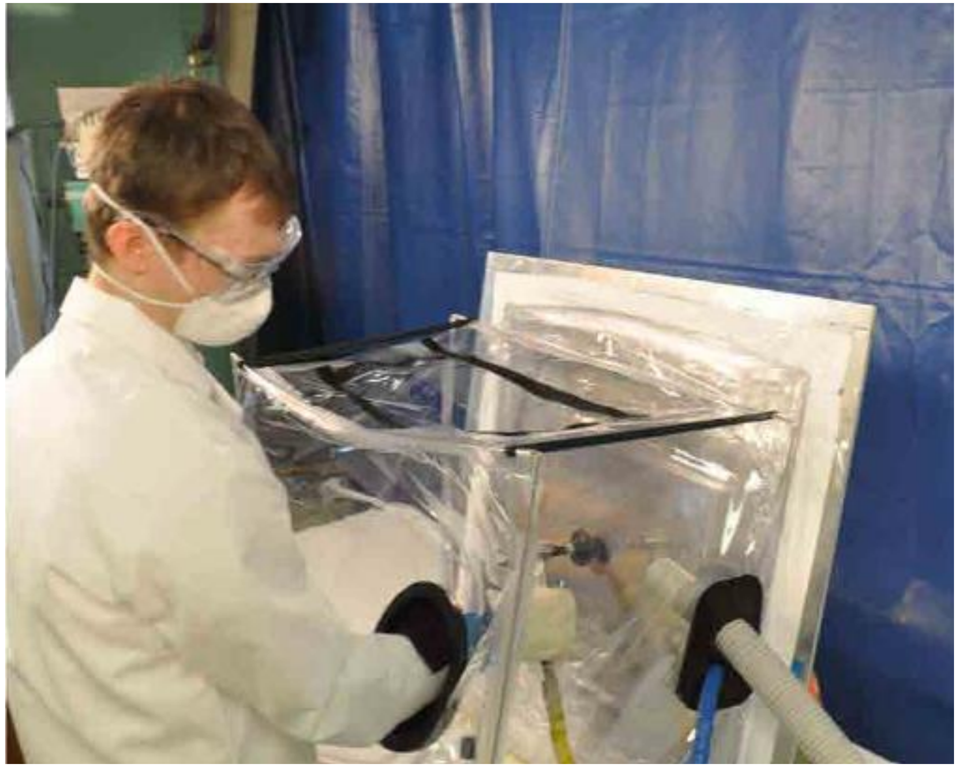


FIGURE 11-59 Resin injection repair for laminate structures.



Composite Patch Bonded to Metal Structures: Repairs to Bonded Metal Structures

- Bonded composite doublers have the ability to slow or stop fatigue crack growth, replace lost structural area due to corrosion grindouts (blendouts), and structurally enhance areas.
- Secondarily bonded pre-cured doublers and in situ cured doublers have been used on a variety of structural geometries ranging from fuselage frames to door cutouts to blade stiffeners.
- Film adhesives using 250°F and above cure are used routinely to bond the doublers to the metallic structure.
- Surface treatment of metal for bonding is critical and care must be taken to create an active surface for bonding in addition to removing contaminates.
- An alternative method is to perform bolted repair.



Example of a portable "cleanroom" for the curing of metal composite
Extracted from Boeing B787 SRM Training Material

Repairs Using Pre-cured Patches*

- Pre-cured patch are composite patches made in advance and then cut to shape to be used in bonded repair.
- This is not restricted to repair to metal structure.*
- Advantages are:
  - Can be made in advance in the autoclave
  - Have the same mechanical properties as the original aircraft structure
  - Easier to bond the patch to the aircraft.
- Disadvantages are:
  - Difficulty of using pre-cured patch to fit to the contour of the damage area.
- Pre-cured patch of the same material of the original structure is preferred as a repair doubler for a bolted repair. It will reduce internal stresses due to different in thermal expansion.*

VIII. Bolted Repair

Bolted Repair VS Bonded Repair*

Bolted Repair*	Bonded Repair*
Repair typically used on composite thicker than 0.125" to ensure sufficient bearing area for load transfer.	Thick composite repair requires several curing sessions or the use of autoclave.
Faster processing time.	Slower processing time.
Repair may be Category B (Require inspections).	Category A (Permanent Repair)
Repair material not sensitive.	Repair material is time, temperature, and moisture sensitive
Heavier repair (additional weight)	Lightweight.
Hole drilling must be done with care and caution. Introduction of new holes reduce cross-sectional area.	No protruding parts or additional fastener holes.
Have to pot the core for sandwich structure before fasteners can be installed.	More suitable for sandwich structure as there is no potential for moisture ingression and no bearing load on the thin face sheets.
No risk of heat damage due to curing.	Risk of heat damage on original structure and porosity forming during the curing of repair plies

Extracted from Boeing Aero QTR 4 2014

VIII. Bolted Repairs

Introduction*

- Bolted repair is typically done on thick composite laminate structures.
- Although composite bolted repair is similar to sheet metal repair, there are differences to be aware like fasteners used, drills, etc.

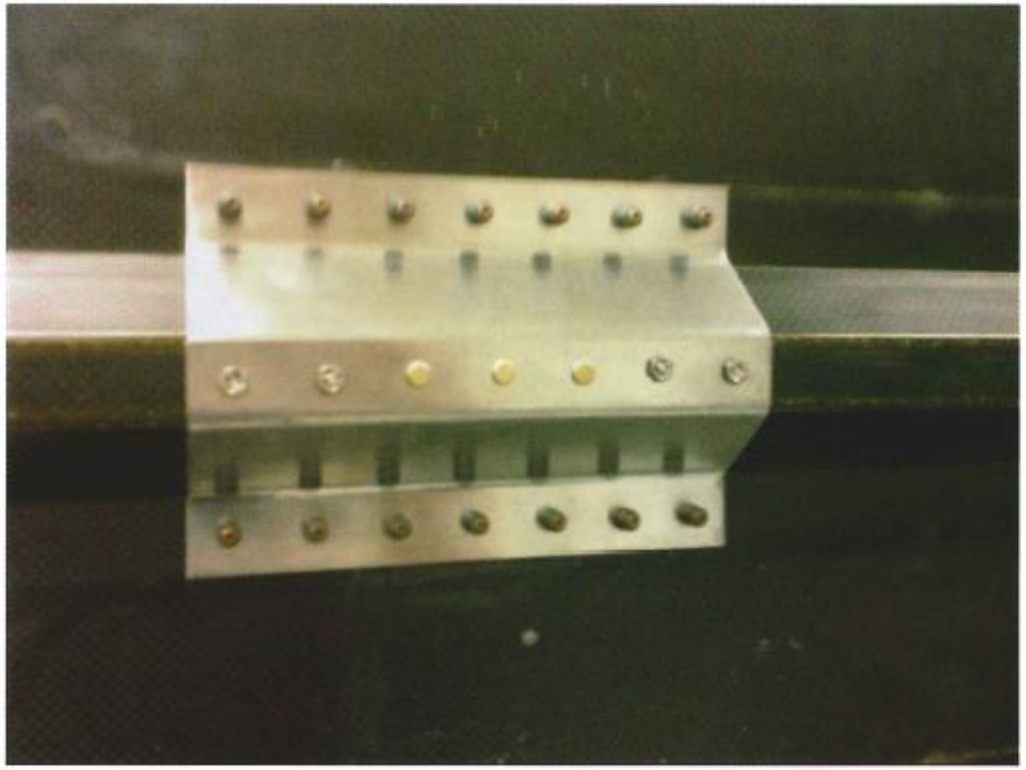


FIGURE 11-63 Stringer repair with titanium formed channel.

Fasteners Used with Composite Laminates & Fastener Materials*

- Specialty fasteners are used for composite structures: threaded fasteners, lock bolts, blind bolts, blind rivets, and specialty fasteners for soft structures honeycomb panels.
- Riveting in composite is restricted due to bucking of rivets can cause delamination. Solid rivets may be used in composites but washers are required on the driven head side to prevent delaminating of the laminate.*
- Main differences between fasteners for metal and composite structures are:
  - Materials: Limited to Titanium alloy, Monel or Stainless Steel alloy
  - The footprint diameter of nuts and collars.
- Carbon fibers composites are cathodic when used with materials such as aluminum or cadmium (a common plating used on fasteners for corrosion protection).

Drilling, Drill Type & Countersinking*

- Drilling holes in composite materials is different from drilling holes in metal aircraft structures:
  - Higher drill speeds (to avoid heat generation)* and lower feeds (to avoid splintering at drill entrance and exit)* are required to drill precision holes.
  - Defects like delamination, separation of bonded composite plies, or fiber breakage may occur at the drill entrance or exit of composite materials.
  - During trimming or drilling of the composites, it should be well supported and firmly backed up with wood or an aluminium plate to prevent backside breakout.
  - Standard tool steels will dull rapidly in the trimming and drilling of composites. As a standard drill dulls, it tends to push against the material rather than cut, causing layer delamination. Diamond-tipped or carbide* cutting and drilling tools will allow more cuts to be taken per tool.
  - Carbon composite are very hard and abrasive. Drill bit for carbon composite is shaped in a spade form or a long tapered form sometimes referred to as a dagger drill.

VIII. Bolted Repairs

Drilling, Drill Type & Countersinking

- Drilling holes in composite materials is different from drilling holes in metal aircraft structures:
  - Sickle-shaped Klenk drill bits have been developed to first pull on Kevlar fibers and then shear them to minimises fiber fuzz. If the Kevlar/epoxy part is sandwiched between two metal parts, standard twist drills can be used.
  - After drilling, each hole is to be visually inspected to ensure correct hole size and location and proper normality, as well as the absence of unacceptable defects.
  - Hole saws may also be used to cut through the core for damage removal, but only when the damage removal cut is to be made perpendicular with the skin.
  - In composite structures, two types of countersink fasteners are commonly used: a 100° included angle tension head fastener or a 130° included angle head fastener.
  - For carbon/epoxy structure, the countersink cutters usually have straight flutes similar to those used on metals. For Kevlar fiber/epoxy composites, S-shaped positive rake cutting flutes are used.

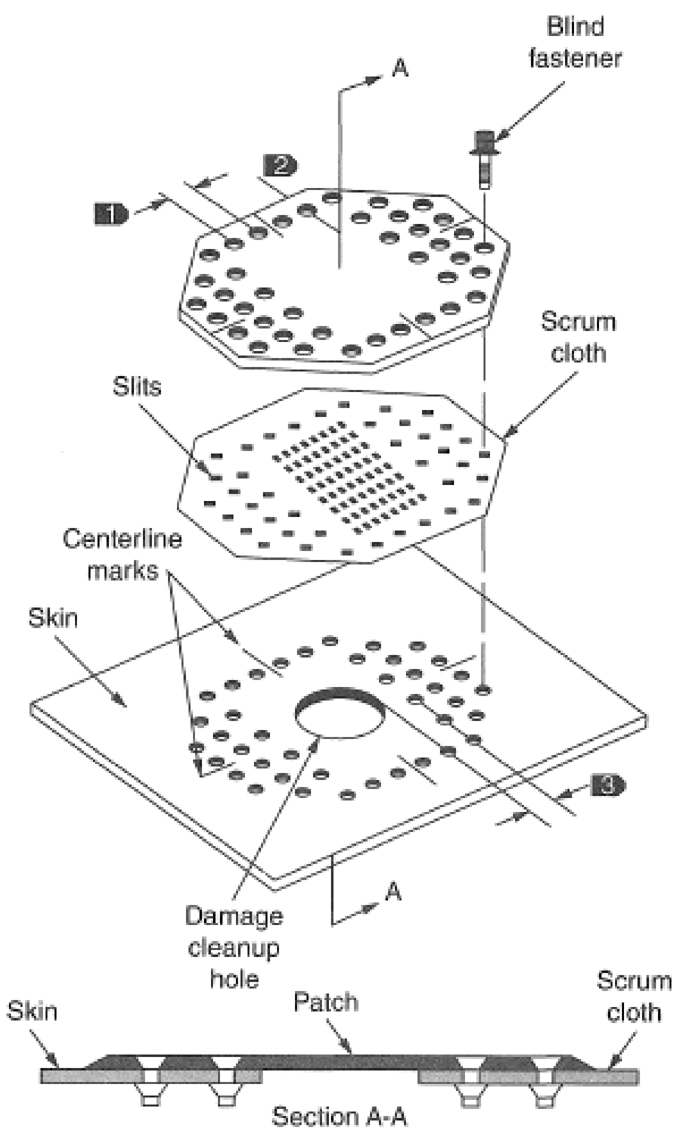
Drill Types

Types of Drill Bits for Composite drilling

http://www.gandtrack.co.uk/products/solid-carbide-tooling

Doubler Material and Edge Margin*

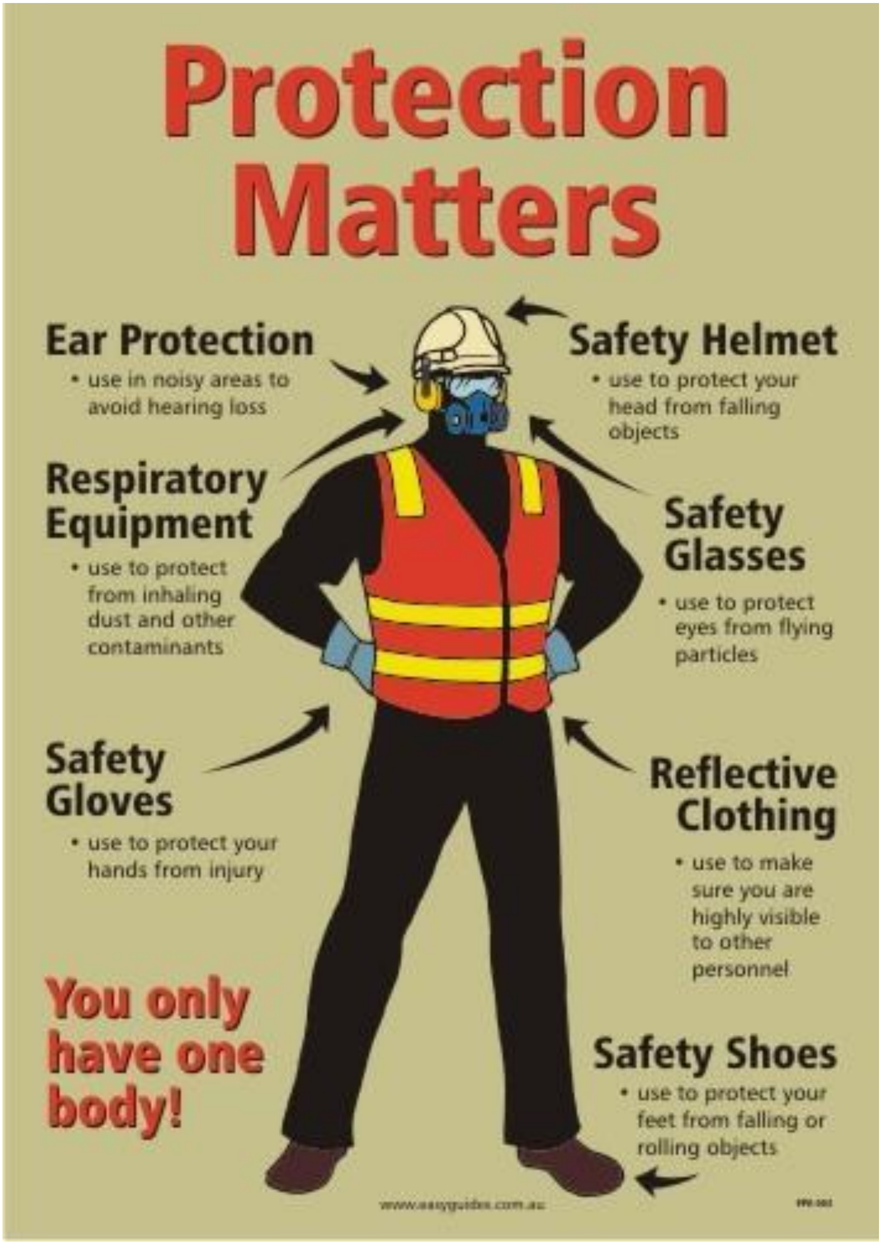
- Doubler material can be any of the three materials : Titanium alloy, Monel or Stainless Steel alloy
- The patch/doubler material is preferred to be titanium alloy as most structural composites are carbon fiber.*
- Aluminium doubler is feasible provided there is a fiberglass ply separating direct contact between the doubler and the carbon composite.*
- A pre-cured patch of the same composite material is preferred to be used as a doubler as it eliminates thermal expansion difference.*
- Edge margin for composite material is to be at least 3D.*



VIII. Bolted Repairs

Repair Safety & Fire Protection

- When performing composite bolted repair, it is important to wear the proper protection equipment as required.
- For example, wearing the correct respiratory equipment during the sanding of composite.
- Attention to fire protection is also important as there is the possibility of powder explosion in the upcoming video upcoming due to the change of material (Non-flammable metal to flammable composite)



Repair Safety & Fire Protection

https://www.youtube.com/watch?v=jENpGlJ0dzA&ab_channel=SpanglerScienceTV

Quiz

Click the **Quiz** button to edit this object

The End